# Energy System Models

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October 21, 2026

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# 1 Introduction

This note deals with arguably *the* main tool that researchers, governments, AI agents etc. reach for when asked to quantitatively assess practically any aspect of the green transition: Energy system models. We use it to construct detailed pathways to net zero emissions, to analyse the effects of carbon pricing, evaluate the value of offshore wind, whether there is a case to invest in more interconnectors, how much a climate target costs and so on.

While detailed energy system models are inherently deeply dependent on technologies and insights from engineering, it can be fascinating and indeed incredibly useful to look at it through the lens of economics. Each model is a central planner's problem: minimise the cost of serving energy demand, subject to physics (e.g. energy balances, capacity limits, weather patterns) and policy (e.g. carbon budgets, land constraints). For economists, the level of technical detail may at times feel discomforting, but by relying on the good old fundamental welfare theorems, the planner's solution can be interpreted as a competitive market equilibrium.[1]

The note presumes that the reader is somewhat familiar with various optimisation techniques and types of problems including Lagrange, Karush-Kuhn-Tucker, and linear programming. Appendix A provides a refresher on the basics. As the note focuses on the models and applications, it also presumes that the reader has access to some supplementary material e.g. textbooks like Cretì and Fontini (2019) or Mulder (2023). All data, figures, tables, and models are collected in a repository here. The repository is equipped with instructions for how to use the models to replicate tables and figures from the note and how to explore further. It is also equipped with a set of instructions that allow you to engage with the repository using your favourite AI assistant. However, **nothing in this note requires you to run anything.**

## 1.1 Where these models sit

Before we start building, it is worth knowing what kind of animal we are dealing with, because you have met its rivals in other courses.

Energy economics is served by two modelling traditions. A **top-down** model writes aggregate energy use as a behavioural relationship, $E_t = F(y_t; \gamma_t)$, where $y_t$ collects prices and income and the parameters $\gamma_t$ are *estimated from observed behaviour* (the computable general equilibrium and econometric models you have seen elsewhere). A **bottom-up** model, which is what this note is about, writes the same quantity as a sum over individual pieces of equipment, $E_t = \sum_i E_{i,t}(x_t; \theta_{i,t})$, where the parameters $\theta_{i,t}$ (efficiencies, capacities, costs) come from *engineering data* rather than from regressions, and the decision rule $E_{i,t}(\cdot)$ comes from a central optimisation rather than from estimated behaviour.

Each buys something and pays for it. Bottom-up models carry enormous technological detail: they know that a combined-cycle plant is 58% efficient, that the wind blew hard on 14 January,

[1] For students with an engineering background, the note may feel rudimentary in terms of technical detail, but hopefully provide plenty of new insights from the economics interpretation.

and that a cable to Norway has 1.7 GW of capacity. That detail is exactly what you need to answer a question about *technologies*, and it is forward-looking in a way estimated models cannot be: A technology that has never been deployed has no history to estimate from. What they pay is scope and discipline. There is no individual optimisation in a planner's problem, so there is no misaligned incentive, no strategic behaviour, no market power and no policy failure; and because the parameters are not identified from observed behaviour, validating the model is genuinely hard. Throughout the note, we will attempt to validate things empirically holding model output against market data or refer to academic literature that does.

## 1.2 How the note is built

The note is organised as one model growing up. Section 2 is the seed: a single hour, ten technologies, and the entire dual vocabulary: merit order, price, rents, the value of lost load, tax versus cap. Section 3 adds time and weather, and with them the economics of intermittency: prices become distributions, and a technology's value comes to depend on *when* it produces. Section 4 couples the hours through a battery and through a ramping constraint; Section 5 couples electricity to heat through a heat pump; Section 6 couples places through interconnectors. Every coupling is one new component and one new dual, never a new theory. Section 7 then makes capacity itself a choice: investment, screening curves, and the long-run zero-profit condition that ties the rents of earlier sections to cost recovery. Section 8 runs the same machinery over Denmark's whole neighbourhood: one carbon budget across twelve zones, interconnectors that can be built, flexible vehicle charging and a hydrogen sector, and asks which of the Danish lessons survive the change of scope. Section 9 closes by turning the tools on the model itself: how much of a capacity expansion answer is an artefact of the weather year it saw, and what the linear planner structurally cannot say.

Four appendices support the text. Appendix A is a refresher on the toolkit of Lagrangians, Karush–Kuhn–Tucker (KKT) conditions and duality that the main sections use freely; Appendix B records where every number comes from; Appendix C is an inventory of what ingredients the models cover and what they do not; and Appendix D places the note's results against the recent literature on the European system.

Finally, some comments on the structure:

i. Most sections open with a complete optimisation problem: objective, constraints, and every multiplier named. Because each model is the previous one plus a little, whatever is new is printed in blue.

ii. Each model section closes with the same small table: constraint, dual, what it measures, and its market counterpart. By Section 9 the tables stack into one mental map from optimisation to market. Learn to fill this table in for any model you meet and you have learned what these models are for.

iii. A handful of important results are set off and numbered (the merit order, tax-cap equivalence, the arbitrage band, the ramping wedge, long-run zero profit).

iv. Every model simplifies, and it is important to be able to know which direction it pushes the model. Most sections set up boxes that take one assumption at a time and reason out the sign of its error, usually without solving anything.

v. Wherever the model has a real-world counterpart we put the two side by side: model prices against Nord Pool prices, model congestion rents against the bottleneck income the transmission system operators (TSOs) actually collected. The disagreements teach as much as the fits, and each one is traceable to a named modelling assumption.

# 2 The Economic Dispatch

The first model is incredibly simple: Given a fixed set of power plants and a known level of demand for electricity, what is the cost-effective way to serve this demand. This is the *economic dispatch* problem. We will use this to very briefly talk about the marginal costs of operating different power plants, then practice setting up the linear program (LP), and interpreting the planner solution in terms of more familiar terms like equilibrium prices, operating profits, and the cost of carbon constraints.

> **Reading alongside this section.** This section assumes only that you can set up a Lagrangian and read KKT conditions; Appendix A is a refresher if one is needed. For institutional background on how a wholesale electricity market is actually organised, who bids what, when, and into which auction, see e.g. Mulder (2023, ch. 1–3) or Cretì and Fontini (2019, ch. 3–5). The lecture note "Generation Technologies" (Berg 2026b) provides an introduction to different power plants and their characteristics. For the interested reader, Dechezleprêtre et al. (2023) holds some interesting evidence on carbon-pricing in practice (focus on the EU Emissions Trading System, the EU ETS).

## 2.1 Technologies and marginal cost

In this simple linear model, we characterize power plants – here indexed $g$ – using only a handful of fixed parameters: First, they have a fixed size, represented by their *generating capacity* $\bar{q}_g$ expressed in MW, which states what the total amount of power (MWh) the plant can deliver in a single hour. Second, we assume that they have a constant *conversion efficiency* $\eta_g$, meaning that for each MWh of primary energy inputs, it produces $\eta_g$ units of MWh electricity. Third, they have constant other non-energy *variable operating costs* $o_g$ per MWh of power output (measured in €/MWh). Fourth, if the power plant relies on fuel injection as the primary energy input and $\varphi_g$ measures the $CO_2$ content in tonnes/MWh, the plant emits $e_g = \varphi_g/\eta_g$ tonnes $CO_2$ per MWh of electricity it produces; This is the plant's *emission intensity*.

Formally, this means that the *marginal costs* $c_g$ and the *emission intensity* $e_g$ are given by:

$$c_g = \frac{p_g^F}{\eta_g} + o_g. \tag{1}$$

$$e_g = \frac{\varphi_g}{\eta_g}. \tag{2}$$

Constant marginal costs up to a hard capacity limit is the modelling decision that makes everything in this note linear, and it is a good first approximation for some technologies.[2]

To get a sense of how different technologies compare, Table 1 presents main technologies from the Danish Energy Agency (DEA).[3] The table is deliberately small enough to redo every calculation in this section by hand, and it is generated from the same file the model reads, so the numbers here are the numbers behind every figure. Its abbreviations are PV for photovoltaic, CHP for combined heat and power (plants that also deliver district heat), and CCGT and OCGT for combined-cycle and open-cycle gas turbines.

Wind and solar, naturally, have no fuel inputs and generally near-zero marginal costs. In our table, waste incineration plants also feature very low marginal costs, because they are essentially paid to handle the waste. Beyond these generator types, marginal costs ranks then (from low-to-high) coal, then gas, biomass, and finally oil peaker; unfortunately, this is the same ranking in terms of emission intensity (except for the oil peaker). That tension – the cheap technologies are either capacity-limited or dirty – is what makes the dispatch problem economically interesting, even in this highly stylized framework.

## 2.2 The planner's problem

We now consider a static model of the electricity wholesale market, with a fixed set of generators $g \in \mathcal{G}$ with characteristics as outlined above, and a fixed demand or *load* $D$ to serve. The planner's optimisation problem is to choose how much each generator produces $q_g \geq 0$ to minimise costs:

$$\min_{q} \quad \sum_{g \in \mathcal{G}} c_g \, q_g \tag{3a}$$

$$\text{s.t.} \quad \sum_{g \in \mathcal{G}} q_g = D \qquad (\lambda) \tag{3b}$$

$$0 \leq q_g \leq \bar{q}_g \quad \forall g \qquad (\underline{\mu}_g, \overline{\mu}_g), \tag{3c}$$

Here (3a) is the total system costs, which at this stage only includes the sum of marginal costs of generation. (3b) is the market clearing condition. (3c) is the capacity and non-negativity constraints for each $g$. Variables in parenthesis ($\lambda$, $\overline{\mu}_g$, $\underline{\mu}_g$) are dual variables/shadow variables/Lagrange multipliers on the relevant constraint; they are the subject of Section 2.3.

[2] It is worst for plants with minimum load levels, start-up costs and ramping limits — the *unit-commitment* features that make the true problem non-convex. We return to what this omission does to prices in Section 9.2.

[3] Parameters are drawn from PyPSA, Danish Energy Agency, et al. (2025), which draws chiefly on the Danish Energy Agency's catalogue (Danish Energy Agency and Energinet 2024), at a pinned version; the fuel prices and carbon contents are the note's own assumptions, and are used unchanged throughout the note.

Table 1: The small technology table. Marginal costs and emission intensities are derived, not assumed; fuel prices are stylised 2025 forward-market levels.

| Technology | Fuel | $p_g^F$ €/MWh$_{th}$ | $\eta_g$ | $o_g$ €/MWh | $\varphi_g$ t/MWh$_{th}$ | $c_g$ €/MWh | $e_g$ t/MWh |
|---|---|---|---|---|---|---|---|
| Solar PV[a] | – | – | – | 0.0 | – | 0.0 | 0.00 |
| Onshore wind[b] | – | – | – | 1.5 | – | 1.5 | 0.00 |
| Offshore wind[c] | – | – | – | 0.0 | – | 0.0 | 0.00 |
| Waste CHP[d] | waste | 0 | 0.21 | 28.5 | 0.15 | 28.5 | 0.73 |
| Coal[e] | coal | 12 | 0.36 | 3.3 | 0.34 | 37.0 | 0.94 |
| Wood-chip CHP[f] | solid biomass | 25 | 0.27 | 4.9 | 0.00 | 97.7 | 0.00 |
| Wood-pellet CHP[g] | wood pellets | 38 | 0.27 | 4.9 | 0.00 | 145.9 | 0.00 |
| Gas CCGT[h] | gas | 35 | 0.57 | 4.6 | 0.20 | 66.0 | 0.35 |
| Gas OCGT[i] | gas | 35 | 0.41 | 4.8 | 0.20 | 91.2 | 0.49 |
| Oil peaker[j] | oil | 55 | 0.35 | 6.3 | 0.26 | 163.5 | 0.73 |

[a] Utility-scale photovoltaics. No fuel, and no variable operating cost in the source. Technology-data key `solar-utility`.
[b] Onshore wind turbines. Technology-data key `onwind`.
[c] Offshore wind turbines. The variable operating cost is nominal. Technology-data key `offwind`.
[d] Waste-to-energy plant. The fuel is free at the gate (gate-fee revenue ignored) but the plant carries a large variable operating cost; only the fossil share of municipal waste is counted as emitting. Technology-data key `waste CHP`.
[e] Hard-coal plant. Technology-data key `coal`.
[f] District-heating biomass plant, priced on wood chips; biogenic $CO_2$ counted as zero. Technology-data key `central solid biomass CHP`.
[g] The same plant burning pellets: identical machine, more expensive fuel. Technology-data key `central solid biomass CHP`.
[h] Combined-cycle gas turbine. Technology-data key `CCGT`.
[i] Open-cycle gas turbine – the same fuel, a cheaper and less efficient machine. Technology-data key `OCGT`.
[j] Oil-fired peaking plant. Technology-data key `oil`.

We note that because we take demand $D$ as given (i.e. demand is perfectly inelastic), the objective of minimising the system costs follows trivially. However, once we start dealing with where this demand comes from, in particular if there are different types of consumers, it is no longer trivial to identify one "correct" objective function. Below, we discuss the simplest adjustment with a consumer that has a *value of lost load*; later we also return to this issue again to discuss winners and losers of various policies.

Before we turn to the formal part, it can be instructive to start with a visualisation of this model in Figure 2.1, as it becomes clearly evident what the optimal solution is. The demand side of this market is straightforward: There is simply a fixed load of $D = 5$ GW represented by the vertical dashed line. On the supply side, order the generators by marginal cost, $c_{(1)} \leq c_{(2)} \leq \ldots$ and fill up demand from the bottom: run the cheapest plant at capacity, then the next cheapest, until load is met. This generation schedule is called *merit order dispatching*. In our case, solar, wind and waste run first, then coal, and the CCGT fleet tops up: the merit order dispatch is exactly the supply curve of a competitive market, and the intersection with the vertical demand curve at the marginal generator's cost is the market equilibrium. Let us now look at this formally.

## 2.3 Duality: the price and the rents

The multipliers named in parentheses in (3) are where the economics live, and extracting them is a routine application of the Kuhn–Tucker technique. Following the convention of Appendix A

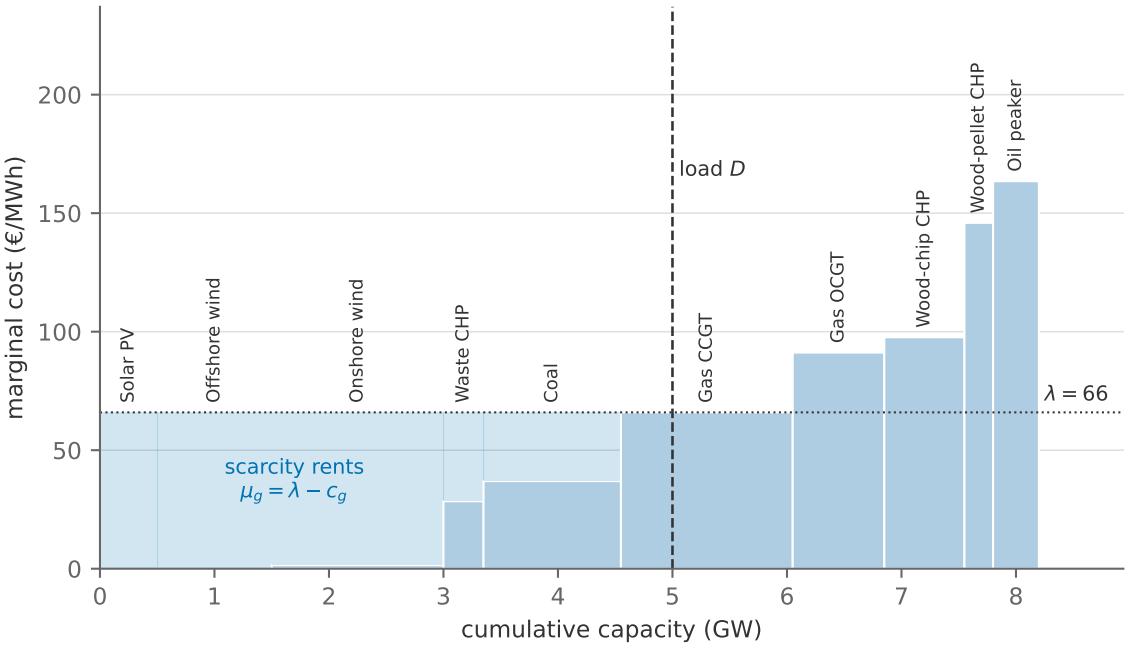



Figure 2.1: The merit order for the instance of Table 1, with load $D = 5$ GW. Each block is one technology, ordered by marginal cost; block width is capacity. The marginal generator is the CCGT, and the price is its marginal cost, $\lambda = 66.0$ €/MWh. The shaded area between the price and the supply curve is the sum of scarcity rents $\sum_g \overline{\mu}_g \bar{q}_g$ — the producer surplus.

— the Lagrangian is always set up for a *maximisation* — we restate the problem as maximising *minus* the system costs and define the Lagrangian

$$\mathcal{L}^{\text{disp}} = -\sum_g c_g q_g + \lambda\Big(\sum_g q_g - D\Big) + \sum_g \overline{\mu}_g \left(\bar{q}_g - q_g\right) + \sum_g \underline{\mu}_g q_g, \tag{4}$$

where $\lambda, \overline{\mu}_g \geq 0, \underline{\mu}_g \geq 0$ are the shadow variables associated with the market clearing condition, the capacity constraint, and the non-negativity constraint, respectively.

A word on the superscript, which is a convention we keep for the rest of the note. Each section adds one piece of machinery to the model, and each therefore has its own Lagrangian; the superscript names which one. Because the Lagrangian is *linear* in the constraints, a section's Lagrangian is always the previous one plus the terms its new machinery contributes — never a fresh derivation. $\mathcal{L}^{\text{disp}}$ here is the base case; Section 3 repeats it over the hours of a year as $\mathcal{L}^{\text{vre}}$, and Sections 4 to 8 each add to that. When we say "the generators' conditions are unchanged", this additive structure is why.

The conditions for optimality are then

$$\lambda - c_g + \underline{\mu}_g - \overline{\mu}_g = 0 \qquad \forall g, \tag{5a}$$

as well as the complementary slackness conditions

$$\overline{\mu}_g(\bar{q}_g - q_g) = 0, \qquad \bar{q}_g - q_g \geq 0, \qquad \overline{\mu}_g \geq 0, \qquad \forall g, \tag{5b}$$

$$\underline{\mu}_g q_g = 0, \qquad q_g \geq 0, \qquad \underline{\mu}_g \geq 0, \qquad \forall g, \tag{5c}$$

One important and general insight when interpreting KKT conditions is related to the pairs of shadow variables of upper ($\overline{\mu}_g$) and lower ($\underline{\mu}_g$) bounds: Only one of the pairs of shadow variables can be active ($> 0$) at any point. To see this, consider our small example here: If $\overline{\mu}_g > 0$ the complementary slackness condition (5b) states that $q_g = \bar{q}_g > 0$. This in turn means that $\underline{\mu}_g = 0$ for the second complementary slackness condition (5c) to hold.

These conditions are worth dwelling on a bit more, at least as good practice, as they are the formalisation of the merit order dispatch solution.

**Result 1** (Merit order and the price)**.** *At the cost-minimising dispatch, generators run in order of marginal cost: every generator with $c_g < \lambda$ runs at capacity, every generator with $c_g > \lambda$ is idle, and the multiplier on market clearing equals the marginal cost of the marginal generator, $\lambda = c_m$. Moreover, this dispatch and this $\lambda$ are a competitive equilibrium: if every generator takes $\lambda$ as a price and maximises profit, markets clear.*

To see it, run through the three positions a generator can occupy:

- $0 < q_g < \bar{q}_g$: both slackness conditions force $\overline{\mu}_g = \underline{\mu}_g = 0$, so eq. (5a) reads $c_g = \lambda$. A partially dispatched generator must be *marginal*, and it pins down $\lambda$.
- $q_g = \bar{q}_g$: now $\underline{\mu}_g = 0$ and $\overline{\mu}_g = \lambda - c_g \geq 0$. A generator runs at capacity exactly when it is cheaper than the marginal one.
- $q_g = 0$: now $\overline{\mu}_g = 0$ and $\underline{\mu}_g = c_g - \lambda \geq 0$. A generator sits idle exactly when it is more costly.

Collecting the multipliers also delivers the LP dual of (3),[4]

$$\min_{\lambda,\, \overline{\mu} \geq 0} \quad \sum_{g \in \mathcal{G}} \overline{\mu}_g \bar{q}_g - \lambda D \tag{6a}$$

$$\text{s.t.} \quad \lambda - \overline{\mu}_g \leq c_g \quad \forall g, \tag{6b}$$

whose optimum equals the primal's maximised objective — minus total system costs (strong duality; Appendix A). The dual objective will earn its keep below: its two terms are producer surplus and consumer expenditure. The economics enter through the interpretation of the multipliers.

**$\lambda$ is the price.** Formally, $\lambda$ is the shadow price of eq. (3b): the increase in minimised system cost from one additional MWh of load. By the argument above it equals the marginal generator's cost, $\lambda = c_m$. But it is also the market-clearing price of a competitive market. Suppose each generator takes a price $\lambda$ as given and chooses output to maximise profit, $\max_{0 \leq q_g \leq \bar{q}_g} (\lambda - c_g)\, q_g$. Its supply correspondence is: nothing if $c_g > \lambda$, everything if $c_g < \lambda$, anything if $c_g = \lambda$. At $\lambda = c_m$ these choices can be made to sum exactly to $D$ — the market clears, and the resulting dispatch is the

[4] Note that because lower bound on production is zero, $\underline{q}_g = 0$, the full dual problem actually includes the sum $\sum_g \underline{\mu}_g \underline{q}_g$, which by construction is always zero and can thus be dropped here.

planner's. This is the first welfare theorem in its smallest habitat, and it is the licence for the phrase used throughout this note: *the model's dual variable is the market's price*. A single LP therefore answers both a planning question (what should run) and a market question (what will the price be).

**$\overline{\mu}_g$ is a scarcity rent.** The multiplier on eq. (3c) is the value of one more MW of capacity of type $g$: relaxing the constraint saves $\lambda - c_g$ by displacing the marginal generator. For every inframarginal plant, $\overline{\mu}_g = \lambda - c_g$ is exactly its operating profit per MWh — the margin it earns selling at $\lambda$ what it produces at $c_g$. In our base run the onshore wind fleet earns $\mu = 64.4$ €/MWh and coal earns $\mu = 29.0$ €/MWh, while the marginal CCGT earns nothing. Summing over plants, producer surplus is $\sum_g \overline{\mu}_g \bar{q}_g$ — the shaded area in Figure 2.1, and precisely the first term of the dual objective (6a).

Two remarks on the rents. First, they are *rents* in the classical sense: returns to a scarce factor (capacity), not profits from any productive cleverness, and taxing them away would not change today's dispatch. But today's rents are tomorrow's investment signal — in Section 7 the condition that a technology's rents, summed over the year, exactly cover its annualised capital cost is the long-run equilibrium condition, an idea going back to Boiteux (1960). A plant that never earns a rent is a plant nobody would build. Second, the marginal plant earning zero operating profit in the single period is not a pathology of the model; it is the knife-edge that makes the level of the price determinate.

## 2.4 The value of lost load

There is an assumption buried in eq. (3b) that we should drag into the open: Writing market clearing as an *equality* says that the load will be served — full stop, whatever it costs. If total capacity were below $D$, the problem would simply have no feasible solution. In effect we have told the planner that electricity is infinitely valuable.

Real systems do not work that way. When supply cannot meet demand, some demand is cut: interruptible industrial contracts are called, then rolling blackouts. So let us put that possibility into the model and see what it does. Suppose consumers are willing to pay $v$ euros per MWh to avoid having their load cut — the *value of lost load* (VoLL). Let $d \in [0, D]$ be the load actually served, so that $D - d$ is the load shed. The planner now maximises welfare rather than minimising cost:

$$\max_{q,\, d} \quad v\, d - \sum_g c_g\, q_g \tag{7a}$$

$$\text{s.t.} \quad \sum_g q_g = d \qquad (\lambda) \tag{7b}$$

$$0 \le q_g \le \bar{q}_g \quad \forall g \qquad (\underline{\mu}_g, \overline{\mu}_g)$$

$$0 \le d \le D \qquad (\underline{\theta},\ \overline{\theta}). \tag{7c}$$

Everything in blue is new; everything else is equations (3) unchanged. The generators' first-order condition (5a) is untouched. The new one is for $d$:

$$v - \lambda + \underline{\theta} - \overline{\theta} = 0, \tag{8}$$

with the usual complementary slackness on the two bounds. Read it exactly as we read the generators' condition — by asking, one at a time, where $d$ can sit:

- Upper bound: no load shedding. There is enough cheap capacity to serve everything. Then $d = D$, $\underline{\theta} = 0$, and $\overline{\theta} = v - \lambda > 0$: consumers value the served load above what it cost to serve. The price is set by the marginal *generator*, exactly as before, and everything in Section 2.3 goes through word for word.
- Interior: partial load shedding. Some load is cut but not all. Then neither bound binds, $\underline{\theta} = \overline{\theta} = 0$, and eq. (8) gives $\boldsymbol{\lambda = v}$. The price is no longer any generator's marginal cost — it is the consumers' willingness to pay. This is **scarcity pricing**.
- Lower bound: everything is shed. Only if consumers value electricity below even the cheapest generator's cost, $v < \min_g c_g$, which is not a case anyone worries about in practice.

Two important consequences are worth noting here:

**VoLL is the price cap, and the scarcity price.** The price can never exceed $v$: if it tried, shedding load would be cheaper than serving it. And in any hour where capacity runs short, the price is *exactly* $v$, regardless of what any generator costs. This means that the demand curve of Figure 2.1 shifts from vertical to horizontal at $v$. Empirical estimates of VoLL for European systems run to roughly 10,000–25,000 €/MWh (Mulder 2023, Box 7.9) — two orders of magnitude above the 66.0 €/MWh of our base case. It is important to note, however, that this measures the costs that consumers are willing to pay to avoid having unexpected black-outs. The more general idea of having "flexible demand", i.e. where consumers react to real time prices throughout the year, is a completely different story that we will touch upon later in the course.

**This is how capacity gets paid.** Look again at the scarcity rent $\overline{\mu}_g = \lambda - c_g$. A peaking plant with high marginal cost earns nothing in a normal hour, because it is not running. It earns $v - c_g$ in a scarcity hour — an enormous margin, but collected rarely. That is not a market failure; it is the mechanism by which an energy-only market finances capacity that exists to be available rather than to generate.

> **Which way does it bias the answer?** *Assuming the load is always served ($v = \infty$).*
>
> The rest of Section 2 and all of Section 3 keep market clearing as an equality — the case $v \to \infty$. Which way does that bend the results? It removes the top of the price distribution: the model's highest possible price becomes the most costly plant's marginal cost instead of VoLL. So a fixed-demand model **understates** average prices, **understates** the revenue of peaking capacity, and

**understates** the profitability of storage and demand flexibility, all of which earn disproportionately in the extreme hours. It does not bias the dispatch at all, as long as capacity is sufficient — which is why we can safely postpone it. The moment capacity becomes a *choice* (Section 7), the omission stops being harmless: a model that cannot price scarcity cannot explain why anyone builds a peaker. Watch this bias reappear, measured, in Section 3.6, where the real day-ahead market of western Denmark (DK1) spikes to many times any plant's marginal cost and the model's prices cannot follow.

## 2.5 A carbon tax

Suppose the planner (or a government legislating on top of the market) prices carbon at $\tau$ euros per tonne. Each generator's effective marginal cost becomes

$$c_g(\tau) \;=\; \frac{p_g^F}{\eta_g} + o_g + \tau\, e_g, \tag{9}$$

and the dispatch problem is otherwise unchanged: the tax works entirely by re-sorting the merit order. Technologies are re-ranked by $c_g(\tau)$, and because the tax loads on $e_g$, the re-ranking is systematic: dirty-and-cheap technologies slide up the order, clean-and-costly ones slide down relative to them.

Figure 2.2 shows the experiment at $\tau = 85$ €/t, a level in the range of recent EU ETS prices: coal's marginal cost rises from 37.0 to 117.2 €/MWh, the CCGT's only from 66.0 to 95.5 €/MWh. System emissions fall from 1545 t to 777 t, by 50%, without any change in demand or in the capital stock.

The price effects repay close reading:

i. *Pass-through.* The electricity price rises by 29.3 €/MWh, which is exactly $\tau \cdot e_m$ for the marginal technology (the CCGT, $85 \times 0.34$). Consumers pay the carbon cost *of the marginal unit*, not the average carbon content of what they consume (with a clean marginal technology the pass-through would be zero even if inframarginal generation were emission intensive).

ii. *Windfalls.* Every clean inframarginal generator pockets the price increase: onshore wind's rent rises from 64.4 to 96.2 €/MWh without wind doing anything differently. Carbon pricing is, among other things, a transfer to clean capacity, which is precisely the investment signal at work, and a recurring political-economy complaint about the instrument.

iii. *The tax falls on rents too.* Coal, which previously earned 29.0 €/MWh, now earns nothing. The burden of the tax is shared between consumers (higher $\lambda$) and dirty producers (lost rents), in proportions the merit order determines.

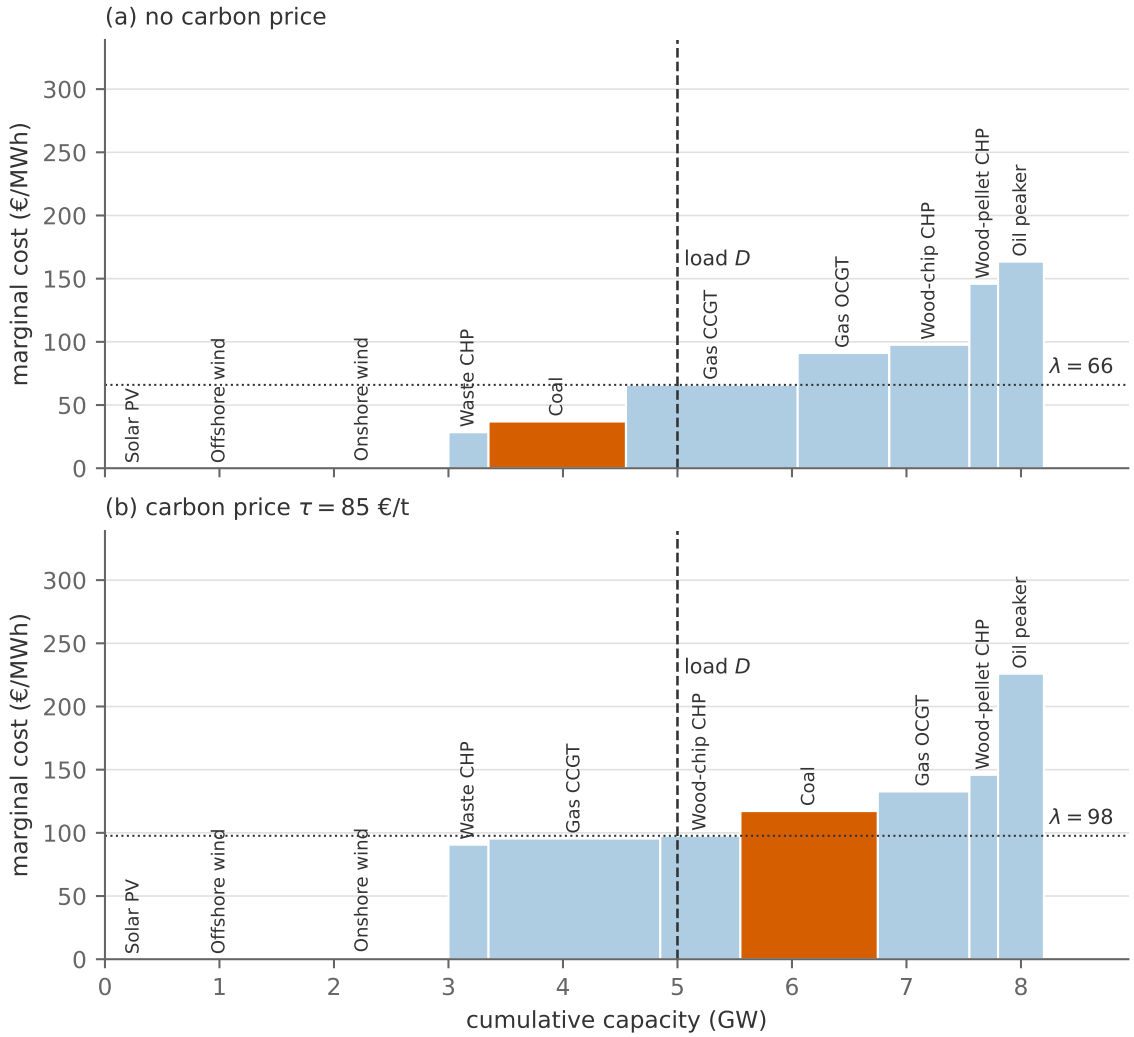



Figure 2.2: The merit order without (a) and with (b) a carbon price of $\tau = 85$ €/t

## 2.6 An emissions cap

The other canonical instrument fixes the quantity instead of the price: total emissions may not exceed a cap $E$. Append to the planner's problem (3) the constraint

$$\sum_{g \in \mathcal{G}} e_g \, q_g \;\leq\; E \qquad (\sigma), \tag{10}$$

with multiplier $\sigma \geq 0$. This is the additive structure at work for the first time: The new constraint adds the term $\sigma \left( E - \sum_g e_g q_g \right)$ to $\mathcal{L}^{\text{disp}}$ in (4), contributing $-\sigma e_g$ to its derivative with respect to $q_g$, so for any active generator, the optimality condition now reads

$$\lambda \;=\; c_g + \sigma e_g + \overline{\mu}_g, \tag{11}$$

which is eq. (9) with $\sigma$ in the role of $\tau$: the shadow price of the cap is an *endogenous carbon price*. Equally useful is the reading of $\sigma$ as *marginal abatement cost*: it measures what the last tonne of abatement cost the system, i.e. the cost premium of the cleaner generation that replaced dirtier

generation at the margin.

**An old friend, arriving from a new direction.** It is worth pausing on that last sentence, because you have met $\sigma$ before. In the lecture note "A Simple Model of Abatement Costs" (Berg 2026a), we started from an aggregate relationship $F(E)$ with emissions $M = \varphi E$, and derived the marginal abatement cost (MAC) curve

$$MAC \equiv -\frac{\partial C}{\partial A} = \frac{F'(E) - p^E}{\varphi}, \tag{12}$$

where $A = M_0 - M$ is abatement and $C$ is net output. The resulting MAC curve is upward sloping, and the optimal policy ensures that this equals the marginal damage, $MAC = D'(M)$.

What eq. (11) says is that a bottom-up model *derives* the same object: We can get the marginal abatement cost curve by resolving the system model for a range of different caps $E$ and recording $\sigma(E)$. Each point on the curve is a named pair of technologies swapping places in the merit order. Figure 2.3 presents the two side by side (so you don't have to fetch the other lecture).

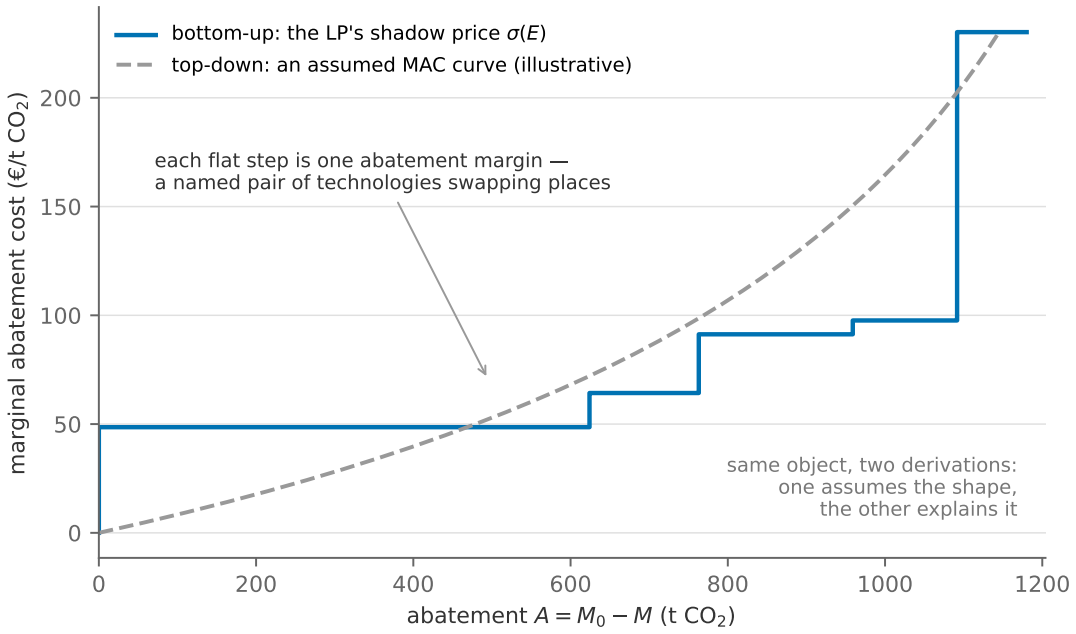



Figure 2.3: The marginal abatement cost curve, derived and assumed. The staircase is this section's LP: $\sigma(E)$ from the cap sweep, plotted against abatement $A = M_0 - M$. The dashed curve is the smooth, convex MAC of the one-good top-down model (12), calibrated only so that the two span the same range.

We note that in this illustrative example next to a very stylized bottom-up model, the two shapes seem very different. In reality, however, the more realism we add to the bottom-up curve, the MAC curve often converge to a more smooth function. Thus, the *shape of the two* are not necessarily the big result here, but rather where they come from and how they may be interpreted.

**Result 2** (Tax–cap equivalence, and its limits)**.** *A cap at level $E$ with shadow price $\sigma(E)$ induces the same dispatch, electricity price and emissions as a carbon tax $\tau = \sigma(E)$; and a cap set at the realised emissions of any tax replicates that tax's outcome. However, it is important to note up*

*front that the equivalence breaks in many different settings, for instance by introducing uncertainty in the model.*

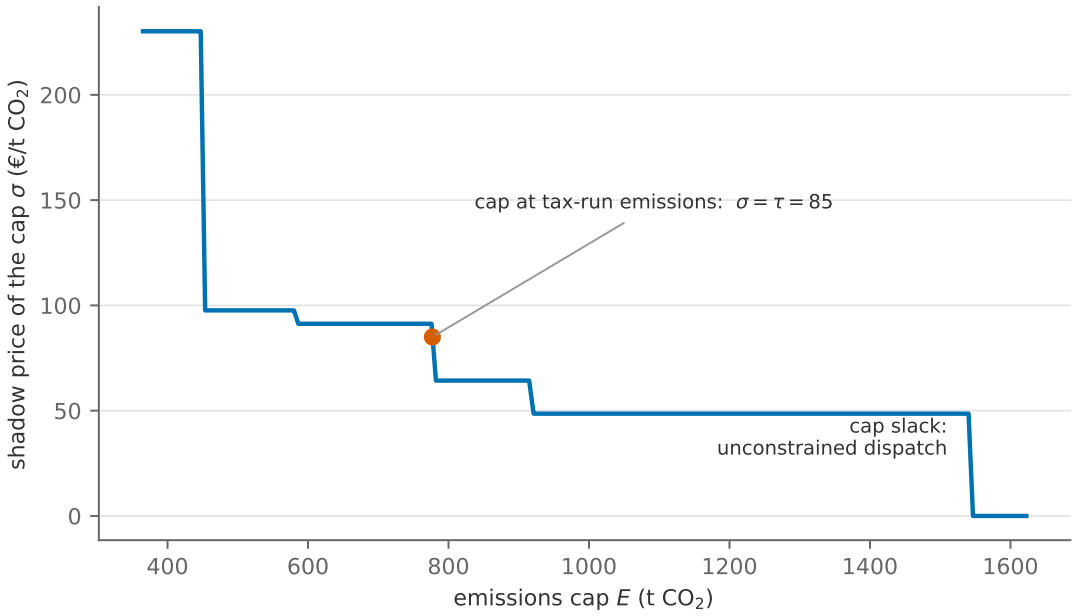



Figure 2.4: The shadow price of the emissions cap as the cap is tightened from unconstrained emissions (1545 t) towards the feasibility floor (365 t, the cleanest way to serve load at all). Each flat segment is one margin of abatement — one pair of technologies trading places in the dispatch; each vertical segment is a cap level at which that margin is exhausted.

The first point about equivalence is straightforward, while the second point on uncertainty deserves a short note: If demand or fuel prices are uncertain, a tax fixes the marginal cost of abatement and lets emissions fluctuate, while a cap fixes emissions and lets the carbon price fluctuate; which is better depends on the relative slopes of marginal abatement cost and marginal damage, the classic argument of Weitzman (1974).

**From duals to markets.** Every model section of this note closes with the same small table: each constraint, its dual, what the dual measures, and the real-world object that measures the same thing:

| Constraint | Dual | What it measures | Market counterpart |
|---|---|---|---|
| market clearing (3b) | $\lambda$ | marginal cost of serving one more MWh | the day-ahead spot price |
| capacity limit (3c) | $\overline{\mu}_g$ | value of one more MW of $g$; scarcity rent | an inframarginal plant's operating margin |
| served load $d \leq D$ (7c) | $\overline{\theta}$ | surplus on load that is served; $\lambda = v$ when it is not | scarcity pricing; the market's price cap |
| emissions cap (10) | $\sigma$ | marginal abatement cost | the EU ETS allowance price |

The current section shows very clearly that given existing capacities, nothing beats running renewables at full capacity; solar and wind has by far the lowest marginal costs of operation. The

setup is very instructive in terms of understanding the basics and being able to read the economics from the duals of the planner solution, but very clearly requires a lot more details before we inevitable still have to conclude that solar and onshore wind are really cost-effective technologies (but we'll learn a lot from the journey there).

# 3 Variable generation and demand

Section 2 presented an incredibly stylized energy system, and so will this section. However, we will introduce an essential feature of the energy system, in particular for a course that deals with the green transition: Hourly variability in generation capacities and demand. This section introduces a time dimension $h$ and adds an *availability profile* to each technology — the fraction of its capacity the weather offers in each hour. Because nothing yet links the hours together, the model simply remains one merit order per hour. However, even though the formal setup is virtually identical, this small change produces plenty of insights from model results and empirical evidence: the price is no longer a number but a distribution, and the value of a technology stops being a property of the technology alone and becomes a property of correlations between availability of generators and demand.

> **Reading alongside this section.** This section requires no supplementary reading to understand the concepts presented throughout. Instead, we point to relevant literature that deals with the same topics that we touch upon here. Hirth (2013) defines the value factor we derive below and estimates it across European systems. Joskow (2011) is three pages on why comparing technologies by levelised cost is a mistake. Borenstein (2012) is a gentle full treatment of the economics of renewable generation. López Prol et al. (2020) redo Hirth's exercise and draw the relative-versus-absolute distinction we take up in Section 3.4; Heptonstall and Gross (2021) is a systematic review that puts a number on total integration costs; and Reichenberg et al. (2023) recasts cannibalisation as *revenue risk*, which is where Section 9 picks it up.

## 3.1 The case of Denmark

Before we turn to the model and empirical evidence, let us briefly take a look at the relevant data and use Denmark as an example.

Figure 3.1 collects hourly data on demand, onshore wind, offshore wind, and solar generation.[5] Panels (a) and (b) shows how availability of particularly solar changes between a week in January and July, and panel (c) summarizes the three technologies' yearly availability with *duration curves*. Duration curves shows for what share of hours is availability above $x$ for the three (we'll return to this below). Importantly, panels (a) and (b) also add the shape of hourly demand in the same hours so we can inspect how demand and variable generation correlates.

[5] All data are from the Danish TSO Energinet's portal Energi Data Service Openly licensed (CC-BY 4.0). Appendix B records the dataset. The note's profiles are the production of each fleet divided by its maximum over the year. That normalisation is a classroom proxy for the true availability per unit of capacity (it ignores capacity additions within the year and assumes the fleet's best hour reaches nameplate), but it preserves the shape.

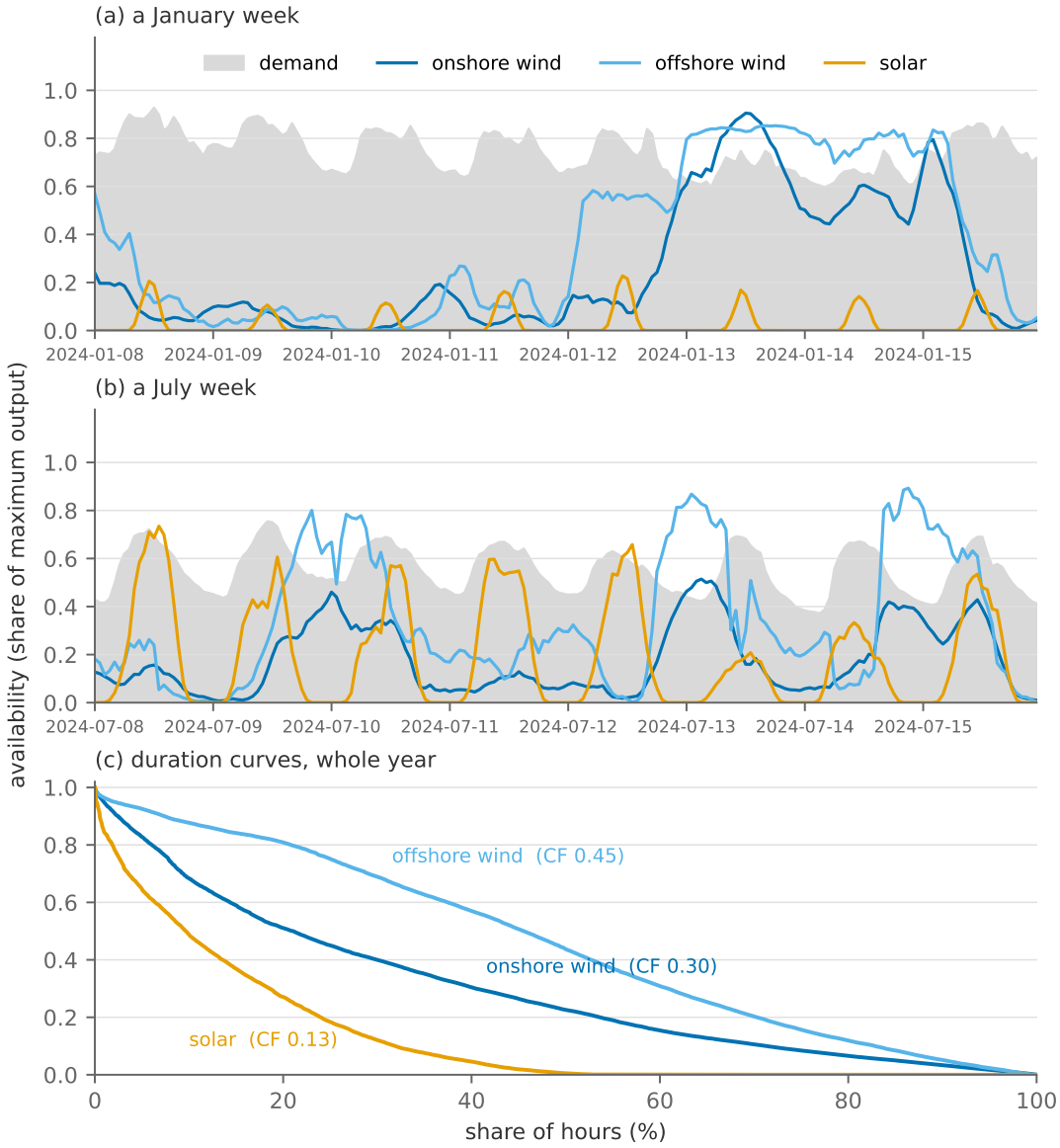



Figure 3.1: DK1 availability profiles and demand, 2024. (a): a winter week — wind-dominated, storms and lulls lasting days, solar nearly absent. (b): a summer week — a strong daily solar rhythm, weaker wind. Demand is the grey band, scaled by its own annual maximum so that it shares the axis. (c): the year as duration curves; the area under each curve is the capacity factor (CF). Source: Energi Data Service, CC-BY 4.0.

The panels show that (i) offshore and onshore wind are strongly correlated, (ii) solar is strongly cyclical both daily and seasonally, (iii) wind is less cyclical (spread out), but comes in multi-day weather system, and (iv) the capacity factors (average yearly availability) are far below one in particular for solar. These patterns are crucial, in particular when we start looking into deeply decarbonised systems, because they govern what kind of complementary technologies have to be deployed for each of them (section 7 returns to this).

On the demand side, DK1 features clear daily and seasonal patterns, with higher levels of demand during winter and the middle of the day. It is worth noting that these swings are strong and varies by a factor of nearly three, from 1.5 GW to 4.2 GW in the the our base year 2024.

Table 2 presents the evidence in a different way by computing correlations between availability

profiles and hourly demand. The positive correlation of the first column tells a positive story about

| Correlation of availability with demand | hour by hour | across months | across hours of the day |
|---|---|---|---|
| Onshore wind | +0.44 | +0.89 | +0.58 |
| Offshore wind | +0.29 | +0.92 | -0.80 |
| Solar | +0.25 | -0.93 | +0.94 |

Table 2: Correlation between availability profiles and hourly demand. Source: Energi Data Service, CC-BY 4.0.

wind and solar resources being available when demand is high. The other two columns nuance this a bit: Danish wind and demand are both generally higher during winter, reflected by a strong seasonal correlation (+0.89 onshore, +0.92 offshore). Solar is the opposite at that scale (-0.93), and nearly perfect at the daily one (+0.94): it peaks at midday, and so, at 11:00, does demand. Solar's mild hourly correlation of +0.25 is not mild alignment. It is one strong alignment and one strong misalignment, cancelling.

Now one additional factor complicates the picture here: We are using *observed* consumption as a proxy for the pure demand variation. However, it is worth noting here that to the extend that demand *can* respond to price signals, the correlations above (in particular the "hour by hour" column) overestimates the true positive correlation: When wind and solar generation is large, the equilibrium price drops (all things equal), which drives up demand.

Finally, before we proceed with the model and simulations comparing it to empirical evidence, it is worth dwelling shortly on how the Danish merit order curve looks like as this will help us understand the results later on. Figure 3.2 collects the fleet of 2024: panel (a) is what stands installed in DK1, panel (b) what each plant costs to run — which is to say, the order they are called in.[6] Denmark has no oil or open-cycle peakers to speak of, and none appear here.

That fleet of power plants has a feature worth stopping at: its firm capacity — everything that is not weather-driven — comes to 2.6 GW, against a peak demand of 4.2 GW. This means that without either imports or significant variable generation, DK1 cannot serve its own peak demand. As a relatively small open economy in a coupled European market, imports and exports are essential for the Danish system: in 2024 DK1 imported 3.9 TWh and exported 3.8 TWh across its six interconnectors. Note what those two numbers do *not* say: gross trade of 32% of consumption nets out to almost nothing (0.1 TWh, or 0.2% of consumption). The wires are busy in both directions — DK1 was a net importer in 53% of hours and a net exporter in the rest — and it is that hour-by-hour swinging, not the annual balance, that the rest of this section is about.

The implication is that, as an artefact of us trying to keep the model simple and ignore transmission lines and exports/imports, we have essentially created an infeasible problem in roughly 450 hours of the year where demand cannot be met with domestic generation. Thus, for the model to run and give *somewhat* reasonable results, we fix this by adding one synthetic "import" plant

[6] Wind and solar capacities are the maxima the profiles were normalised by, so profile times capacity reproduces DK1's own hourly production. The thermal capacities come from PyPSA-Eur, an open model of the European power system, in the same snapshot of the zone that Section 6 uses, so the two sections describe the same country; Appendix B documents both.

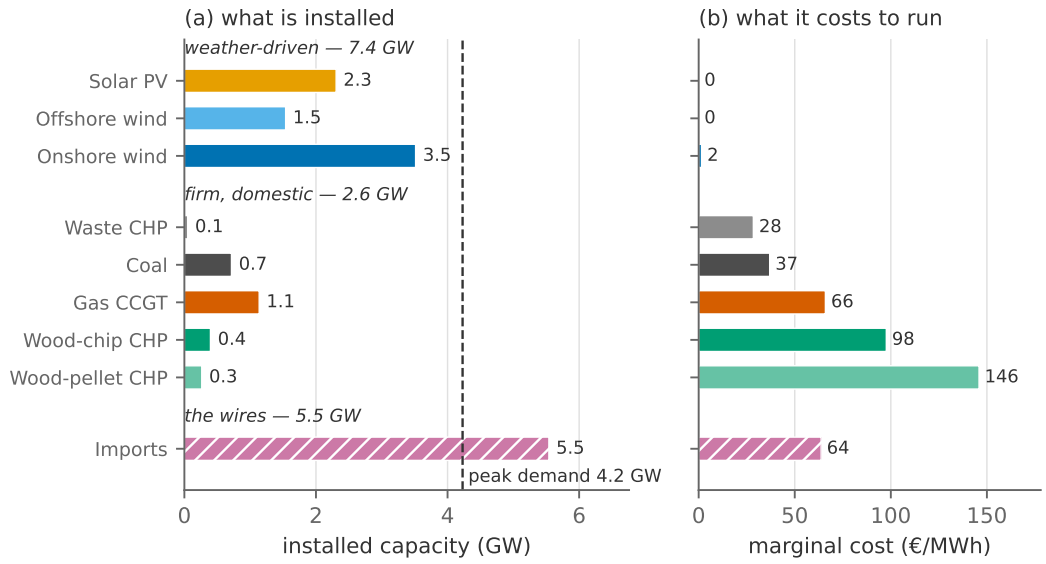



Figure 3.2: The DK1 fleet of 2024 that this section dispatches, grouped as the text reasons about it and ordered within each group by marginal cost. (a): installed capacity, against the year's peak demand. (b): marginal cost. The hatched row is not a Danish power station but the synthetic import plant introduced at the end of this subsection; its cost is the hourly price of the neighbouring zones, and panel (b) shows the annual mean of that price. Sources: Energi Data Service and PyPSA-Eur; Appendix B documents both.

on top of the true Danish fleet of power plants. In our results later on, this "import" plant will be presented as a part of the Danish merit order curve and be important for the price formation.[7]

## 3.2 The model, repeated over the year

Index hours by $h = 1, \dots, T$ and let $\gamma_{g,h} \in [0, 1]$ be the availability of technology $g$ in hour $h$. With fixed load $D_h$, the planner now solves

$$\min_{q} \quad \sum_{h} \sum_{g \in \mathcal{G}} c_{g,h}\, q_{g,h} \tag{13a}$$

$$\text{s.t.} \quad \sum_{g \in \mathcal{G}} q_{g,h} = D_h \qquad (\lambda_h) \qquad \forall h \tag{13b}$$

$$0 \le q_{g,h} \le \gamma_{g,h}\, \bar{q}_g \qquad (\underline{\mu}_{g,h}, \overline{\mu}_{g,h}) \qquad \forall g, h \tag{13c}$$

This is identical to the model in Section 2, except for most symbols now carrying an hourly index $h$ as well.[8]

Because neither the objective nor any constraint links two different hours, the problem *separates*:

[7] Specifically, we set $\bar{q} = 5.5$ GW — the sum of DK1's interconnectors to Norway, Sweden, Germany and eastern Denmark (DK2) — available in every hour at a marginal cost equal to the capacity-weighted average of those zones' day-ahead prices, which averaged 63.7 €/MWh over the year. Four of the six wires, then: the cables to the Netherlands and to Great Britain are left out because this section prices imports off its neighbours' published day-ahead prices, and the twelve-zone network of Section 6 does not extend that far west. It is a stand-in, and a deliberately crude one: Denmark may *buy* at what its neighbours charge but may not *sell* to them, and their prices are read from the market rather than computed. Section 6 replaces it with a network of twelve zones that computes those prices, and the difference between the two is worth keeping in mind for the rest of this section.

[8] The marginal cost $c_{g,h}$ only varies because of our synthetic "import" plant, where this cost reflects the actual import price for DK1 in a given hour in 2024.

it is $T$ copies of Section 2's LP, and every result from there applies hour by hour. The Lagrangian says the same thing: It is Section 2's, one copy per hour,

$$\mathcal{L}^{\text{vre}} \;=\; \sum_{h} \mathcal{L}_h^{\text{disp}} \;=\; \sum_{h} \Big[ -\sum_{g} c_{g,h}\, q_{g,h} + \lambda_h \Big( \sum_{g} q_{g,h} - D_h \Big) + \sum_{g} \overline{\mu}_{g,h} \big( \gamma_{g,h} \bar{q}_g - q_{g,h} \big) + \sum_{g} \underline{\mu}_{g,h}\, q_{g,h} \Big], \tag{14}$$

and differentiating it hour by hour returns eq. (5) with a $h$ subscript throughout; $\lambda_h$ is the price in hour $h$, equal to the marginal cost of that hour's marginal technology; $\overline{\mu}_{g,h}$ is generator $g$'s scarcity rent in hour $h$. This is the object the rest of the note builds on: Every section from here adds terms to $\mathcal{L}^{\text{vre}}$ rather than starting again. The supply and demand curves now move with weather and habits: Wind entering at near-zero marginal cost shifts the whole curve rightward in windy hours and vanishes in calm ones, so the same load can meet a cheap curve or an expensive one depending on the hour.

## 3.3 Residual load

Since variable renewable energy (VRE, wind and solar) bids in at near zero marginal cost, it is dispatched whenever available, and the conventional fleet faces what is left: the *residual load*

$$R_h \;=\; D_h - \sum_{g \in \text{VRE}} \gamma_{g,h}\, \bar{q}_g. \tag{15}$$

This curve represents what other plans and imports need to cover in different hours across the year. Figure 3.3 sorts $R_h$ into a duration curve, for the actual fleet and for counterfactual fleets with wind capacity doubled and quadrupled.

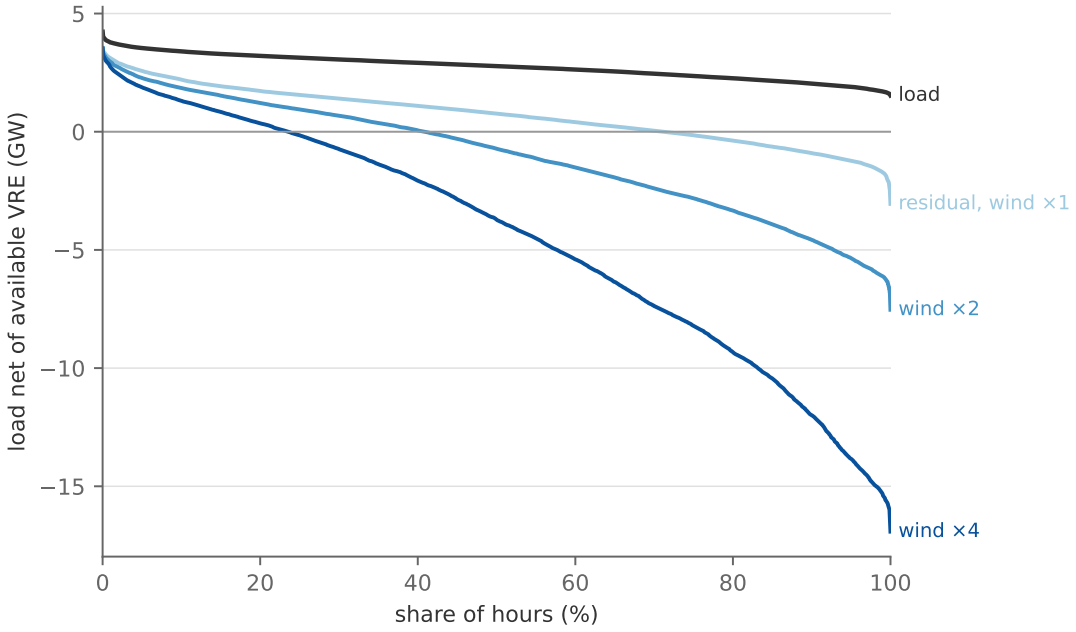



Figure 3.3: Load duration curve and residual load duration curves for DK1 2024, at actual wind capacity and with wind scaled ×2 and ×4 (solar at actual capacity throughout).

As more VRE is added, the overall levels of residual demand naturally drop, but the curve also steepens: For the hours where residual load is largest, new wind resources barely has any impact (left tail), whereas the hours that already has plenty of wind generation becomes oversaturated (right tail). The curve shows that at current DK1 capacity levels, a significant number of hours feature negative residual demand, i.e. surplus production of VRE; at ×4 the surplus exceeds total load for most of the year and to a severe degree. These hours of structural surplus require some mechanism to soak up the excess energy: exports, storage, flexible demand, or nothing at all. When the available renewable energy produces more than the system can absorb on the demand side, is has to be *curtailed* for the system to balance. It is not a malfunction: it is the optimal response to energy that is worth less than nothing to the system in that hour, and a well-designed renewable system may curtail a great deal. Curtailment lowers a technology's realised capacity factor below what the weather would allow, which raises its levelized cost, and in real markets it is the mechanism behind negative prices and the reason power purchase agreements (PPAs) are written the way they are.

## 3.4 Price distributions and value definitions

**The price distribution.** Figure 3.4 sorts the equilibrium prices. Absent, the synthetic "import" plant that has time-varying marginal costs to reflect the state of neighbouring markets, this curve would simply be a staircase with each step reflecting the different plant's marginal costs. The smooth parts represent hours where imports are price setting.

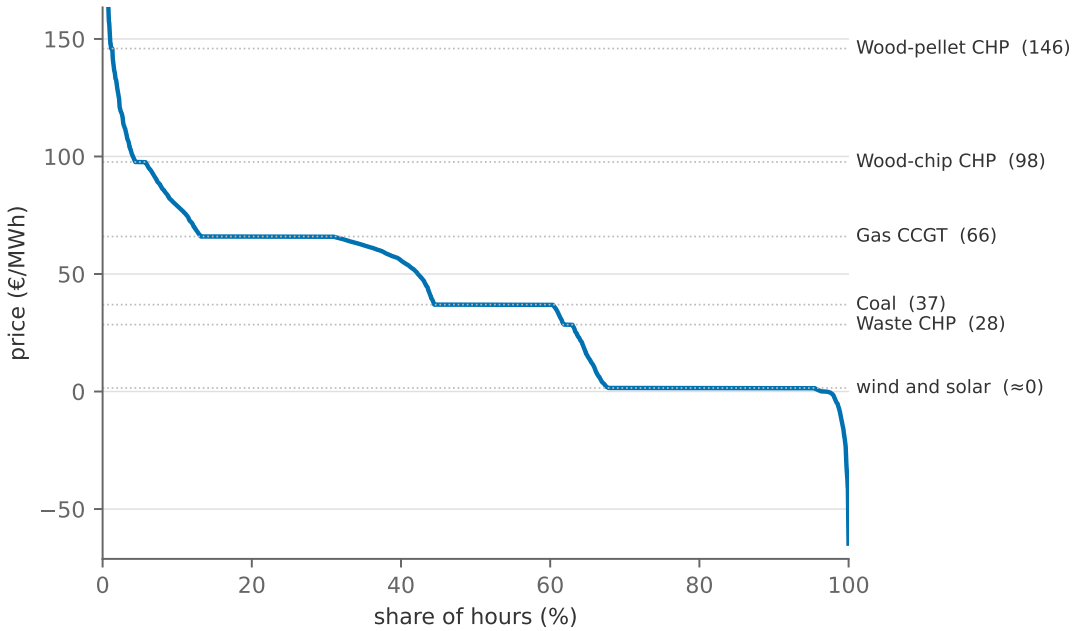



Figure 3.4: The price duration curve: the year's $\lambda_h$, sorted. The price only ever takes values in the fleet's menu of marginal costs (right margin), because in every hour some technology is marginal and eq. (5a) sets $\lambda_h$ equal to its cost. The axis is cut just above the most expensive domestic plant so that the staircase stays legible: the tail is not shown, and it runs to 894 €/MWh in the year's scarcest hour, where the price has left the domestic menu behind and the wires are marginal.

Which technology depends on the hour's residual load: in the surplus hours at the right the marginal unit is *wind itself*, sending the price to its near-zero marginal cost (onshore wind alone sets it in 28% of hours); coal sets it in the mid-residual hours (16%), gas in the tighter ones (18%), and wood in the tightest few.

**Value definitions.** Three statistics summarise how a technology meets the price distribution. The *capacity factor* is average output per unit of capacity,

$$CF_g = \frac{1}{T\,\bar{q}_g} \sum_h q_{g,h}, \tag{16}$$

a purely physical number. The *capture price* is the average price the technology actually sells at — the generation-weighted mean,

$$\bar{p}_g = \frac{\sum_h \lambda_h\, q_{g,h}}{\sum_h q_{g,h}}, \tag{17}$$

and the *value factor* normalises it by the *base price* $\bar{p}$,

$$VF_g = \frac{\bar{p}_g}{\bar{p}}, \qquad \bar{p} \equiv \frac{\sum_h \lambda_h D_h}{\sum_h D_h}, \tag{18}$$

so that $VF_g > 1$ means the technology produces disproportionately in expensive hours and $VF_g < 1$ the opposite (Hirth 2013). The value factor therefore captures a revenue channel that tells us something important about the technology beyond its cost structure: It measures the relative value of the MWh that a technology produces relative to the average technology and thus prices the intertemporal scarcity of the system (see also e.g. Joskow 2011).

**Absolute and relative cannibalisation in Danish data.** Figure 3.5 simulates our small toy model with the Danish weather, demand patterns, fleet, and import prices. Panel (a) shows the capture prices across different energy shares for wind and solar and panel (b) shows the corresponding value factors. It is worth stressing that the shapes of the two curves naturally depends on the system configuration, but they replicate the empirical evidence we have so far (see e.g. Hirth 2013; López Prol et al. 2020): Solar's value factor is above one for very low market shares and drops sharply, and wind's starts out around roughly 1, but drops more slowly across the market share.

Figure 3.6 shows the interaction between the wind and value and the solar market share. Panel (a) shows wind's value factors as a function of solar's energy share and panel (b) the same thing only in levels. Here, it shows that the value factor of wind clearly increases in the solar market share, while the level of the wind capture price clearly decreases.

The figures present (at least) three important conclusions worth carrying with us: First, for a wind developer: the revenue per MWh is not the market price but the capture price, and the capture price is a *decreasing function of everyone else's wind*. Second, for a policy maker: the first units of a VRE technology are the most valuable ones, so support schemes calibrated to today's

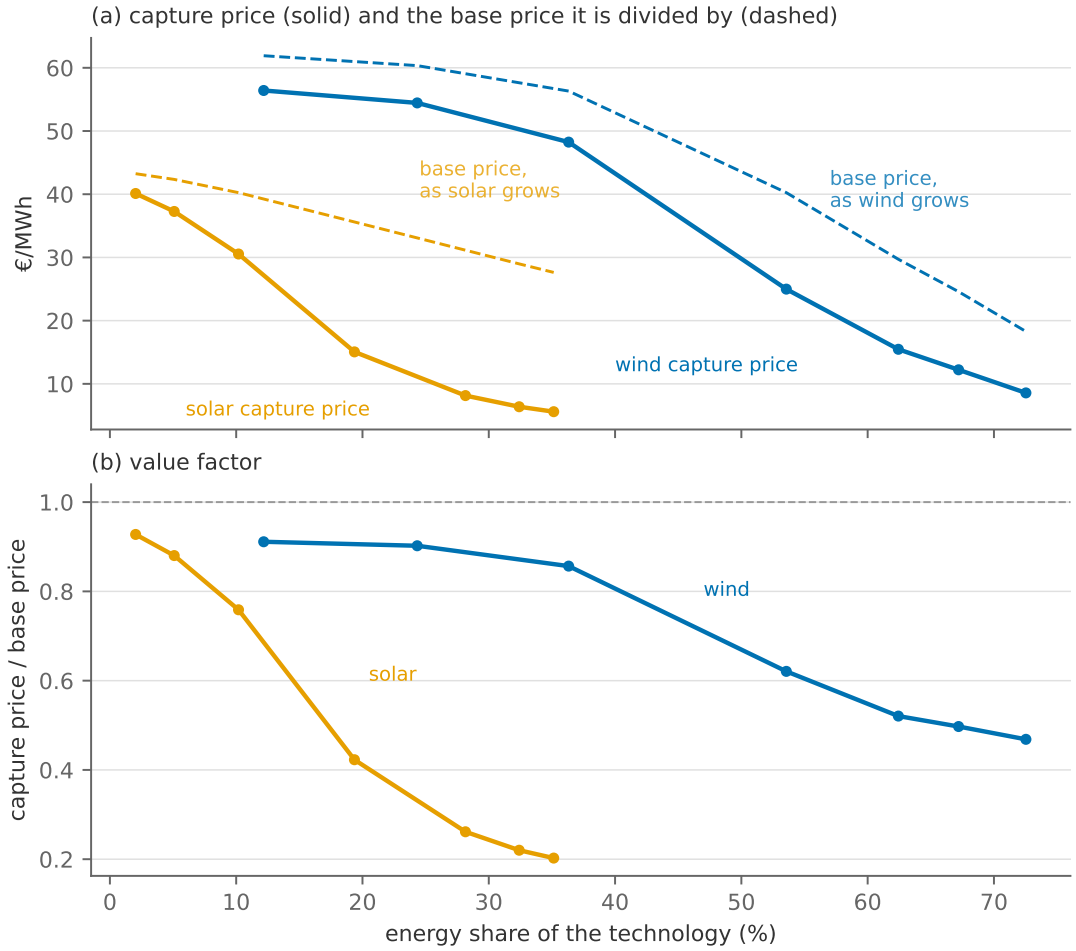



Figure 3.5: Cannibalisation as a curve: capture price (a) and value factor (b) against the technology's own energy share, scaling wind and solar capacity separately. Each point is a full re-solve of the year. The dashed line in (a) is the base price $\bar{p}$ *along that same sweep* — the denominator panel (b) divides by. It is drawn because it does not stand still: as zero-cost capacity enters, the whole price level falls with it (the merit-order effect), so the value factor in (b) is a ratio of two quantities that are both moving. Panel (b) is therefore what is left of the decline once the fall in the general price level is divided out: the part specific to the technology's own timing.

prices systematically overpay tomorrow's entrants unless they are indexed to something like the capture price. Both readings, however, take the fleet as fixed. Nothing in this section can absorb surplus energy or shift it into the tight hours — the value collapse is partly a statement about missing *flexibility*, not only about wind. The following sections discuss this further.

The third and final conclusion we want to stress here is the warning in López Prol et al. (2020) about using the value factor alone to measure the profitability and complementarity of technologies: On the surface, the value factor in Figure 3.6 in panel (a) indicates that wind and solar may be complementary technologies in the sense that using more solar increases the value of wind (at least the value *factor*). However, panel (b) clearly rejects this story, showing that the capture price of wind still decreases with more solar in the system, only slower than the *average* price. This is precisely the distinction López Prol et al. (2020) draw between *relative* and *absolute* cannibalisation on Californian data, and it is why they insist on reporting both.

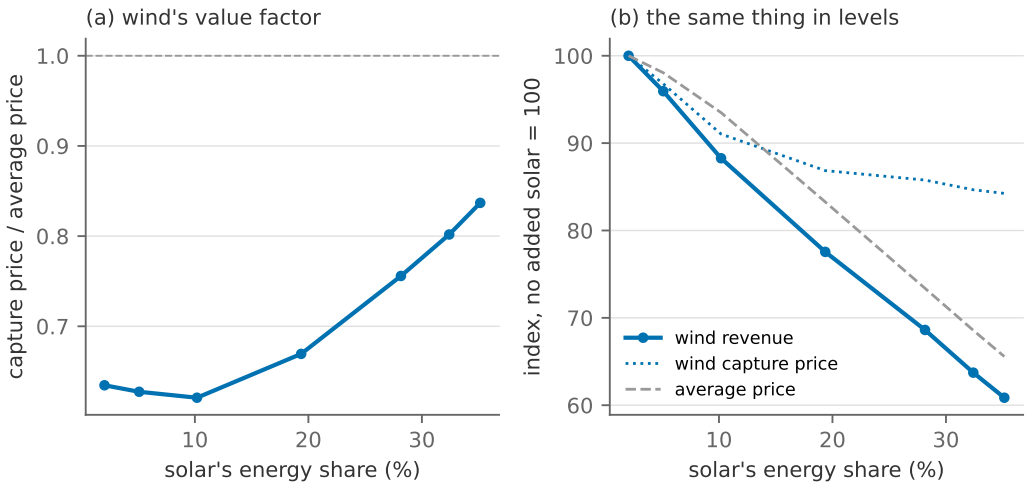



Figure 3.6: The same six runs, read two ways. Solar capacity is scaled up; wind is untouched. (a): wind's value factor *rises*, from 0.63 to 0.84. (b): wind's capture price and its total revenue both *fall* — revenue by 39%. Nothing good has happened to wind. The value factor improved only because its denominator, the base price, fell faster than its numerator.

## 3.5 Is cannibalisation a market failure?

We have just derived a result that sounds alarming: build more wind and wind becomes less valuable. It is tempting to jump from there to "therefore renewables need support to overcome cannibalisation". While there are real arguments for support schemes for renewables, this one does not hold up to scrutiny.

Recall the standard rule for when regulation is warranted on *efficiency* grounds: intervene when private actions impose costs on others *that do not pass through prices*. Economists call a cost onto others that travels *through* the price system a **pecuniary externality**, and a cost that travels outside it a **technological** (or real) externality. Only the second is a market failure. When a new bakery opens and drives down the price of bread, the incumbent baker is harmed, but nothing inefficient has happened — the price fell because bread became genuinely more abundant, and the price change is the mechanism by which that information reaches everybody. Compensating the incumbent would make things worse, not better.

Before we end with a real confrontation with data, let us dwell on this rather important topic for just a moment longer and reflect on how the variable nature of certain renewables may be a good reason for interventions:

**Pecuniary — no case for intervention.**

- *Low prices in windy hours.* Electricity genuinely *is* less scarce when the wind blows. The low price is correct information, not a distortion, and it is what tells a heat pump, a battery or a smelter to consume then.
- *Cannibalisation itself.* A wind farm that depresses the price other wind farms receive is doing exactly what the new bakery does. The efficient amount of wind is precisely the amount at

which the marginal farm's capture price covers its cost — which is Result 6, and which the market reaches on its own.

- *Higher price volatility.* Unpleasant for an investor, but risk that can in principle be traded: fixed-price contracts, PPAs and hedges exist. However, if insurance markets are missing or incomplete, this can be a real argument for second-best type of regulation; the market failure is in a separate market, but cannot be addressed, which warrants regulation of volatility in other markets.

**Potentially technological — a case worth examining.**

- *Network congestion.* The grid is a common-pool resource. If a generator connects where it aggravates congestion and does not pay for the congestion it causes — because tariffs are flat, or because zonal pricing hides the constraint (Section 6.1) — that cost genuinely falls on others outside the price it faces.
- *Reliability and reserve requirements.* More forecast error means the system operator must hold more reserve. If reserve is procured centrally and its cost socialised into a flat tariff rather than charged to whoever created the imbalance, the externality is real. Note that this is an argument about *uncertainty*, not variability — a distinction Section 9 takes seriously.
- *Local amenity effects.* Noise, landscape, wildlife. These are genuine external costs — but they attach to *siting*, and have nothing to do with variability.

Two conclusions follow: First, most of what is loosely called "the integration cost of renewables" is not a market failure: Hirth (2013) decomposes the gap between a technology's levelised cost and its market value into *profile* costs (our value factor, in €/MWh), *balancing* costs (the cost of forecast error) and *grid* costs, and is explicit that these represent the genuinely lower value of electricity delivered at the wrong time, in the wrong place, or with uncertainty attached — not a distortion to be corrected. Heptonstall and Gross (2021) put the total at roughly 20–35 €/MWh at high penetration, rising with the renewable share and dominated by the profile component — which is to say, dominated by exactly the mechanism this section derived.

Second, when regulation *is* warranted, the targeting principle applies: one instrument per externality, applied as close to the source as possible. If congestion is the problem, price congestion — that is what nodal pricing does. If reserve is the problem, charge imbalance to those who cause it. If carbon is the problem, price carbon, as Section 2.5 did.

While cannibalisation can be a poor justification for renewable support schemes, other arguments do survive contact with the targeting principle. Here, we briefly touch upon three different arguments.

The first is *innovation*: Deploying a technology teaches the industry how to build it more cheaply, and the firm doing the deploying captures only part of that learning: the rest spills over to its competitors and its successors, so private deployment falls short of what is collectively

worthwhile. Gillingham and Stock (2018) summarise it as follows: the dynamic, innovation-driven case for support is real and hard to quantify, and it is a different case from the static one this section deals with.[9]

The second is *second best*. Everything above assumed carbon is priced at its damage; where it is not, a subsidy to clean generation stands in for the missing carbon price — a poor substitute, but not nothing (Kalkuhl et al. 2013). Abrell et al. (2019) solve for the optimal support in exactly that setting and find that how the subsidy is *financed* matters far more for welfare than how finely it is differentiated across wind and solar. Helm and Mier (2021) add this section's own mechanism: because the value of clean generation falls as the fossil share falls, the optimal subsidy is a declining schedule rather than a rate — support calibrated to today's system overpays tomorrow's.

The third is *risk*. The volatility bullet above assumed hedges can be bought; where they cannot, an investor facing uninsurable revenue risk demands a higher return and the same capacity costs more to build. Newbery (2023) argues that a long-term contract that removes that risk lowers the cost of capital by enough to pay for itself — provided it is written so that it does not also switch off the price signal the generator dispatches against.

## 3.6 Comparing model outcomes and market data

Finally, let us compare how our stylized model fare when compared to market data. Recalling that the shadow value $\lambda_h$ is the market equilibrium price, we put the simulated $\lambda_h$ beside the published day-ahead prices for DK1 in Figure 3.7.

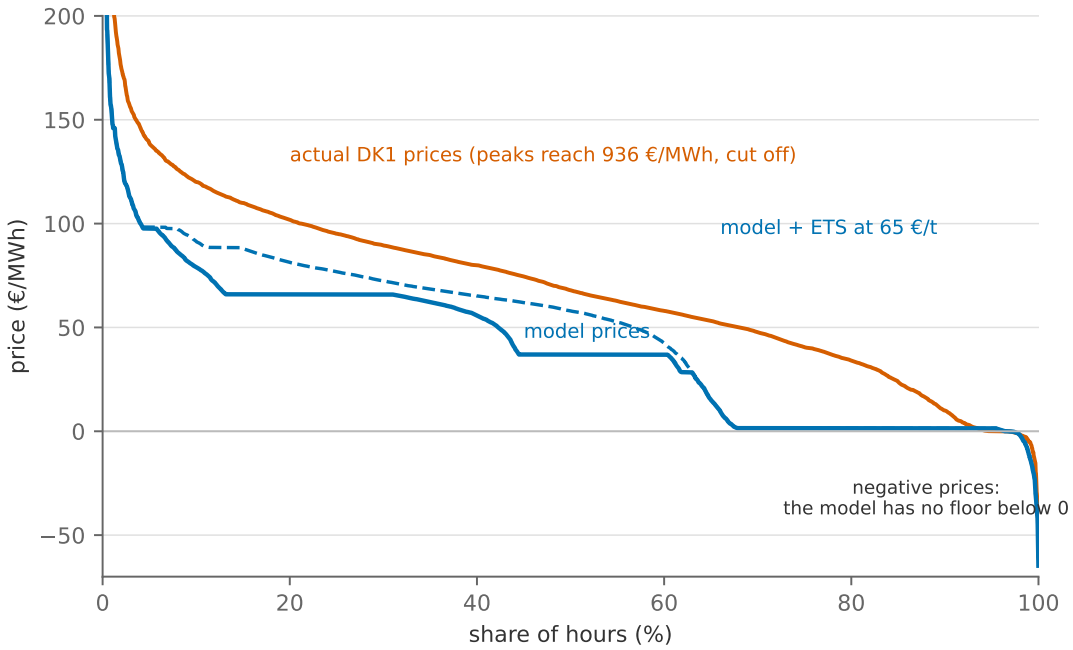



Figure 3.7: The model's price duration curve against the actual DK1 day-ahead prices of 2024. The dashed curve is the same model with the year's average ETS price (65 €/t) charged on every emitting technology. The shape agrees; the level does not, and the gap has one cause.

[9] For other relevant literature on this see e.g. Fischer and Newell (2008), Acemoglu et al. (2012), Popp (2019), and Nemet (2019).

**The shape is right.** The model's price is at or below 37 €/MWh in half of all hours and reaches 98 €/MWh in the top twentieth, against 68 and 139 €/MWh in the market. It is worth noting that the tail with high prices comes directly from our way of accounting for expensive imports.

**The level is not.** The model's base price is 40.2 €/MWh against the market's 70.6, and adding the year's ETS price to every emitting plant lifts it only to 48.5. The model is roughly a third too cheap, and the reason is the boundary rather than the fleet: **Denmark may buy but not sell**. In the real 2024 DK1 exported 3.8 TWh, and every one of those MWh was sold into a market that wanted it at a price. Here they have nowhere to go, so they are curtailed and the price falls to wind's marginal cost instead of settling at what a neighbour would have paid. Cheap hours are therefore too cheap and too frequent, which drags the whole distribution down.

**The market values, and where they miss.** Ratios cancel much of that level gap, so the market-value statistics are the sharper test. In the actual 2024 market the Danish wind fleet earned a capture price of 55.4 €/MWh, a value factor of 0.80; the model produces 0.62 at the same capacity. It misses in the direction the missing exports predict: without an outlet, wind cannibalises itself harder in the model than it does in reality, and the difference between 0.62 and 0.80 is a measurement of what DK1's interconnectors are worth to a Danish wind operator.

Solar is the mirror image, and the more interesting one. The market gave Danish solar a value factor of 0.68; the model gives 0.76, at a domestic solar share of 10.2%. With the neighbours' hourly prices in the merit order, cheap German midday power now undercuts Danish solar in exactly the hours Danish solar produces. The cannibalisation of Danish solar is substantially *foreign* cannibalisation, and it is now inside the model rather than in a paragraph explaining a discrepancy.

**A week, side by side.** Duration curves compare distributions after reordering hours neatly. Before we proceed to more complex models, let us briefly look at the model versus data throughout a chronological week. Figure 3.8 compares the model dispatch one January week with market data, both as a share of that hour's load, so that the comparison is about *composition* rather than levels. The panels are collected into comparable blocks (variable renewables, dispatchables, and net trade) to make categories across Energinet's settlement data and model technologies compatible.

First, because the modell cannot export excess supply, the model stack closes at 100%. The real system exports instead, most visibly during the storm from 15 January, when DK1 generated 160% of its own load from wind and solar alone and sent the difference abroad. The magnitude of this channel is significant: net trade swings from +50% of load to −96% within the single week. Second, dispatchable plants never fully switches off in the market data. In panel (b), it never falls below 24% of load (757 MW), even in hours with huge variable renewable generation and prices at or below zero, where the model has already shut its thermal fleet down entirely. If we dive into the underlying data, this comes from local CHP plants. In the Danish context, this is likely due to the fact that the plants are committed to produce district heating in a cold January month in

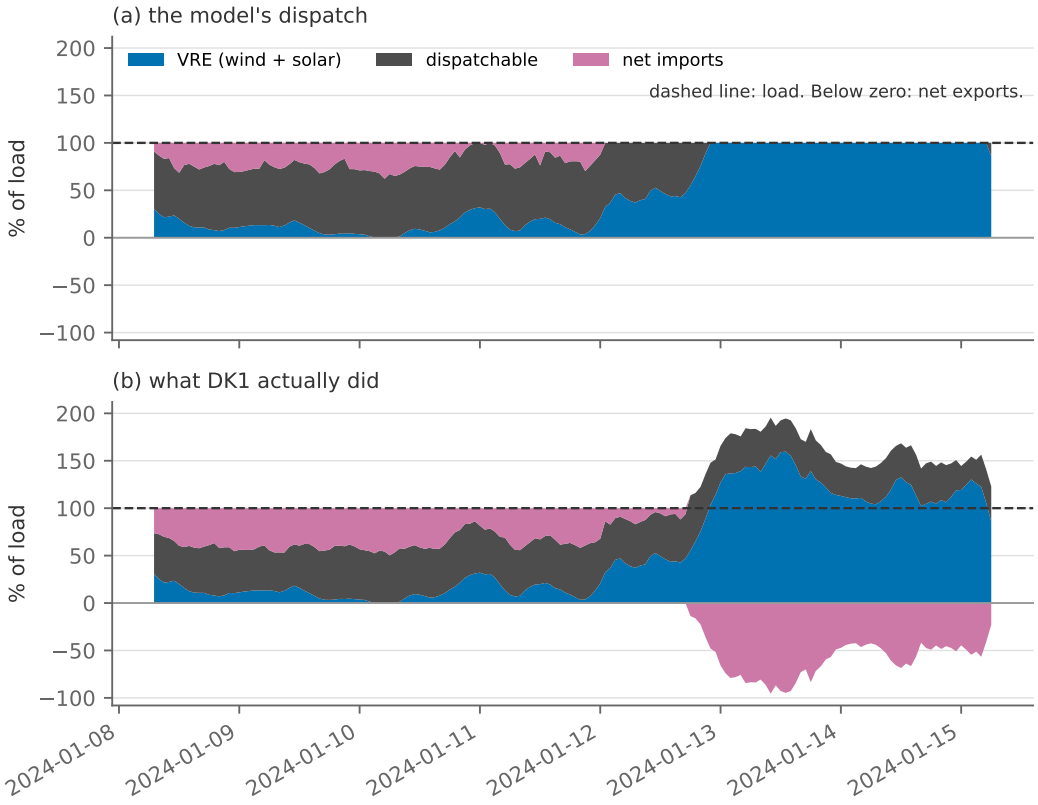



Figure 3.8: One week (8–15 January 2024), modelled and realised, each hour as a percentage of that hour's load, in three blocks: variable renewables (wind and solar), dispatchable plant, and net imports — which stack downwards in the hours the zone is a net exporter. The week is chosen automatically as the year's most variable in residual load.

Denmark.[10] What this link to district heating means for the model more broadly is something we will discuss further in section Section 5.

**What is still missing.** Three things, and each returns later:

i. *No plant here will pay to produce.* The market goes negative in 4% of hours and the model in 3%, but the model's negative hours are borrowed rather than earned: they are hours when the neighbours' price is negative and the import plant carries it in. No Danish generator in this fleet ever bids below zero, because none of the reasons to do so is in the model — subsidised generation, must-run CHP, inflexible plants avoiding a shutdown cost. Section 4.3 puts the first of them in.

ii. *Trade runs one way.* Section 6 makes the neighbours endogenous and lets power flow both directions; comparing that section's prices with this will shed light on what the exporting channel is worth.

iii. *Missing flexibility.* Demand is exogenous in every hour, and there are no intertemporal flexibility that allows supply or demand to move across hours from a surplus hour to a tight

[10] There are other potential reasons for this type of behavior, including generation subsidies or technical constraints such as ramping or unit commitment. We revisit the latter in the next section.

one.

**Which way does it bias the answer?** *No negative bids, one direction of trade, no flexibility.*

The three omissions above do not all push the answer the same way.

- *No negative bids.* A fleet that shuts down the moment the price falls below its marginal cost is more flexible than the real one: **understates** curtailment, **overstates** what wind and solar capture in the surplus hours.
- *Trade one way.* Surplus is curtailed rather than exported, which **understates** the value of Danish wind — the gap between 0.62 and 0.80 above is a measurement of it. But the neighbours' prices are data, not outcomes, so the model cannot let Denmark move them: that **overstates** what the wires are worth.
- *No flexibility.* Nothing carries energy between hours, so a surplus can only be thrown away and a tight hour met by the most expensive plant available: **overstates** the spread of prices and, with it, the cannibalisation derived above.

The second is the largest, which is why this model's wind is worth less than the market's rather than more. Sections 4 and 6 take on all three.

**From duals to markets.** As usual, let us summarise the bridges we have used to cross from model outcomes to market data.

| Constraint | Dual | What it measures | Market counterpart |
|---|---|---|---|
| market clearing per hour (13b) | $\lambda_h$ | that hour's system marginal cost | the hourly day-ahead price (Elspot DK1) |
| availability-scaled capacity (13c) | $\overline{\mu}_{g,h}$ | the hour's scarcity rent on $g$ | a plant's hourly gross margin |
| derived: capture price (17) | | generation-weighted price earned | a wind farm's realised revenue per MWh |
| derived: value factor (18) | | market value relative to what the average MWh costs | the "profile discount" in PPA pricing |

**Where this goes next.** From Section 4 onwards the models grow: storage, a second energy carrier, a network of twelve zones. They stop being small enough to write out constraint by constraint, and are assembled instead from standard components in an established modelling framework. That buys realism — the simulations below are closer to the system they describe than anything in this section — at the price of a little distance between the blocks of algebra we write down and the machinery that solves them. Nothing about the economics changes: every result that follows is still read off the same duals, the same $\lambda_h$ and the same $\overline{\mu}_{g,h}$, and Appendix C.1 translates each component back into the constraints it stands for.

# 4 Storage and flexibility

Previous models treat hours independently: the LP of Section 3 separates into one merit order per hour, and no decision in one hour constrains any other. Storage ends that. A battery that charges at noon and discharges at eight in the evening ties the two hours into a single decision problem, and the economics acquire a genuinely intertemporal object: the value of energy *held*, not consumed.

Section 3 ended with the observation that the collapse in the value of wind was partly a statement about a system with no flexibility at all: nothing could absorb a surplus hour or move energy into a tight one. The next few sections add the standard responses to intermittency one at a time:

i. *Storage*, and the flexibility of the thermal fleet itself (this section);

ii. *Sector coupling and flexible demand*: a consumer that can respond to the price (Section 5);

iii. *Trade*: moving energy in space rather than time (Section 6);

iv. *Backup capacity*, which is an investment question and waits for Section 7.

For each of the first three we will ask the same three questions at the end of the section: what does the new technology do to the two tails of the price distribution, what does it do to the value of wind and solar, and what does it do to system cost. Comparing the answers across the three sections gives a compact summary of the economics of flexibility.

> **Reading alongside this section.** Zerrahn et al. (2018) treat the economics of storage in renewable systems, in particular the question of *how much* storage a decarbonising system needs, which is where the result in Section 4.4 comes from. Butters et al. (2025) study battery investment in market equilibrium, which connects this section forward to Section 7. For the intertemporal optimisation technique itself, Appendix A covers everything used here. The lecture note "Generation Technologies" (Berg 2026b) describes the storage technologies themselves.

## 4.1 The model

Before writing down the model, let us briefly state what characterises a battery; "Generation Technologies" has the longer description. A battery is not a generator: it buys electricity in one hour and sells it back in another, and some of the energy is lost on the way. Four numbers describe it. First, a *power rating* $\bar{s}$ in MW, the most it can charge or discharge in an hour. Second, a *duration* $\bar{\delta}$ in hours; the combination $\bar{\delta}\bar{s}$ measures the energy a battery can hold in MWh. Third, one-way efficiencies $\eta_c$ for charging and $\eta_d$ for discharging, whose product is referenced *round-trip efficiency*. Fourth, a small operating cost $c^s$ per MWh charged or discharged, which is added to capture small wear on battery cells from usage. The stored energy at the end of hour $h$ is the *state of charge*, $soc_h$.

The model is then Section 3's problem with three additions, all of them in blue: a charge variable, a discharge variable, and the bookkeeping that connects them across hours. The planner chooses generation $q_{g,h}$, charging $g^c_h$, discharging $g^d_h$ and the stored energy $soc_h$ to solve

$$\min \quad \sum_h \sum_{g\in\mathcal{G}} c_g\, q_{g,h} \;+\; \sum_h c^s \left(g^c_h + g^d_h\right) \tag{19a}$$

$$\text{s.t.} \quad \sum_g q_{g,h} + g^d_h - g^c_h \;=\; D_h \qquad (\lambda_h) \qquad \forall h \tag{19b}$$

$$0 \le q_{g,h} \;\le\; \gamma_{g,h}\,\bar{q}_g \qquad (\underline{\mu}_{g,h}, \overline{\mu}_{g,h}) \qquad \forall g,h \tag{19c}$$

$$soc_h \;=\; soc_{h-1} + \eta_c\, g^c_h - g^d_h/\eta_d \qquad (\theta_h) \qquad \forall h \tag{19d}$$

$$0 \le soc_h \;\le\; \bar{\delta}\,\bar{s} \qquad (\underline{\psi}_h, \overline{\psi}_h) \qquad \forall h \tag{19e}$$

$$0 \le g^c_h \le \bar{s}, \quad 0 \le g^d_h \le \bar{s} \qquad (\underline{\phi}^c_h, \overline{\phi}^c_h;\ \underline{\phi}^d_h, \overline{\phi}^d_h) \qquad \forall h \tag{19f}$$

$$soc_T \;=\; soc_0 \qquad (\kappa) \tag{19g}$$

Equation (19a) now has marginal costs from the usual generators $g \in \mathcal{G}$ and the battery, where marginal costs of charge and discharge are assumed the same $c^s$. Equation (19b) now has the battery on the supply side when it discharges and the demand side when it charges. The battery is not a generator, it is both a producer and a consumer. Equation (19d) is the "law of motion" for stored energy: charging one MWh from the grid puts $\eta_c$ MWh in the tank, and delivering one MWh to the grid draws $1/\eta_d$ MWh out of it, so a round trip loses $1-\eta_c\eta_d$. Equations (19e) and (19f) are two *separate* capacity limits: storage technology is characterised by a power rating $\bar{s}$ (how fast) and an energy capacity, written as a duration $\bar{\delta}$ (how long). Finally eq. (19g) requires the tank to end where it started, without which the optimiser would cheerfully empty it on the last day and book the proceeds as free energy.

**Duality: the value of stored energy.** Let us run through the technique of Section 2.3 once more. Following the convention of Appendix A we maximise minus the system costs. Every term of Section 3's Lagrangian (14) survives unchanged — the battery's contribution $+g^d_h - g^c_h$ to market clearing splits off as a term of its own, because the Lagrangian is linear in the constraint — so we may write the new one as the old one plus what the battery adds,

$$\begin{aligned}\mathcal{L}^{\text{sto}} = \mathcal{L}^{\text{vre}} &+ \sum_h \lambda_h\left(g^d_h - g^c_h\right) - \sum_h c^s\left(g^c_h + g^d_h\right) \\ &+ \sum_h \theta_h\left(soc_{h-1} + \eta_c g^c_h - g^d_h/\eta_d - soc_h\right) + \sum_h \overline{\psi}_h\left(\bar{\delta}\bar{s} - soc_h\right) + \sum_h \underline{\psi}_h\, soc_h \\ &+ \sum_h \overline{\phi}^c_h\left(\bar{s} - g^c_h\right) + \sum_h \underline{\phi}^c_h\, g^c_h + \sum_h \overline{\phi}^d_h\left(\bar{s} - g^d_h\right) + \sum_h \underline{\phi}^d_h\, g^d_h.\end{aligned} \tag{20}$$

Because $\mathcal{L}^{\text{vre}}$ enters whole, the generators' conditions are unchanged from Section 3 — differentiating with respect to $q_{g,h}$ touches nothing in blue. The three new variables give three new optimality

conditions,

$$g_h^c: \quad -c^s - \lambda_h + \eta_c\,\theta_h + \underline{\phi}_h^c - \overline{\phi}_h^c = 0, \tag{21a}$$

$$g_h^d: \quad -c^s + \lambda_h - \theta_h/\eta_d + \underline{\phi}_h^d - \overline{\phi}_h^d = 0, \tag{21b}$$

$$soc_h: \quad -\theta_h + \theta_{h+1} + \underline{\psi}_h - \overline{\psi}_h = 0, \tag{21c}$$

together with complementary slackness on every bound, exactly as in (5b)–(5c).[11]

The new dual $\theta_h$ on eq. (19d) is the *shadow price of stored energy*: what one more MWh in the tank at the end of hour $h$ is worth to the system. Every dual so far has priced a *flow*; this one prices a *stock*. Read eq. (21a) by asking, as in Section 2, where $g_h^c$ can sit:

- Interior: charging, below the power rating. Both bound multipliers are zero, so $\lambda_h = \eta_c\,\theta_h - c^s$: the price paid for a MWh, plus the operating charge, equals what the $\eta_c$ MWh that reach the tank are worth.
- Upper bound: charging flat out. Now $\overline{\phi}_h^c = \eta_c\theta_h - c^s - \lambda_h > 0$: the price is below the threshold, and the inverter's rating is what stops the battery from buying more. $\overline{\phi}_h^c$ is the scarcity rent on the power rating.
- Lower bound: not charging. Now $\underline{\phi}_h^c = \lambda_h + c^s - \eta_c\theta_h > 0$: the price is above the threshold.

Equation (21b) reads the same way with the thresholds mirrored: discharging pays when $\lambda_h > \theta_h/\eta_d + c^s$. Finally, eq. (21c) says how $\theta_h$ moves through the year: when neither the empty-tank nor the full-tank bound binds, $\theta_{h+1} = \theta_h$. Moving a marginal MWh of stored energy from one hour to the next is then feasible in both directions, so its value cannot differ between them. The value steps only in hours where a storage constraint is active. Collecting the three conditions gives the following optimality rule:

**Result 3** (The arbitrage band)**.** *Optimal storage operation is a threshold rule around the stored-energy value: charge when the market price is below $\eta_c\,\theta_h - c^s$, discharge when it is above $\theta_h/\eta_d + c^s$, and hold in between. The width of the band is the round-trip loss plus twice the operating charge $c^s$, and $\theta_h$ itself moves only in hours where a storage constraint binds: a full or empty tank, or the power rating.*

Storage is thus an arbitrageur with a cost wedge, and $\theta_h$ is its private valuation of inventory. For a hydro reservoir this valuation has a name in the industry, the *water value*, and Section 6 will meet it as an equilibrium object.

## 4.2 What arbitrage does to prices

To investigate the effect of adding batteries to the model, the following section reuses the model from Section 3 and adds a single battery with a power rating of 1000 MW and a duration of 4 hours, so that it can charge or discharge at full power for four hours before it is full or empty. Its

[11] In the last hour the cyclic condition (19g) takes the place of $\theta_{T+1}$; its multiplier $\kappa$ plays that role.

one-way efficiencies give a round-trip efficiency of 90%, a standard value for lithium-ion batteries (see e.g. the short note on technologies, Berg 2026b), and its operating cost $c^s$ is set so small that it only stops the battery from cycling between hours of equal price. Because the state of charge links every hour to the next, the model is solved over all hours of the year in sequence rather than hour by hour.

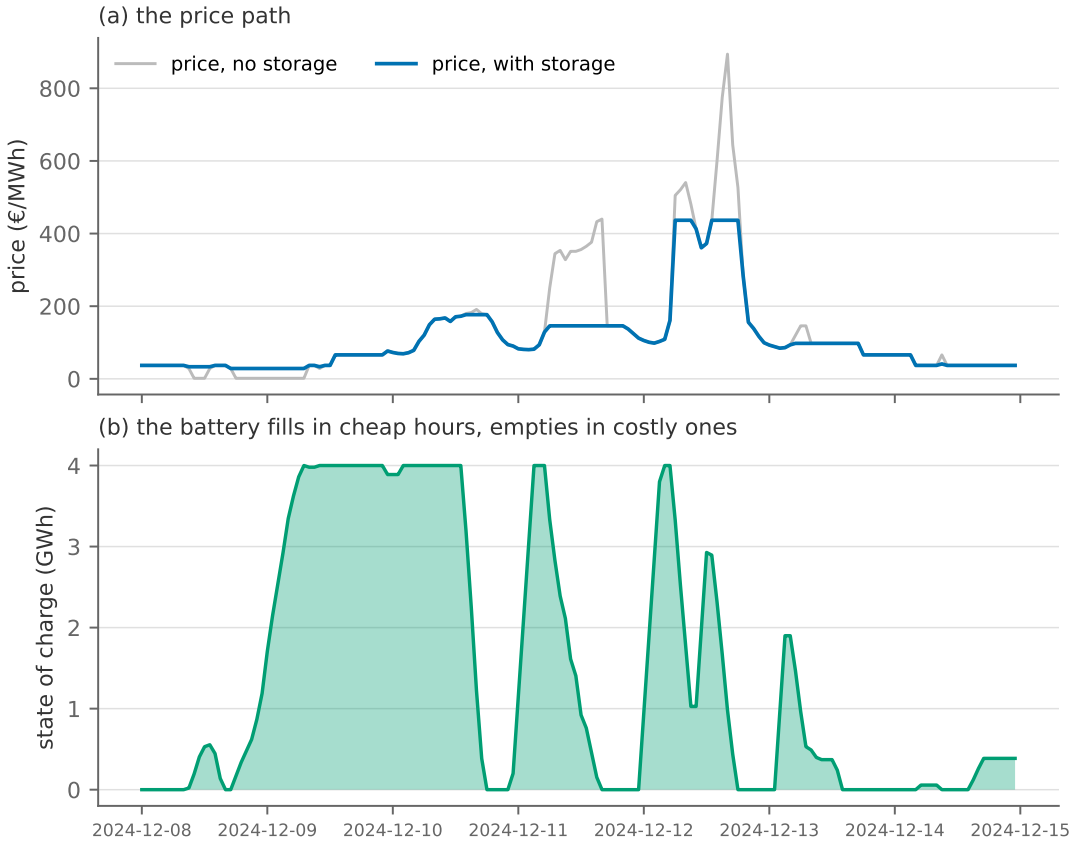



Figure 4.1: The week of the year with the most volatile prices, without and with the battery. The state of charge (b) is the mirror image of the price path (a): full just before the expensive hours, empty just after the cheap ones.

Figure 4.1 shows the battery at work in the most volatile week of the year. The pattern is exactly the threshold rule above: the state of charge rises in the cheap hours and is spent into the expensive ones. And because charging adds demand in cheap hours and discharging adds supply in costly ones, the price path itself is compressed — arbitrage moves the prices it trades on. Figure 4.2 shows the same effect for the whole year at once, zoomed in on the two tails of the price duration curve. Two things stand out. First, in the expensive tail the curve with storage sits below the no-storage curve, and the difference runs through the 10–15% most expensive hours of the year, not just the few scarcity peaks. Second, in the cheap tail the two curves coincide. In these hours the price is set either by curtailed wind, at its variable cost of about zero, or by imports at the neighbours' price, which is data in this model; the battery adds 1000 MW of demand in such an hour, but it exhausts neither the surplus nor the import capacity, so the price does not move. Over the year, the average price falls a little (41.0 to 38.6 €/MWh) while its standard deviation falls from 43.1 to 35.5 €/MWh. Storage does not necessarily make electricity cheaper on average, but

it makes the price *distribution* narrower, and in this system it does so from the top.

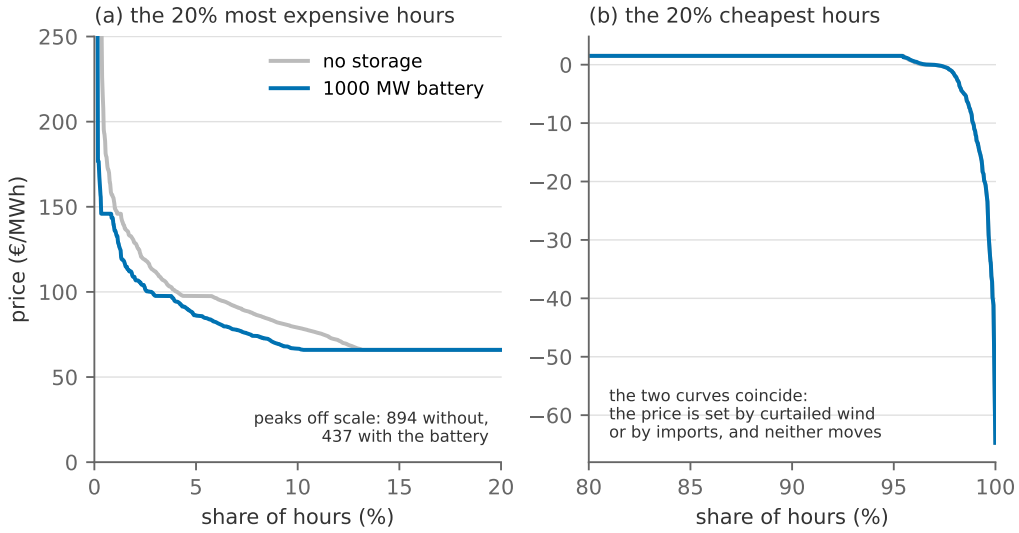



Figure 4.2: The two tails of the price duration curve for the year, with and without the battery: (a) the 20% most expensive hours, with the scarcity peaks off scale, and (b) the 20% cheapest hours, where the two curves coincide.

Storage's own profit is the flip side of that compression. The battery's arbitrage revenue over the year is 37.8 M€, while total system cost falls by 51.2 M€. However, this measure of system costs does not cover the investments required to stand up the battery. Priced at the investment costs of Section 7, with the inverter and the cells each annuitised over their own lifetime (Appendix B documents the sources), the battery costs 65.3 M€ a year to own, so this year's spreads alone would not pay for it.

Two things qualify that conclusion. First, batteries are the technology whose cost has fallen fastest: the price of lithium-ion cells has dropped by more than 90% since their commercial introduction in the early 1990s, at a learning rate of around 20% for every doubling of cumulative production (Ziegler and Trancik 2021; Way et al. 2022), and the investment cost above is a 2030 vintage that may well prove conservative. Second, the spread the battery earns on depends on the system it sits in. We will revisit the profitability of batteries in Section 7 in a 2050 system with far more solar and wind than DK1 has today, and there the optimal plan builds 3.2 GW of batteries at the 8-hour duration; the European system of Section 8 builds far more still.

In the next few subsections, we will explore further how the overall system configuration affects the profitability of batteries and what this tells us about complementarities between different technologies.

## 4.3 Adding more rigidities to the system – ramping constraints

We note that beyond the technical specifications of the battery, including the investment costs, what matters for the profitability of the battery is how flexible the system already is as it determines the price spread that it can take advantage of. So, our first extension here is to add some realistic rigidities to the system, to see if this changes our conclusion for the batteries – ramping constraints:

A large thermal plant cannot jump from zero to full output between two hours: boilers have thermal inertia, turbines have mechanical limits, and pushing them hard costs money and lifetime. The simplest way to model this is to include simple limits on how much a generator can adjust its output from hour to hour.

Let generator $g$ face a *ramp rate* $\rho_g \in (0, 1]$: the share of its capacity by which it can increase output from one hour to the next. Add to the model

$$q_{g,h} - q_{g,h-1} \;\leq\; \rho_g \, \bar{q}_g, \qquad (\xi_{g,h}), \tag{22}$$

with $\xi_{g,h} \geq 0$ its multiplier. Now the generation $q_{g,h}$ appears in two ramping constraints, once as the new level in hour $h$, and once as the old level in hour $h+1$; the optimality condition now reads:

$$-c_g + \lambda_h + \underline{\mu}_{g,h} - \overline{\mu}_{g,h} - \xi_{g,h} + \xi_{g,h+1} \;=\; 0. \tag{23}$$

Rearranging gives the section's second result.

**Result 4** (The ramping wedge)**.** *With ramping constraints, generator $g$ enters the merit order in hour $h$ not at its marginal cost $c_g$ but at an* effective *marginal cost*

$$c_{g,h} \;=\; c_g \;+\; \underbrace{\xi_{g,h} - \xi_{g,h+1}}_{\textit{flexibility cost}}. \tag{24}$$

*The merit order therefore becomes* time-dependent*: the same two plants can swap places from one hour to the next without any cost or price changing. A plant may run when $c_g > \lambda_h$ (accepting a loss now to be positioned for an expensive hour later), and may withhold when $c_g < \lambda_h$ (because ramping up now would strand it in a cheap hour).*

Here, $\xi_{g,h}$ is what the plant paid in hour $h-1$ in terms of output it had to produce, at a loss, purely to be high enough to reach its hour-$h$ level. $\xi_{g,h+1}$ is what it is willing to pay *now* in order to reach a desired level in hour $h+1$. A peaker anticipating an expensive evening starts climbing in the afternoon and takes a loss doing it. Let us dig a bit deeper into what this tells us:

i. *"Flexibility" is now a measurable cost:* The quantity $\xi_{g,h} - \xi_{g,h+1}$ is in €/MWh and can be summed over the year. This is the missing third column in any comparison of generating technologies: investment cost, marginal cost, *and* flexibility cost.

ii. *It reorders which technologies are attractive.* A coal or nuclear plant with a low $\rho$ is penalised precisely in the volatile hours that a renewable-heavy system produces in abundance, while a gas peaker with $\rho \approx 1$ is not. This is a large part of the reason Section 7's optimal system builds OCGT rather than anything cheaper per MWh, and a large part of the reason nuclear and variable renewables tend to be *substitutes* rather than complements: an inflexible baseload plant and a fluctuating one compete for the same residual hours.

iii. *The price no longer equals anybody's marginal cost.* In an hour where a ramping constraint

binds, $\lambda_h = c_{g,h} \neq c_g$ for the marginal plant. Section 2's crisp identity survives — the price is still the marginal cost of serving one more MWh — but the marginal cost is now a shadow-price-adjusted object.

Figure 4.3 puts numbers on the wedge for DK1's own fleet. One ramp rate is imposed on the slow thermal plant – the coal, wood and waste CHP boilers, 1.4 GW in all – and tightened from none to 0.1 per hour, while the gas turbine and the 5.5 GW of interconnectors stay free. At the catalogue rate of 0.9 per hour, which is what the network sections of Sections 6 to 8 impose, the summed flexibility cost is 0.9 M€ a year and the battery's value does not move; at 0.1 the rigidity costs the system 15.4 M€ and the battery's value rises by 15%. The rigidity that matters in DK1 is not in its boilers: flexibility arrives through the wires, and it is the wires the battery competes with.

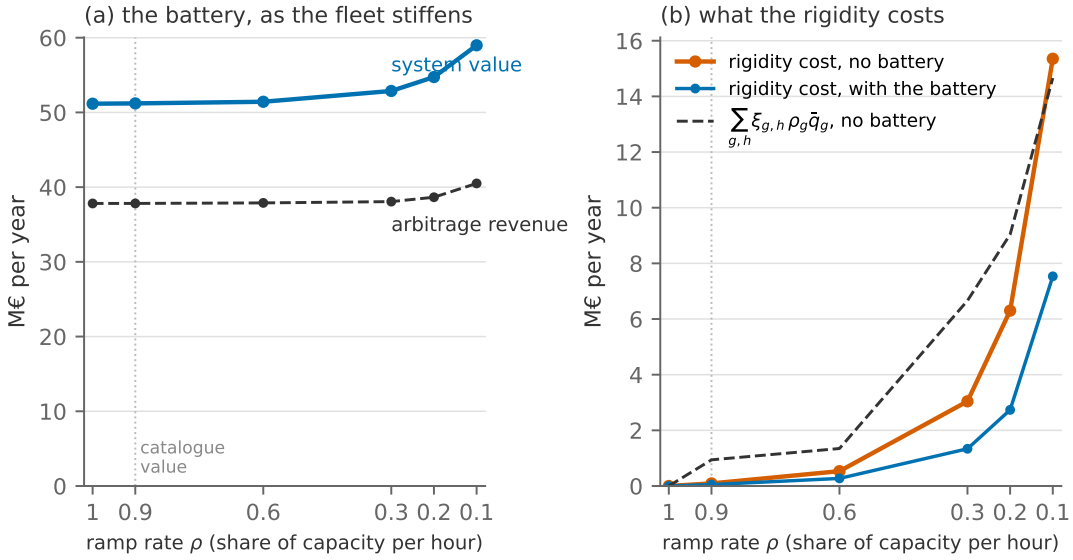



Figure 4.3: The ramping wedge in DK1. A ramp rate $\rho$ on the slow thermal plant, tightened from none ($\rho = 1$) to 0.1 per hour. (a) The 1000 MW battery's arbitrage revenue and its system value. (b) What the rigidity costs the system relative to no limit, with and without the battery, and the summed shadow value of the ramp constraints $\sum_{g,h} \xi_{g,h}\, \rho_g\, \bar{q}_g$. The dotted line marks the catalogue rate.

**Which way does it bias the answer?** *Frictionless ramping.*

- *No ramp limits.* A fleet that can follow any residual load shape **overstates** the flexibility of the conventional plants, and therefore **understates** the value of everything that competes to provide flexibility: storage, interconnection, demand response, and gas peakers relative to baseload. It also **understates** price volatility, because a fleet that can follow any residual load shape produces fewer extreme hours than a real one.
- *Perfect foresight.* The planner never ramps in the wrong direction, which a real operator sometimes does, so this too **overstates** how easily the system can be run.

## 4.4 Storage and the value of wind

Now we turn to exploring the complementarity of wind and storage. Figure 4.4 re-runs the deep-cannibalisation world — wind at 1.5 times DK1's own capacity — with the battery swept from zero to 8 GW.

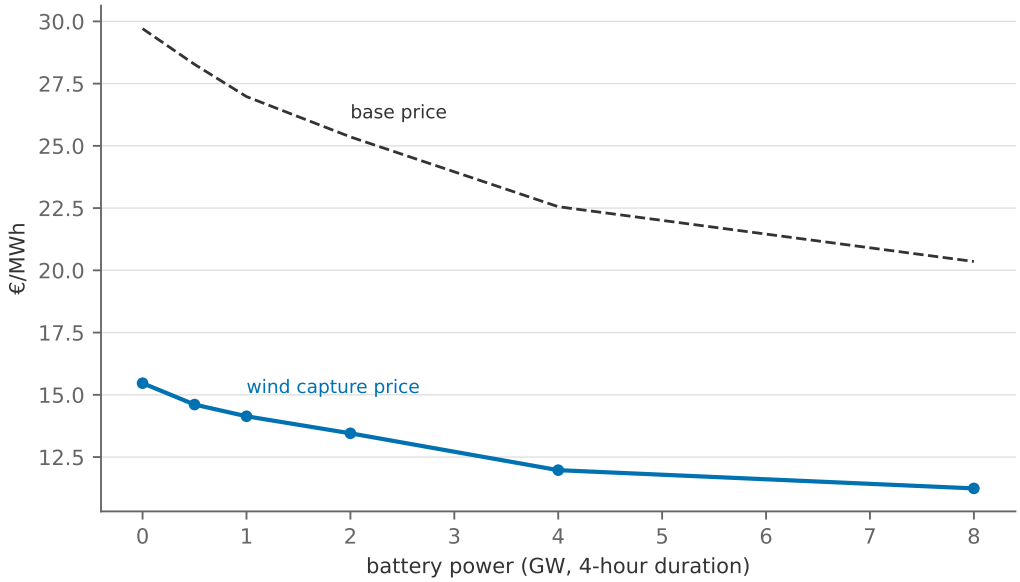



Figure 4.4: Wind's capture price and the base price $\bar{p}$ (eq. (18)) as battery capacity grows, with wind at 1.5 times DK1's actual capacity. Even 8 GW of 4-hour batteries barely moves wind's relative value: the surpluses it would need to absorb last for days, not hours.

The figure shows that the effect of the battery, even at the huge scale of 8 GW, is moderate: While prices generally drop, the value factor for wind only lifts from 0.52 to 0.55, with the capture price actually dropping. The main reason for this is a mismatch of time scales between the 4-hour battery and the wind with availability profiles with peaks and calms lasting potentially for days: Against a week of surplus the battery fills in its first four hours and then stands full and useless; against a week of calm it is empty by breakfast. (Solar is the opposite case: its cycle is exactly diurnal, so the same battery is a near-perfect complement to solar — which is why the two arrive as a pair in Section 7's optimal mixes.) The general lesson: "storage" is not one technology but a family indexed by *duration*, and each VRE technology needs the flexibility that matches its own spectrum of fluctuations. What can absorb a Danish wind week is long-duration storage (hydrogen, reservoirs) — or the trade of Section 6, which is storage in space rather than time, with Norwegian reservoirs playing the tank.

One more lesson hides in the exercise, to be collected in Section 9: everything above assumes the battery knows the week's prices in advance. The LP's operator has perfect foresight, and part of the arbitrage value computed here is the value of that clairvoyance; Section 9.2 revisits this.

**From duals to markets.**

| Constraint | Dual | What it measures | Market counterpart |
|---|---|---|---|
| storage balance (19d) | $\theta_h$ | value of a stored MWh at hour $h$ | a hydro reservoir's *water value*; a battery trader's inventory valuation |
| energy capacity $soc_h \le \bar{\delta}\bar{s}$ | $\overline{\psi}_h$ | value of a larger tank | the price premium of storage *duration* |
| power rating $g^c_h \le \bar{s}$, $g^d_h \le \bar{s}$ | $\overline{\phi}^c_h, \overline{\phi}^d_h$ | value of faster charge/discharge | an inverter's capacity value |
| ramp limit (22) | $\xi_{g,h}$ | the flexibility cost of moving output | the loss a plant accepts to be positioned for a peak |

**The three questions.** As promised at the start of the section, the same three questions will be answered at the end of Sections 5 and 6.

| **Adding storage to the system. . .** | |
|---|---|
| . . . does what to the two tails of the price distribution? | Compresses the expensive tail; the cheap tail does not move, because its price is set by curtailed wind or by imports at the neighbours' price. The standard deviation falls from 43.1 to 35.5 €/MWh while the mean barely moves. |
| . . . does what to the value of wind? | Very little at 4-hour duration (0.52 → 0.55 even at 8 GW): the timescales do not match. It would do a great deal for solar. |
| . . . does what to system cost? | Lowers it, by 51.2 M€ a year at the base battery — and with strongly decreasing returns, since each MW narrows the spread the next would earn on. This *does not*, however, include investment costs. |

# 5 Electrification and sector coupling

The green transition is not only a story about electricity. To a large extent it is a story about everything else becoming electricity: heating, transport, and large parts of industry. Heat is the natural first case for this note. Denmark's district heating systems are large, centrally dispatched, and increasingly served by heat pumps drawing power from the grid. This section adds a heat sector to the power model and asks what electrification does to load, prices and emissions.

> **Reading alongside this section.** This section requires no supplementary reading. Ruhnau et al. (2020) study heat pumps, wind and the interaction between them in a model of five north-western European power markets, and Section 5.3 ends with their two findings: what a heat pump's consumption costs the power system, and what heat pumps do, and do not do, for the value of wind. The heat pump is also this note's first price-responsive consumer, which makes it the natural place to meet the demand-response literature: Ambec and Crampes (2021) on how real-time pricing can balance

intermittent supply in theory, and Fabra et al. (2021) on how much consumers actually respond in practice. Andersen and Dietrich (2026) is the Danish counterpart to the latter, on household smart meter data, and Section 5.2 takes its central finding — that flexibility follows the appliance — as the qualification to everything this section makes the heat pump do. Joskow and Wolfram (2012) is a short overview of dynamic pricing if the topic is new to you.

## 5.1 A heat market and a heat pump

The model adds a second market, for heat, with a district heating load $D_h^H$ and two technologies to serve it. The first is a gas boiler, which produces heat at a constant marginal cost $c^B = p^{gas}/\eta^B + o^B$ per MWh of heat. It is the marginal heat technology throughout, so whenever it runs it sets the heat price: merit order logic transplanted to a new carrier. The second is the heat pump, and this is where the coupling lives. It is a conversion from the electricity market to the heat market, whose efficiency is the hour's **coefficient of performance** (COP),

$$q_h^H \;=\; COP_h \cdot q_h^E, \qquad COP_h = 6.81 - 0.121\,\Delta T_h + 0.00063\,\Delta T_h^2, \tag{25}$$

where $q_h^E$ is the electricity the pump draws, $q_h^H$ the heat it delivers, and $\Delta T_h$ the temperature lift from the outdoor air to the 50 °C supply of the district heating network. The curve is the empirical air-source relation of Staffell et al. (2012). For a reader meeting the COP for the first time: A COP of 3 means that one MWh of electricity delivers three MWh of heat. No thermodynamic laws are harmed, because a heat pump does not *create* heat, it *moves* it from outdoors to indoors, and moving energy is cheaper than making it. The catch is that the cost of moving heat rises with the temperature lift, which is exactly what eq. (25) encodes.

Figure 5.1 shows the outdoor temperature over the year and the COP it implies.

The figure shows that (i) the COP averages a comfortable 3.0 over the year, a shade below the 3.2–3.6 the Danish Energy Agency's catalogue assumes for large district-heating heat pumps (Danish Energy Agency and Energinet 2024), which exploit lower supply temperatures and better heat sources than our deliberately simple setup; (ii) it ranges from 1.8 on the coldest hours to 4.3 in summer; and (iii) it falls to its lowest values exactly when it is coldest, which is precisely when heat demand peaks. This adverse correlation is the signature fact of heating electrification: The technology is least efficient exactly when it is most needed. Electrified heat therefore adds a load that is not just large but *peaked and weather-correlated*, arriving on top of a power system whose tight hours are the cold, dark, calm ones (Section 3).

The instance is the Section 3 power system plus a stylised 20 TWh/yr heat load whose shape follows Danish degree-hours (Aarhus temperature, 2024) with a flat hot-water base; the base case gives the heat pump 800 MW of electric capacity.[12] Written out in full, with the heat side in blue,

[12] The heat side is deliberately coarse: no combined heat and power plants, no thermal storage in the network, one representative temperature. Each would add realism, Danish CHP in particular, without changing the mechanism this section teaches; Section 5.4 discusses the first of them.

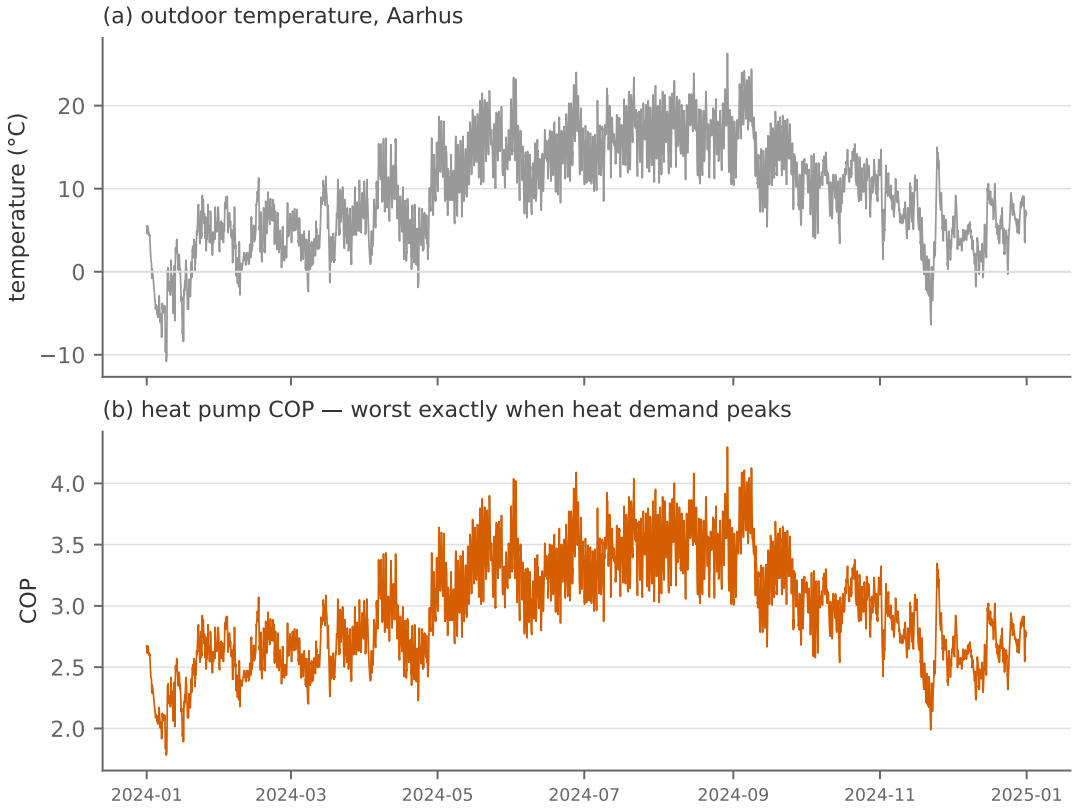



Figure 5.1: Outdoor temperature (a) and the heat pump COP it implies (b), hourly over 2024.

the planner chooses generation $q_{g,h}$, boiler heat $q^B_h$ and the heat pump's electricity intake $q^E_h$ to solve

$$\min \quad \sum_h \sum_g c_g\, q_{g,h} \;+\; \sum_h c^B q^B_h \tag{26a}$$

$$\text{s.t.} \quad \sum_g q_{g,h} - q^E_h \;=\; D_h \qquad (\lambda^E_h) \qquad \forall h \tag{26b}$$

$$q^B_h + COP_h\, q^E_h \;=\; D^H_h \qquad (\lambda^H_h) \qquad \forall h \tag{26c}$$

$$0 \le q_{g,h} \le \gamma_{g,h}\, \bar{q}_g \qquad (\underline{\mu}_{g,h}, \overline{\mu}_{g,h}) \qquad \forall g,h$$

$$0 \le q^E_h \;\le\; \bar{q}^{HP} \qquad (\underline{\mu}^{HP}_h, \overline{\mu}^{HP}_h) \qquad \forall h \tag{26d}$$

$$0 \le q^B_h \;\le\; \bar{q}^B \qquad (\underline{\mu}^B_h, \overline{\mu}^B_h) \qquad \forall h \tag{26e}$$

This is the problem of Section 3 with a second market bolted onto it. Two features are worth dwelling on before we solve it. First, the heat pump appears with a *minus* sign in the electricity balance (26b) and a *plus* sign in the heat balance (26c): It is a consumer in one market and a producer in the other, which is what **sector coupling** means operationally. Second, notice where the COP sits: multiplying $q^E_h$ in eq. (26c). It is a *conversion factor between two markets*, and because it is time-varying, the exchange rate between electricity and heat moves hour by hour.

Note also that the heat pump does *not* need a cost of electricity in its own objective term, although it is tempting to put one there. It is already accounted for: The pump's electricity

consumption enters eq. (26b), so it is priced at $\lambda_h^E$ automatically by the market clearing constraint. Adding a cost term as well would charge for the same electricity twice.

**Duality: the heat price and the fuel-switching margin.** The Lagrangian follows the same additive pattern as Section 4's, and it is worth noticing that it branches from the same place: The heat sector is built on Section 3's problem, not on the battery, so it is $\mathcal{L}^{\text{vre}}$ that reappears here, with the pump's withdrawal $-\,q_h^E$ from market clearing splitting off as its own term and $\lambda_h \equiv \lambda_h^E$,

$$\begin{aligned}\mathcal{L}^{\text{heat}} = \mathcal{L}^{\text{vre}} &- \sum_h \lambda_h^E\, q_h^E - \sum_h c^B q_h^B + \sum_h \lambda_h^H \left(q_h^B + COP_h\, q_h^E - D_h^H\right) \\ &+ \sum_h \overline{\mu}_h^{HP}\left(\bar{q}^{HP} - q_h^E\right) + \sum_h \underline{\mu}_h^{HP}\, q_h^E + \sum_h \overline{\mu}_h^B \left(\bar{q}^B - q_h^B\right) + \sum_h \underline{\mu}_h^B\, q_h^B. \end{aligned} \tag{27}$$

The generators' conditions are untouched, for the same reason as in Section 4: $\mathcal{L}^{\text{vre}}$ enters whole. The two new variables give

$$q_h^E: \quad -\lambda_h^E + COP_h\, \lambda_h^H + \underline{\mu}_h^{HP} - \overline{\mu}_h^{HP} = 0, \tag{28a}$$

$$q_h^B: \quad -c^B + \lambda_h^H + \underline{\mu}_h^B - \overline{\mu}_h^B = 0, \tag{28b}$$

with complementary slackness on each bound. Equation (28b) is the merit order of Section 2 on the heat market: Whenever the boiler runs below capacity, $\lambda_h^H = c^B$. Equation (28a) is the coupling, and we read it as usual by asking where $q_h^E$ can sit:

- Interior: the pump is marginal in the heat market. Both bounds are slack and $\lambda_h^H = \lambda_h^E / COP_h$: The heat price is the electricity price converted through the hour's COP.
- Upper bound: the pump runs flat out. Then the boiler is marginal, $\lambda_h^H = c^B$, and $\overline{\mu}_h^{HP} = COP_h\, c^B - \lambda_h^E > 0$ is the pump's scarcity rent: One more MW of pump would displace boiler heat worth $COP_h\, c^B$ at an electricity cost of $\lambda_h^E$.
- Lower bound: the pump is off. Then $\underline{\mu}_h^{HP} = \lambda_h^E - COP_h\, \lambda_h^H > 0$: Electricity is too costly, in the hour's exchange rate, to beat the boiler.

The dispatch rule is therefore a comparison of $\lambda_h^E / COP_h$ with $c^B$. Because $COP_h$ moves with the temperature, the same electricity price can put the pump on one side of the boiler in a mild hour and on the other side in a cold one. Sector coupling thus means that the two prices are linked by an arbitrage condition, eq. (28a), exactly as two zones will be linked by an interconnector in Section 6.

## 5.2 The coupled dispatch

Figure 5.2 shows how the heat load is served in the coldest week of the year, and the electricity price the heat pump faces.

The dispatch follows the rule derived above: The pump serves heat whenever $\lambda_h^E / COP_h$ is below the boiler's cost, and the boiler covers what the pump's capacity cannot, or takes over

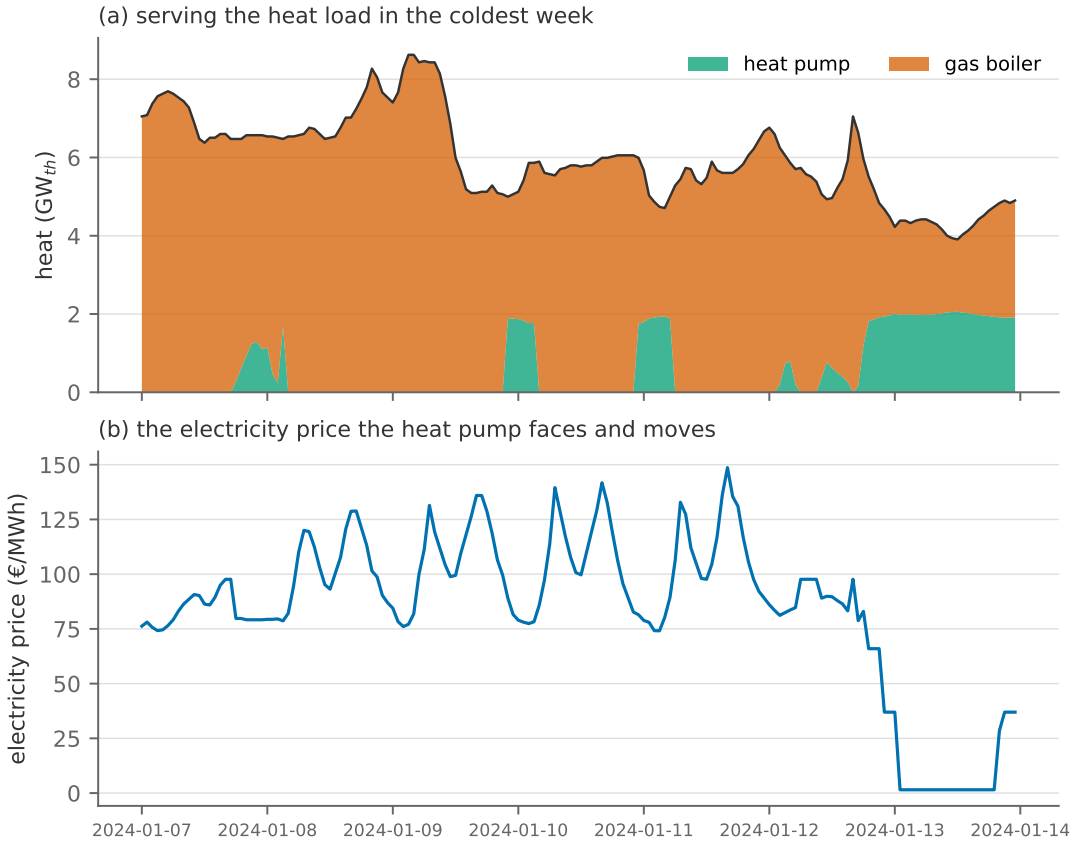



Figure 5.2: The coldest week of the year: how the heat load is served (a) and the electricity price the heat pump faces (b).

entirely when power is costly. Over the year the pump serves 59% of the heat.

**This is demand response.** It is worth naming what we have just built. The heat pump is the first *price-elastic electricity consumer* in this note. Every load we have met until now was a fixed number $D_h$ that had to be served (Section 2.4 showed what it takes to relax even that, and it took the value of lost load to do it). The heat pump is different: It looks at $\lambda_h^E$ and switches off when power is expensive, because it has an outside option, the boiler. That is precisely what **demand-side flexibility** means, and it is the second item on the menu of responses to intermittency that Section 4 set out. One qualification decides what it can do for the two tails of the price distribution. The heat pump is a *new* load, and its flexibility keeps it out of the costly hours, which is not the same as taking consumption that was already there out of them. Demand that existed before and becomes flexible can relieve both tails: It leaves the scarce hours and fills the cheap ones. A new load, however flexible, can at best fill the cheap tail.

It is worth briefly noting that demand flexibility comes in many shapes and forms, and they differ quite a lot in *how* flexible they actually are. Our current model with district heating, the heat pump can be very flexible: It can be turned on/off at any given hour, in particular because the backup boiler is always there to kick in if needed. Other types of "flexible" demand may in fact be much less flexible. Recent Danish evidence makes the point sharply: Andersen and Dietrich (2026) match household smart meter records to the hourly price and find that most households do

not respond to it at all, while nearly a third cut consumption significantly when the price rises. Averaged over everyone, a DKK 1/kWh increase in the hourly price — some 130 €/MWh, a very large move — lowers demand by 2.6%, an implied price elasticity of about −0.06. What separates the responsive third from the rest is the instructive part. Of every household characteristic they measure — income, education, age, total consumption — the largest single effect is owning an electric vehicle: It adds some 1.4 percentage points to a household's response per DKK/kWh, against that sample average of 2.6, and it survives controlling for all the others.

That is the same mechanism this subsection has just derived, met in the data. A household's lighting and cooking have no outside option and no store, so a price signal has nothing to work on; a car battery has both, exactly as our heat pump has the boiler behind it. Flexibility is not really a property of consumers, then, but of what they are consuming for: It comes from having somewhere else to put the demand, or some later hour to put it in. This is worth carrying into the sections ahead, where "flexible demand" will appear as a modelling assumption rather than an estimate. We return to it in more detail later in the course.

## 5.3 The rollout experiment

Let us now use the model to ask the policy question: What happens to emissions and prices as more and more heat is electrified? Figure 5.3 sweeps the heat pump's capacity from zero to 2 GW, counting emissions across *both* sectors.

The figure shows the following:

i. Without a carbon price, emissions fall from 8.26 to 6.09 Mt over the sweep, a decrease of 26%.

ii. If we instead combine the rollout of the heat pumps with a carbon tax around the level of the EU ETS, we reach a reduction of 53.2% after 800 MW of pumps.

iii. Beyond 2.0 GW the taxed line turns back *up*: Further rollout raises emissions.

iv. The average electricity price rises (from 41.0 to 51.5 €/MWh), while the heat price falls toward the pump's cost as pump capacity stops being scarce.

**Emission reductions in the model.** The main mechanism of the rollout is the following: A MWh of heat moved from the heat boiler to the pump avoids the boiler's gas, and costs $1/COP$ MWh of electricity. While this clearly reduces emissions from the heat boiler, the full emission reduction depends on how emission intensive the *marginal* generator in the electricity market is; in the current model system, this is coal 16% of hours. At a winter COP of around two, the emissions avoided in the boiler and the emissions added at the coal plant nearly cancel. Heat pumps are only as clean as the marginal electricity behind them: The right emissions number for an electrification decision is the marginal intensity, not the grid average, and the two can differ by a factor of several. This is also why the two curves in panel (a) behave as they do. With the carbon price pushing

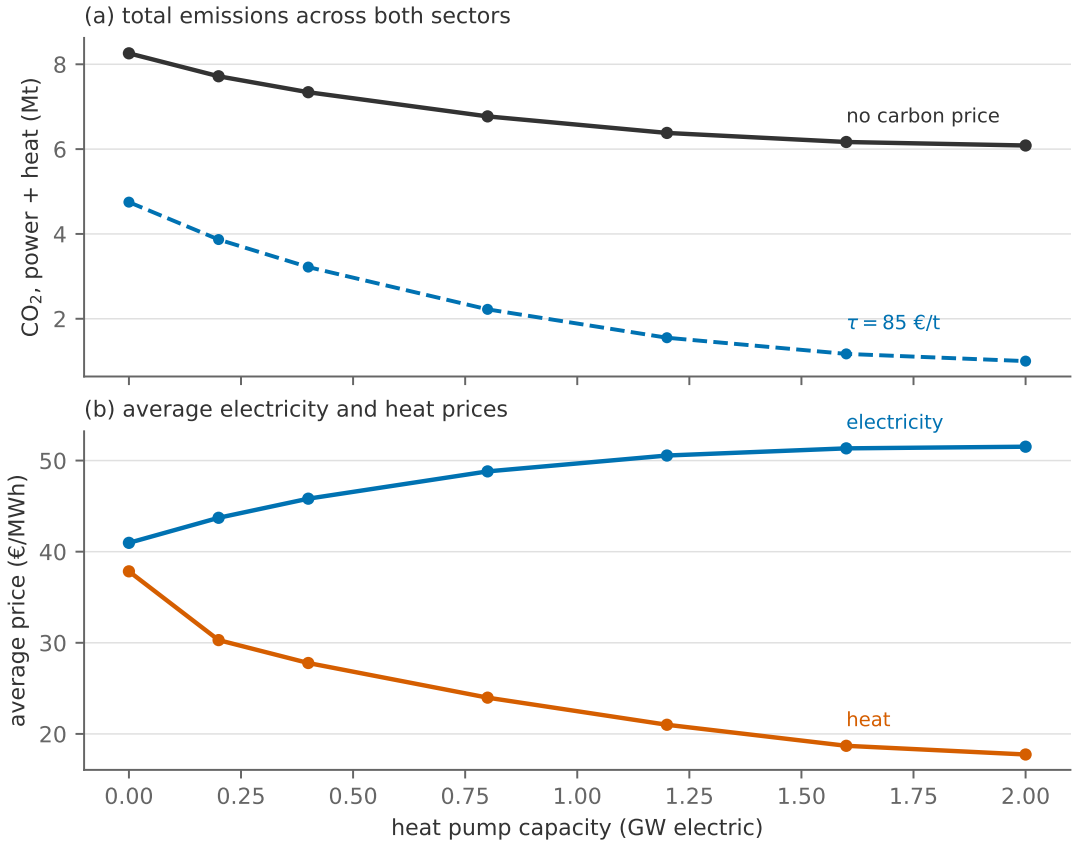



Figure 5.3: Heat pump capacity swept from zero to 2 GW electric: (a) total $CO_2$ across power and heat, without and with the carbon price of 85 €/t from Section 2; (b) average electricity and heat prices.

coal off the margin, the same pumps do far more good; electrification pays off *jointly* with clean marginal power.

**Carbon leakage.** Another perspective that our calculations above do not catch is that in 35% of hours, the marginal unit supplying electricity is not a Danish plant at all, but the import plant. We do not count those emissions in our scheme (following the territorial convention that counts what is emitted inside the country). As we rely on more imported electricity, however, foreign emissions are likely to increase as well; we refer to this as carbon leakage. This is an interesting topic that is not unique to this experiment. We will not elaborate much more on this point here, except to note that this is *the* main reason why coordinated carbon policies e.g. like the EU ETS are more efficient at reducing global emissions than national policies.

**Complementarity of heat pumps and VRE.** It would be easy to leave this section believing that heat pumps and wind are natural allies: Heat demand and wind are both higher in winter, the pump can consume when power is cheap, cheap hours are windy hours, so electrification should lift wind's capture price. Ruhnau et al. (2020) test that proposition in a model of five north-western European power markets in which the planner builds everything except the wind turbines and the heat pumps, and their answer is more sober. It comes in two parts.

The first is a new metric, the capture price of eq. (17) turned around to the demand side. A heat

pump's **load cost** is the average electricity price weighted by the pump's own consumption: what the system charges, per MWh, for serving that load. In our base case the pump pays 47.0 €/MWh against an average price of 48.8, a little less, because the boiler lets it skip the costly hours. Weighted instead by the shape of the heat load, which is the electricity a pump with neither a boiler nor a heat store behind it would have to draw, the same hourly prices average 60.6 €/MWh, and that is before such a load, heaviest in the cold hours, has pushed those prices up itself. This is the case Ruhnau et al. (2020) study: Their heat pumps follow the heat load, with an electric heater as backup, and their load cost rises by some 21 €/MWh as heat pumps grow from nothing to 15% of electricity consumption. An inflexible heating load arrives in the cold hours that already set the peak, the planner builds peaking plants to serve it, and the pumps consume in exactly the scarce hours whose prices pay for those plants. Wind brings the load cost down again, and so do heat pumps with a smaller and steadier load (ground-source pumps feeding floor heating) and thermal storage that lets the pumps move their consumption by a few hours; together they more than offset the increase.

The second part is the question we started with, and there the effect on wind's market value is small: a few euros per MWh at most, and nothing at all at a 30% wind share. They give two reasons, and our own model suggests a third:

i. *More heat pumps need more wind.* At a given wind capacity, heat pumps do raise wind's value, by 2–3 €/MWh, because they add demand in the windy season. But the paper holds wind's *share* of consumption fixed, as renewable targets are written, so every heat pump brings more wind with it, and the extra wind cannibalises its own value (Section 3.4). At a 30% wind share the two effects cancel.

ii. *Flexibility crowds out flexibility.* Thermal storage lets the pumps shift their load into windy hours, and at moderate wind shares it does reduce curtailment. Yet it leaves wind's value where it was, and at 30% wind lowers it slightly: By flattening the price spreads that interconnectors and pumped hydro earn their keep from, it makes the planner build fewer of them. With the pumped hydro held at its old size, the thermal storage does give wind a small gain. And the storage itself runs into the time-scale mismatch that defeated the battery in Section 4.4: Heat demand varies *seasonally*, while hot water tanks and building mass hold hours, so the storage lowers the pumps' load cost by less the more heat pumps there are.

iii. *Prices move in steps.* This one is ours, although the paper notes the case behind it: Shifted consumption need not change a price at all if it only absorbs energy that would otherwise have been curtailed at a price of zero. More generally, in a linear model, moving a modest amount of demand into a cheap hour does not move that hour's price, because the price is pinned at the marginal technology's cost until the shift is large enough to change *which* technology is marginal. The step-shaped supply curve of Figure 3.4 is exactly this: Prices jump between plateaus rather than responding smoothly. So a model can simultaneously report that heat pumps consume in cheap hours and that wind gains nothing. That is not a

contradiction but a property of the step function.

The paper's policy conclusion follows from the first metric. Most heat pump owners pay a retail tariff whose electricity is bought on a *standard load profile*, so the cost of a peaky load and the value of a flexible one are averaged over all customers, and neither reaches the owner who could act on it. Settling heat pumps on hourly wholesale prices through smart meters would let each pump face its own load cost, which is the point of Andersen and Dietrich (2026) again: A price signal works on a consumer with somewhere else to put the demand, and a heat pump with a heat store has exactly that.

## 5.4 What is left out: combined heat and power

Danish district heating is historically built on **combined heat and power** (CHP): plants that generate electricity and capture the waste heat instead of throwing it away. This type of coupling creates some interesting dynamics: Recall from Figure 3.8, the local CHP plants kept producing electricity through hours of storm-level wind and prices at or below zero, and Section 3.6 attributed this to a commitment to deliver district heat in a cold month. That is the joint-output constraint described below at work, seen from the electricity side: A plant whose heat is needed produces electricity whether or not the electricity price covers its cost.

A regular condensing power plant converts perhaps 25–35% of its fuel into electricity and rejects the rest as warm water. A *back-pressure* plant produces electricity and heat in a roughly fixed ratio $\nu = q^E/q^H$, so with $\eta^E$ its electrical efficiency, its total conversion efficiency is

$$\frac{q_h^E + q_h^H}{\text{fuel in}} \;=\; \left(1 + \frac{1}{\nu}\right)\eta^E, \tag{29}$$

which lands in the range 85–95%.

Modelling it would a fixed proportion between two outputs exactly like eq. (26c) but with $\nu$ in place of the COP, and one genuinely new economic result. A back-pressure plant's value depends on *both* prices at once: It is worth running when $\lambda_h^E + \lambda_h^H/\nu$ exceeds its marginal cost. Where a heat pump *correlates* the two prices (it buys power to make heat, so it pushes power prices up and heat prices down at the same time), CHP *decorrelates* them: A back-pressure plant is profitable on high electricity prices even when heat is cheap, and vice versa. Two coupling technologies, opposite signs on the same statistic, which is a good reminder that "sector coupling" is not one thing with one effect.

**Which way does it bias the answer?** *A constant coefficient of performance, and a fixed fleet.*

- *Had we used one year-round COP of 3.0 instead of the hourly curve.* The pump runs hardest when it is coldest, and cold hours are low-COP hours, so the hours that carry the most weight are systematically the hours where the true COP is below the mean. A constant COP therefore **overstates** the pump's efficiency where it matters and **overstates** its value: It understates the electricity drawn in the tightest hours, the peak load electrification adds, and

the backup capacity the system needs.

- *The power fleet is held fixed.* This stacks the deck against electrification, and **overstates** its emissions and its price effect: A planner adding 2 GW of heat pumps would likely also add wind to serve them. What the fleet *should* look like once investment responds is exactly the question of Section 7.

**From duals to markets.**

| Constraint | Dual | What it measures | Market counterpart |
|---|---|---|---|
| electricity clearing (26b) | $\lambda_h^E$ | power system marginal cost | the day-ahead spot price |
| heat clearing (26c) | $\lambda_h^H$ | marginal cost of a MWh of heat | the marginal-cost basis of a district heating tariff |
| heat pump capacity (26d) | $\overline{\mu}_h^{HP}$ | value of one more MW of pump | the scarcity premium in cold snaps |
| derived: $\lambda_h^E / COP_h$ vs boiler cost | | the fuel-switching margin | heat-pump-versus-gas dispatch parity, the heat sector's "spark spread" |

**The three questions.**

| **Adding heat pumps to the system. . .** | |
|---|---|
| . . . does what to the two tails? | Fills in the *cheap* tail only. The pump adds demand when power is cheap and withdraws it when power is costly, so surplus hours shrink while scarcity hours are untouched: As a new load it cannot *supply* anything on a cold calm evening. Without the boiler or a heat store behind it, it would add to the scarce hours instead, which is the case Ruhnau et al. (2020) study. Only demand that was already there and becomes flexible relieves both tails. The average power price rises (41.0 → 51.5 €/MWh). |
| . . . does what to the value of wind? | Raises it at a given wind capacity, by putting demand into the windy season. But see Ruhnau et al. (2020) above: Once wind grows with the new load and other flexibility investment responds, the effect is small, and nil at high wind shares. |
| . . . does what to system cost? | Lowers the cost of *heat* substantially, raises the cost of power, and cuts total emissions by only 26% against this fleet, because the marginal electricity is often coal. Electrification pays off jointly with clean marginal power, not before it. |

One forward pointer closes the section. Flexible heat is storage in disguise: Buildings and district heating networks hold heat cheaply for hours, so an electrified heat sector is not only new load

but potentially the cheapest flexibility on the system, the thermal analogue of Section 4. We have left it out here deliberately, to teach the coupling mechanism with a single technology and a single new equation. The next section leaves the single node behind and asks what trade with the neighbours can do.

# 6 Transmission and trade

Denmark's electricity system cannot be understood as an island: On a windy day DK1 exports most of what it produces, and on a calm one it lives off Norwegian water and German gas. This section puts the single node of Sections 3 to 5 into its geography, and adds the third item on the flexibility menu of Section 4: trade, which moves energy in space rather than in time. Trade needs wires, and wires have physics that no previous section required. So the section first describes the market's geography and the grid that carries it, at the level an economist needs, and then extends the model with zones and the lines between them. From the model we derive what a congested line is worth, confront the prices and rents with the real market once more, and ask what a bigger cable is worth.

> **Reading alongside this section.** This section requires no supplementary reading. Cretì and Fontini (2019, ch. 14–16) cover the ground of Section 6.2 in more detail: the basic principles of transmission, meshed networks and congestion, and transmission pricing in practice. Gonzales et al. (2023) is a good answer to the question "do economists actually use these models?". They study two Chilean grid expansions connecting a solar-abundant desert region to demand centres, using a cost-based dispatch model with ramping and hydro storage (essentially this note's model with the extensions of Section 4) plus endogenous renewable investment, and find solar generation up 180%, generation costs down 8% and emissions down 5%. Bjørndal and Jörnsten (2001) is the Nordic origin of the zonal-versus-nodal question of Section 6.1. On who should pay for a cable, Joskow and Tirole (2005) is the reference on merchant transmission and Hogan (2018) a short primer on benefits and cost allocation; Section 6.7 draws on both.

## 6.1 The geography: bidding zones

There are different ways of arranging wholesale electricity markets. In Europe, generators and consumers are collected in **bidding zones**: an area within which the market pretends there is no grid at all. Any generator in the zone can serve any consumer in it, every trade inside the zone clears at one price, and only *between* zones is transmission capacity rationed. Most zones are countries, but a few countries are split, because a bottleneck inside them is too large to ignore: Norway has five zones, Sweden four, Italy seven, and Denmark two, DK1 for Jutland and Funen and DK2 for Zealand. Figure 6.1 shows the map, with the twelve zones this section models highlighted.

This zonal design can be computationally and politically convenient, but it comes at a cost: congestion *inside* a zone has not gone away; it has only been hidden from the price. When the market's outcome would overload a line inside a zone, the system operator must fix it afterwards

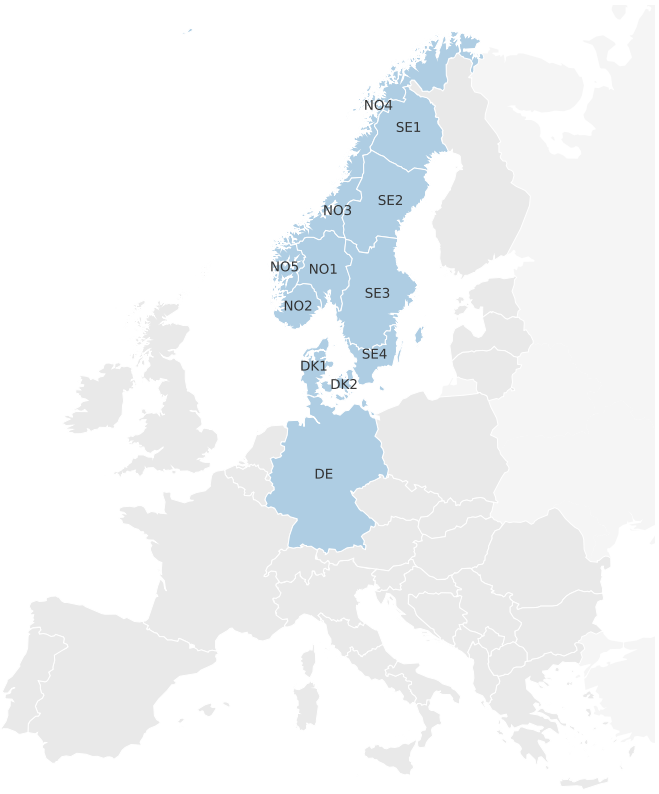



Figure 6.1: The bidding zones of Europe's day-ahead market, with the twelve zones of this section's network highlighted. Zone borders inside Norway, Sweden and Denmark follow county borders and are approximate.

by **redispatch**: paying a generator on the wrong side of the bottleneck to produce less and another on the right side to produce more, at cost, after the market has cleared. German redispatch costs run into billions of euros a year. That cost is real, it is absorbed by network tariffs. The alternative is **nodal pricing**, the design of most American markets, in which every node of the network gets its own price computed from a model that includes the full grid. Internal congestion is then priced rather than redispatched: A generator behind a bottleneck sees a low price and responds to it, and a consumer in a constrained area sees a high one. This is the textbook-efficient answer, and it goes back to Schweppe et al. (1988); Bjørndal and Jörnsten (2001) is the Nordic statement of what zones cost relative to it. We return to the comparison once the model is on the table.

**The network of this section.** As in the previous models, the following sections build on actual data on networks, power plant fleets, historical weather patterns etc.. To do this, we rely on PyPSA-Eur, the open European model used across research and industry; Appendix B documents this further for the interested reader. We focus on the Nordic part of the European electricity network. The network consists of the 12 **bidding zones** of Denmark and its neighbours: DK1, DK2, Germany, SE1–SE4 and NO1–NO5. This is far as we are prepared to go in a lecture note in terms of model size, but even this is a simplification of the highly integrated European grid.

It is worth noting that the ten stylised plants of Table 1 do not come with us, as this is not

detailed enough for our broadened model scope. The fleet is now 67 generators and 14 hydro reservoirs, inventoried in Appendix C, which adds technologies the small table never had (lignite, nuclear, run-of-river) and gives every plant its own efficiency instead of one representative value per type. What does not change, however, is how we interpret and solve the model in general: Every thermal generator here burns fuel at the prices of Table 1, so a combined-cycle plant costs the same to run in this section as it did in Section 2. Nothing in the vocabulary changes either; the shadow variables are read here exactly as they were read there.

## 6.2 Transmission grids, for economists

The wires between the zones are the new ingredient, and they deserve a few pages before the model, because the model treats two kinds of wire differently and the reason is physical. We keep to what the model needs; Cretì and Fontini (2019, ch. 14–15) go a bit further.

**What a line is to the model.** The transmission grid is the high-voltage part of the electricity network, 220 and 400 kV in continental Europe, which carries power from large plants to the areas that consume it and from one zone to the next; below it, distribution grids at lower voltages bring the power to the individual consumer, and they play no role in this note.

For our purposes a transmission line $\ell$ is characterized by four numbers. First, a *capacity* $\bar{f}_\ell$ in MW: A conductor heats as current flows through it, and its *thermal rating* is the flow it can carry without sagging or damage. The market rarely gets to use all of it. What it trades on is the **net transfer capacity** (NTC), a commercial figure the system operators publish for every border, deliberately below the thermal limit to leave a security margin for the failure of any single element and, on borders crossed by alternating-current lines, for flows the market did not schedule; the paragraphs on alternating current below explain where those come from. On the Jutland–Germany border, for example, the market only sees 2.5 GW where the wires could physically carry several times that. Second, a *efficiency* $\eta_\ell$, the share of the power sent that arrives: Resistance turns part of the flow into heat, and the loss grows with the square of the current, so a heavily loaded line loses proportionally more than a lightly loaded one. The simulations charge every direct-current crossing a flat 3% and approximate the quadratic loss of an alternating-current line piecewise (Appendix C). Third, an *investment cost*, Table 3, which the model of this section does not need because the grid is given, but which Section 6.6 and Section 8 will. Fourth, a property with no number: whether an operator can set the flow on the line directly. That depends on whether the line carries alternating current (AC) or direct current (DC), which is where the physics enters.

**Alternating and direct current.** Almost all of the grid runs on **alternating current** (AC): The voltage swings back and forth fifty times a second, and every generator connected to the grid spins in lockstep with that rhythm. An area in which this holds is a **synchronous area**, and it is one coupled machine: A disturbance anywhere in it is felt everywhere in it within a second, and the frequency is one number for the whole area. Continental Europe, from Portugal to Poland, is one synchronous area; the Nordic countries are another. The two Danish zones sit in different

Table 3: What a wire costs: investment costs of transmission technologies in the technology-data compilation the note prices everything from (Appendix B), all with a 40-year lifetime. A DC line pays for its converter stations once and for its cable per kilometre.

| Technology | Investment | Unit |
|---|---|---|
| AC overhead line | 442 | €/MW per km |
| DC overhead line | 442 | €/MW per km |
| DC submarine cable | 1008 | €/MW per km |
| DC converter station pair | 166 | thousand €/MW |

ones. DK1 is synchronised with Germany and the continent, DK2 with Sweden and Norway, and no AC line joins them. **Direct current** (DC) does not alternate. A DC line takes AC power in at one end through a *converter station*, sends it as direct current, and converts it back at the other end. The converters are power electronics, and they set the flow: A DC line carries exactly what its operator tells it to. Because the two ends need not be synchronised, DC is also the only way to join two synchronous areas, which is why every cable between DK1 and the Nordic zones, and the Great Belt link between the two Danish zones, is a DC cable. Figure 6.2 draws the network with the distinction made. Its unit is the **interconnector**: everything that joins two zones, the AC lines across their border or a DC cable, which the market sees as one capacity. There are 15 AC and 7 DC interconnectors, in two synchronous areas.

**Three differences that matter.** First, *cost*. An AC line needs only the wire, while a DC line pays for a converter station at each end, on the order of 166 thousand euros per MW for the pair, before a single kilometre is built (Table 3). Over short distances AC therefore wins. Over long ones DC does. The compilation behind the table prices the two overhead wires alike, and in practice a DC line is only somewhat cheaper per kilometre, two conductors where AC needs three; what decides the long-distance case is the lower losses and, above all, the sea. Under water DC is the only option for anything beyond a few tens of kilometres: A long AC cable spends its capacity charging its own insulation rather than delivering power. This is why every long sea crossing in the North Sea and the Baltic is a DC cable. For a cable of the Skagerrak's class, 360 km under water, the figures of Table 3 give roughly 529 M€ per GW; the European network planners' own candidates for the next cable on that border are priced at 850–1150 M€ per GW (Appendix B). Second, *losses*: lower per kilometre on DC, but the converters add their own share at each end, which is what the flat 3% per crossing stands for. Third, *control*, the difference that matters most for the model. A DC flow is set by its operator; an AC flow follows from everything else that is happening on the network. We take the two cases in turn.

**Controllable flow: the DC line.** A DC line from zone $a$ to zone $b$ is the simplest object in this section. Its operator picks a flow $f_{\ell,h}$ between zero and the capacity,

$$0 \;\leq\; f_{\ell,h} \;\leq\; \bar{f}_{\ell}, \tag{30}$$

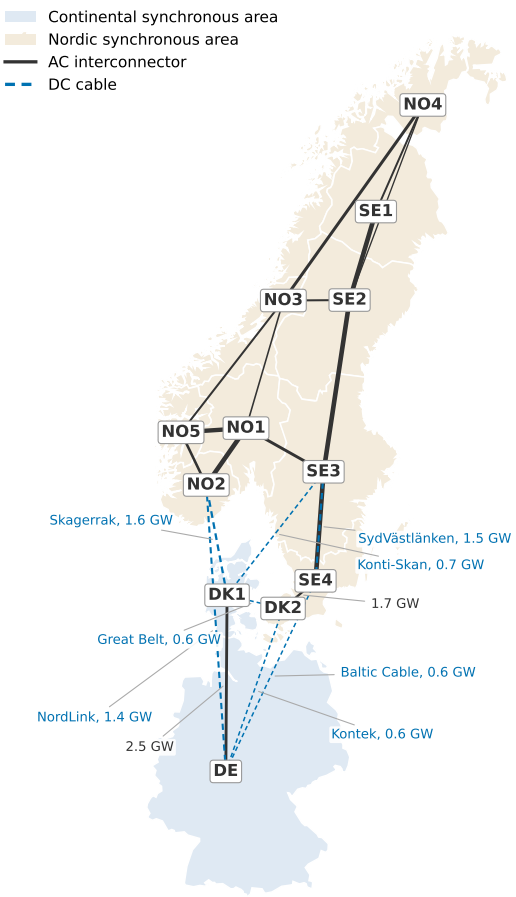



Figure 6.2: The network of this section: twelve zones in two synchronous areas, joined by AC interconnectors (solid) and DC cables (dashed), with line width scaled to the capacity the market trades on. The named cables and the two AC borders facing Denmark are labelled with their capacity.

zone $a$ gives up $f_{\ell,h}$ MWh and zone $b$ receives $\eta_\ell\, f_{\ell,h}$, the rest having been lost on the way. To let power go the other way we simply give the same cable a second flow variable in the opposite direction. Nothing else constrains it. This is the transport model of trade that an economist would write down unprompted, and for a DC cable it is exactly right.

**Flows the operator cannot set line by line: the AC network.** What follows is background. You need not carry the physics with you, only the three consequences the paragraph ends with, and the one constraint the model keeps. On an AC network the operator does not set the flow on each line. Power injected in one zone and withdrawn in another spreads over *every* path between them, in proportions fixed by the lines' electrical properties, and the flow on any one line can be changed only by changing where power is injected and withdrawn. The exact description is a system of nonlinear equations in the voltages and phase angles at every node, which this note, and every market model in practice, avoids by linearising it.[13] In the linearised version the flow

[13]For a line between $a$ and $b$ the exact active power flow is $f_\ell = V_a V_b \sin(\theta_a - \theta_b)/x_\ell$, with $V$ the voltage and $\theta$ the phase angle at each end. Under normal operation the voltages sit near their nominal values and the angle differences are small, so $f_\ell \approx (\theta_a - \theta_b)/x_\ell$ in convenient units. Engineers call this the "DC power flow", an unfortunate name: It is an approximation of AC physics and has nothing to do with DC cables. Stott et al. (2009) review how good it

on a line is proportional to the difference in phase angle across it, divided by the line's *reactance* $x_\ell$, a fixed property of the conductor that grows with its length. Two consequences follow. First, between two zones joined by parallel paths the flow splits in inverse proportion to the paths' reactances: The path with half the reactance carries twice the flow. Second, around every loop the reactance-weighted flows sum to zero,

$$\sum_{\ell \in C} \varepsilon_{C\ell}\, x_\ell\, f_\ell \;=\; 0 \qquad \text{for every loop } C, \tag{31}$$

with $\varepsilon_{C\ell} = \pm 1$ recording whether line $\ell$ is traversed with or against the direction chosen for the loop. This is *Kirchhoff's voltage law*, and eq. (31) is the one line of physics the model carries. A *loop* here is any set of lines along which power can leave a zone and return to it by a different route. NO1 to NO5 to NO2 and back to NO1 is a loop, and so is the larger ring through Sweden. The single interconnector between Germany and DK1 is not, since there is no second route back, and neither is a DC cable, whose flow is set by its converters. In this section's network every loop lies inside the Nordic synchronous area, 5 independent ones in all, and eq. (31) holds on each.

Figure 6.3 shows the consequence with three zones and three identical lines. Send 300 MW from A to C. The direct line has reactance $x$ and the detour through B has $2x$, so 200 MW goes direct and 100 MW through B, and B carries 100 MW of a trade it was no part of. Three lessons carry over to the real network. First, the 100 MW through B is a **loop flow**, capacity used by a trade nobody sold on that line, and the reason the NTC on an AC border sits below the wires' rating. Second, the transfer capacity of a meshed grid is not the sum of its lines: Rate every line in the figure at 250 MW and the direct line fills first, at a transfer well short of the 500 MW of wire touching C. Third, loading one AC line changes the flow on every other line of the loop, which is why an AC system is so much less flexible to operate than a DC one, and why Europe's grid is adding DC cables. The famous continental case is trade between Germany and Austria that loops through Poland and the Czech Republic. Since 2024 the Nordic day-ahead market has rationed its borders by *flow-based* coupling, which puts eq. (31) into the market algorithm itself rather than hiding it behind fixed NTCs.

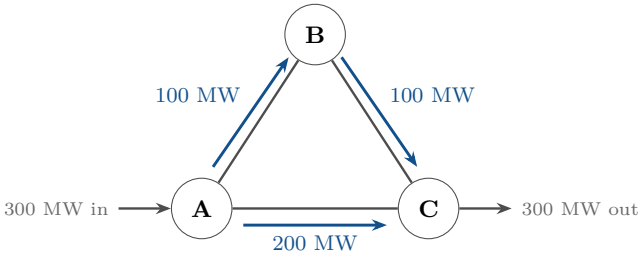



Figure 6.3: The linearised power flow on a loop of three identical lines: A transfer of 300 MW from A to C splits two to one between the direct line and the detour through B.

is.

## 6.3 The model

Each zone $z$ has its own market clearing constraint with its own price $\lambda_{z,h}$, and the interconnectors move power between the zones. We index the interconnectors by direction: $\mathcal{L}$ contains every interconnector twice, once for each way power can flow, and for $\ell \in \mathcal{L}$ we write $a_\ell$ for the zone the flow leaves and $b_\ell$ for the zone it enters. Every flow is then non-negative, the losses are charged where the power arrives, and an interconnector's capacity $\bar{f}_\ell$ is its NTC. Written out, with geography in blue, the planner chooses generation $q_{g,h}$ and flows $f_{\ell,h}$ to solve

$$\min \quad \sum_h \sum_z \sum_{g \in \mathcal{G}_z} c_g \, q_{g,h} \tag{32a}$$

$$\text{s.t.} \quad \sum_{g \in \mathcal{G}_z} q_{g,h} + \sum_{\ell:\, b_\ell = z} \eta_\ell \, f_{\ell,h} \;=\; D_{z,h} + \sum_{\ell:\, a_\ell = z} f_{\ell,h} \qquad (\lambda_{z,h}) \qquad \forall z, h \tag{32b}$$

$$0 \le q_{g,h} \le \gamma_{g,h} \, \bar{q}_g \qquad (\underline{\mu}_{g,h}, \overline{\mu}_{g,h}) \qquad \forall g, h$$

$$0 \;\le\; f_{\ell,h} \;\le\; \bar{f}_\ell \qquad (\underline{\omega}_{\ell,h},\ \overline{\omega}_{\ell,h}) \qquad \forall \ell, h \tag{32c}$$

$$\textstyle\sum_{\ell \in C} \varepsilon_{C\ell} \, x_\ell \, f_{\ell,h} \;=\; 0 \qquad (\zeta_{C,h}) \qquad \forall \text{ AC loops } C \tag{32d}$$

Equation (32b) is Section 3's market clearing, one copy per zone, with the zone's imports net of losses on the supply side and its exports on the demand side. Equation (32c) is eq. (30), the capacity limit on every interconnector in each direction. Equation (32d) is eq. (31) for every loop of the AC network, with $C$ now running over the directed interconnectors along the loop; on a DC cable there is no such constraint, and on the AC interconnector between Germany and DK1, which closes no loop, it is vacuous. Nodal pricing, for the record, is this same problem written with a node in place of a zone and eq. (32d) enforced on every loop of the full grid.

One housekeeping note. The problem above is the one the text reasons about, and it stays that minimal on purpose. The *simulations* behind the figures of this and the following sections carry two further constraints on top of it, in order to get closer to more realistic simulations: ramping limits on the slow thermal plants, outages on the thermal fleets.[14] Appendix C elaborates on the model ingredients.

**Duality: the price spread.** The Lagrangian extends as it has in every section, with one wrinkle: The market clearing constraint is not merely extended here but *replicated*, one copy per zone. So the starting point is $\mathcal{L}^{\text{vre}}$ read zone by zone — write $\mathcal{L}^{\text{vre}}_z$ for (14) restricted to zone $z$'s generators, load and price $\lambda_{z,h}$ — and the wires add their terms on top,

$$\begin{aligned}\mathcal{L}^{\text{net}} = \sum_z \mathcal{L}^{\text{vre}}_z &+ \sum_h \sum_z \lambda_{z,h} \Big[ \sum_{\ell:\, b_\ell = z} \eta_\ell \, f_{\ell,h} - \sum_{\ell:\, a_\ell = z} f_{\ell,h} \Big] + \sum_h \sum_C \zeta_{C,h} \sum_{\ell \in C} \varepsilon_{C\ell} \, x_\ell \, f_{\ell,h} \\ &+ \sum_h \sum_\ell \overline{\omega}_{\ell,h} \big( \bar{f}_\ell - f_{\ell,h} \big) + \sum_h \sum_\ell \underline{\omega}_{\ell,h} \, f_{\ell,h}.\end{aligned} \tag{33}$$

[14] In addition, the model approximates the AC losses piecewise rather than with the flat efficiency $\eta_\ell$, because this is the standard approach in the PyPSA open source framework anyway.

Now consider a DC cable $\ell$ from zone $a$ to zone $b$. Flow carries no cost in the objective and closes no loop, so it appears only in the blue terms of the first line, and differentiating $\mathcal{L}^{\text{net}}$ with respect to $f_{\ell,h}$ gives

$$\eta_\ell \, \lambda_{b,h} - \lambda_{a,h} + \underline{\omega}_{\ell,h} - \overline{\omega}_{\ell,h} \;=\; 0, \tag{34}$$

with complementary slackness on the two bounds. We read it the usual way:

- Lower bound: nothing is sent from $a$ to $b$. Then $\underline{\omega}_{\ell,h} = \lambda_{a,h} - \eta_\ell \lambda_{b,h} \geq 0$: What a MWh fetches at the far end, net of what is lost on the way, does not cover its price at home. Power may well be flowing the other way.
- Interior: the cable is used, below capacity. Both multipliers are zero and $\eta_\ell \lambda_{b,h} = \lambda_{a,h}$. Arbitrage across a slack cable does not equalise the two prices; it brings them within the *loss wedge*, $\lambda_{b,h} - \lambda_{a,h} = (1 - \eta_\ell)\, \lambda_{b,h}$. On a lossless cable the wedge is zero and the prices coincide.
- Upper bound: congested from $a$ to $b$. Then $\overline{\omega}_{\ell,h} = \eta_\ell \lambda_{b,h} - \lambda_{a,h} > 0$. The prices decouple, and the spread net of losses is the shadow price of the cable's capacity: the value of one more MW of wire in that hour.

On an AC interconnector eq. (34) gains a term $\sum_C \zeta_{C,h}\, \varepsilon_{C\ell}\, x_\ell$ for every loop the interconnector lies on, so two zonal prices can differ across a slack AC line when a loop constraint binds somewhere else on the ring: the price of loop flow. The reading is otherwise the same.

**Result 5** (Prices across an interconnector)**.** *In a zonal market, power flows toward the more costly zone, and only when the spread covers the losses. Across a slack interconnector the two zonal prices differ by the loss wedge* $(1 - \eta_\ell)\, \lambda_{b,h}$ *and by nothing else; across a congested one the spread net of losses,* $\eta_\ell \lambda_{b,h} - \lambda_{a,h}$*, is the shadow price of one more MW of capacity, exactly the* $\overline{\mu}_g$ *of Section 2 attached to a wire instead of a plant.*

Zonal pricing is thus the planner's market clearing decentralised by geography. Because the zones here are the real bidding zones, each $\lambda_{z,h}$ has a direct empirical counterpart in the Nord Pool and EPEX day-ahead prices for the same areas.

## 6.4 Prices and rents, in the model and in the market

Let us now solve the year and read the outcome through Result 5: first the zonal prices, then the rents the interconnectors between them collect, and then both against what the market actually did in 2024. Two choices about the instance should be stated first. Every solve in this section charges the 2024 ETS average of 65 €/t on every emitter, because the section is held against the 2024 market and that year's plants paid it; Section 7 charges a higher reference price because it asks about 2030, not about a year that happened. And the network is calibrated to its year in two respects before it is solved: Norwegian reservoir inflow is scaled up from the normal-year level every section uses so that the model's hydro output equals the 140 TWh Norway's plants actually produced in a record year, and the internal Nordic interconnectors are derated from their thermal

ratings to the transfer capacities the market trades on. Appendix B documents both, and what each does to the prices; the sections that follow, which look at 2050 rather than at 2024, keep the normal-year level.

**Zonal prices.** Figure 6.4 shows the year's price duration curves for the five zones that face Denmark.

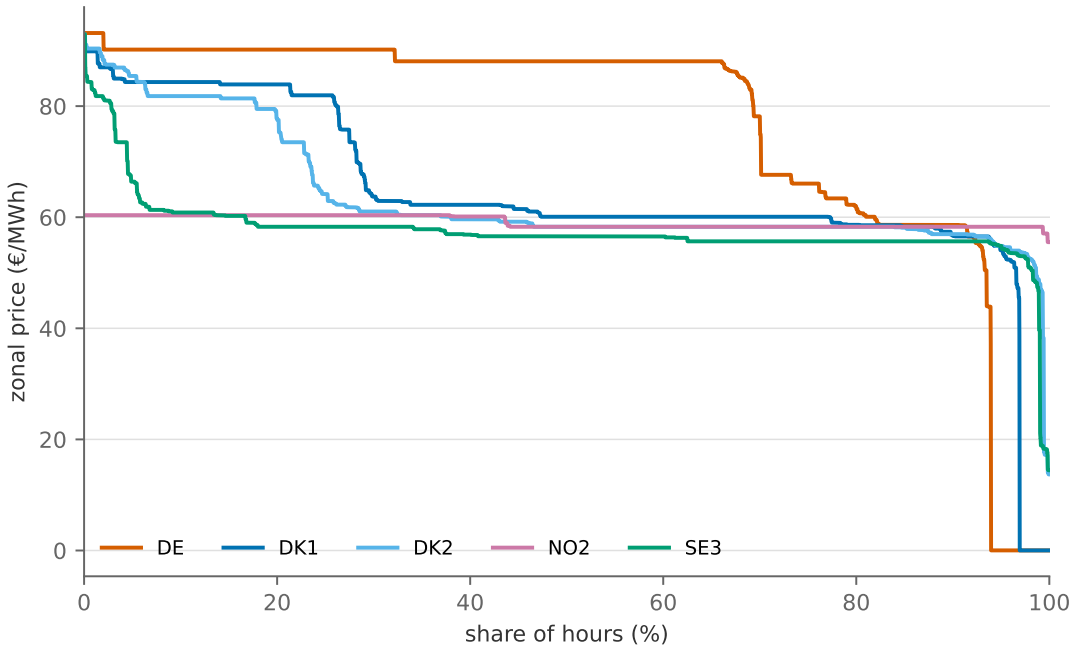



Figure 6.4: Price duration curves for the five zones that face Denmark (the remaining Nordic zones track their neighbours closely).

The figure shows the following:

i. The Nordic zones are almost perfectly flat. Reservoir hydro is the intertemporal arbitrage of Section 4 at national scale, and arbitrage flattens the price it trades on. The plateau *is* the water value, the shadow price of stored water, and it pins prices whenever hydro is marginal, which in the north is most of the year (southern Norway averages 59.2 €/MWh, central Sweden 57.7).

ii. Germany, thermal at the margin, swings: gas plus carbon in the tight third of the year, coal and lignite plus carbon below that, and zero in wind-surplus hours. It averages 76.9, well *above* the hydro plateau: With the carbon price charged, the thermal fleet is the costly side of every cable to the North.

iii. The Danish zones are the instructive case. Glued to the German price when the cables to the North are full and pulled down to the Nordic plateau when they have room, they end the year at 64.8 (DK1) and 63.7 (DK2), between their two neighbours.

The mechanism is what matters. Denmark sits between a costly thermal neighbour and a cheap hydro neighbour, and its price is the German one in the hours when the Nordic cables are congested

and close to the Nordic one in the rest; the German zeros reach it too, through the slack border. A zone between two markets does not average them; it takes a different draw from each tail. The geographical fact of the Danish power market stands: It is a small system whose price is set abroad more often than at home.

**Congestion rent.** A congested interconnector buys at the cheap end and sells at the costly end. The quantity the system operators report as their *bottleneck income*, and the one Figure 6.5 plots, is the flow times the price spread, summed over the year,

$$R_\ell \;=\; \sum_h f_{\ell,h}\,(\lambda_{b,h} - \lambda_{a,h})\,. \tag{35}$$

By eq. (34) and complementary slackness this splits into two parts,

$$R_\ell \;=\; \underbrace{\sum_h \overline{\omega}_{\ell,h}\,\bar{f}_\ell}_{\text{scarcity rent}} \;+\; \underbrace{\sum_h (1-\eta_\ell)\,\lambda_{b,h}\,f_{\ell,h}}_{\text{value of the losses}}, \tag{36}$$

since the bracket in eq. (35) equals $\overline{\omega}_{\ell,h} + (1-\eta_\ell)\lambda_{b,h}$ whenever the cable flows. The first term is the scarcity rent $\overline{\mu}_g \bar{q}_g$ of Section 2 for a wire: the shadow value of the capacity constraint, collected in cash by whoever owns the cable. In Europe that is the TSOs, who must by regulation spend it on more capacity or lower tariffs (European Parliament and Council 2019). The second term is not scarcity at all: On a lossy line the two end prices differ even when the line is slack, and that spread pays for the energy burned in transit. Every term is non-negative, because power flows toward the expensive zone. Figure 6.5 tallies the year by interconnector.

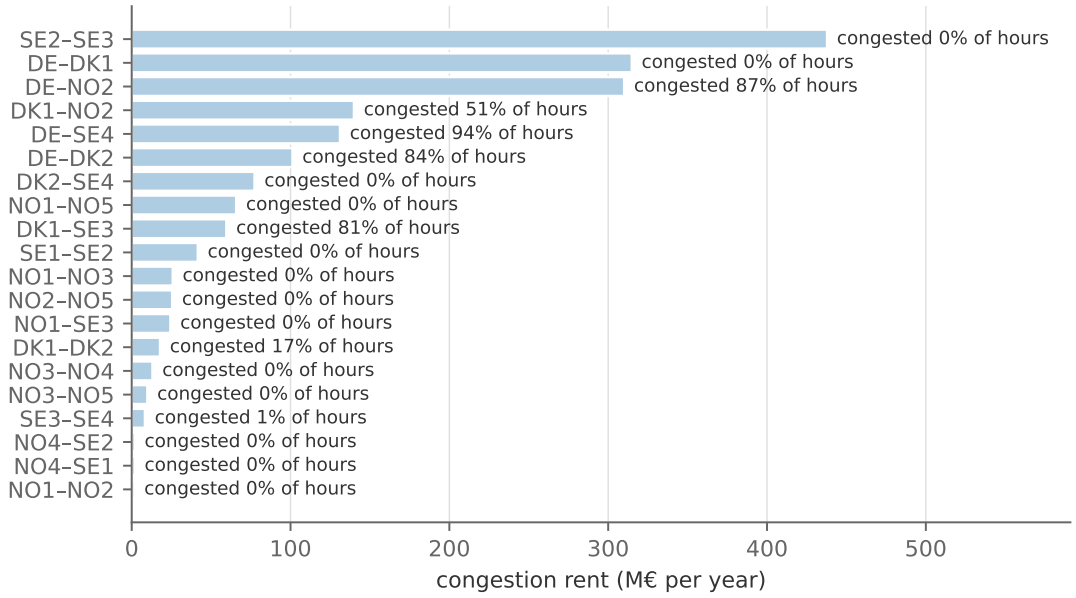



Figure 6.5: Annual congestion rent by interconnector, computed as flow times spread (35), with the share of hours each interconnector runs at its limit.

The figure shows that the yearly congestion rent totals 1809 M€. This is partly due to congested

cables with significant price spreads (in particular From Germany to the Nordics), but also because of non-congested and heavily traded interconnectors' value of losses (e.g. internally in Sweden). It is worth noting that, in reality, what we here represent as income from "non-congested" lines is not always collected as such. For instance, the two largest bars , the interconnector between central and southern Sweden and the Jutland–Germany border, earn their whole "rent" this way. In the real market this happens only on the DC cables: The Nordic day-ahead coupling schedules the Great Belt, Skagerrak and Konti-Skan with an explicit loss factor, so a flow on them is scheduled only when the spread covers the loss, and the TSOs' bottleneck income on them includes that wedge. An AC border between two zones is treated as lossless by the coupling; its two prices are equal whenever the capacity is slack, and the TSOs buy the physical losses separately and recover them through tariffs. The two largest bars are both AC interconnectors that never congest, so in the market they would show no spread and no income at all. The Jutland–Germany interconnector never reaches its limit and loses about three percent of what it carries, yet the two zonal prices differ by a dozen euros in most hours. Two features of the network do that. The network measures every interconnector between the centres of the zones it joins, which makes this border several hundred kilometres long where the real tie is a fraction of that; and the piecewise approximation of the quadratic loss, with two tangents, puts the *marginal* loss rate, which is what enters the flow condition and hence the spread, at up to twice the average rate. The bar credits a slack interconnector with roughly four times the income the Danish TSO reported on it; the validation below returns to it.

**The model against the market.** Because the zones are the real bidding zones and the year is 2024, every number above has a published counterpart: the day-ahead area prices, and the congestion rent actually collected on the Danish borders, computable from public data as hourly exchange flow times the hourly price spread, exactly eq. (35). Figure 6.6 makes both comparisons. The prices are compared for the six zones Energi Data Service publishes, the two Danish zones and the four that border them.

Panel (a) shows that average prices fare relatively well, but with a noticeable difference in Norway and Sweden where the model prices are too high. As mentioned earlier, these differences can likely be attributed to the fact that the model is still simplified in various ways. Some of the relevant items include: (i) We do not include British and Dutch interconnectors, (ii) no "must-run" generators and unit commitment, (iii) an exaggerated loss wedge on the Jutland–Germany interconnector, (iv) and perfect foresight.

Panel (b) is the harder test, since rents live on the hourly *differences* between prices. Summed over the Danish borders the model collects 711 M€ against the 802 M€ of bottleneck income the TSOs actually reported: the right order of magnitude, but with the composition wrong. The model collects far too much on the Jutland–Germany border and too little on the Nordic cables (Skagerrak: 140 M€ modelled against 290 M€ realised). The likely culprits are the same as above, but not something we will dwell more on for now.

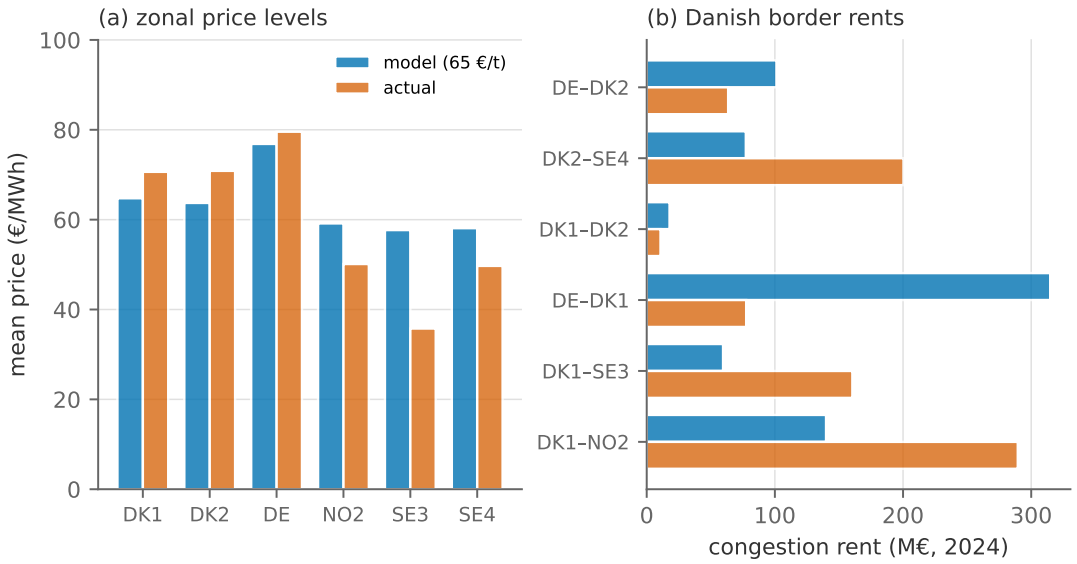



Figure 6.6: Model against market, 2024. (a): mean zonal prices in the model and realised, for the six zones Energi Data Service publishes. (b): congestion rent on the Danish borders, the model against rents computed from Energi Data Service's hourly exchange flows and the zones' price spreads.

**What is still missing.** Of the three boundary errors named in Section 3's box, this section removes two: Denmark can now sell, and the neighbours' prices are computed rather than read from the market. The third, a network inside each zone, is what zonal pricing leaves out by construction, and it is time to return to the targeting principle of Section 3.5. We asked there whether network congestion caused by renewable siting is a genuine externality. Under nodal pricing it largely is not: The congestion is priced, and the generator faces it. Under zonal pricing it largely is: A wind farm sited behind an internal bottleneck imposes redispatch costs on everyone else and sees a price that does not reflect them. The market failure, in other words, is not created by intermittency; it is created by the pricing geography. That is a much more useful diagnosis than "renewables impose system costs", and it points at an instrument, pricing the location, rather than at a subsidy. Our model is zonal, so it inherits this blind spot. Every congestion rent computed here is a *cross-border* rent. Internal Danish, German and Norwegian congestion is invisible by construction, and the model therefore understates the true cost of moving power around. The bias box collects it with the two assumptions the validation just exposed.

**Which way does it bias the answer?** *Zonal prices, fixed transfer capacities, and perfect foresight over the water.*

- *No network inside a zone.* Internal congestion and redispatch are invisible, so the model **understates** the cost of moving power and **overstates** the value of variable generation sited behind internal bottlenecks.
- *Fixed NTCs instead of flow-based coupling.* The right interconnectors congest, at partly the wrong hours, so the model **understates** the rents collected on the Nordic cables.
- *Losses priced on AC borders, over interconnectors measured between zone centres, with two*

*tangents.* A property of how this network was built, not of zonal models. The market couples AC borders without a loss wedge; the model charges one, over a long line and at a marginal rate up to twice the true one, so it **overstates** the price wedge across the Jutland–Germany border, **understates** DK1's price, and **overstates** the "rent" on an interconnector that never congests.

- *Perfect foresight over inflow.* Real reservoir owners hedge both ways: In a wet year they produce ahead of the risk of spilling and the price falls, in a dry one they sell at normal water values until the shortage is visible and then face a spike. The planner never spills and saves from the first week, so the model **overstates** Nordic prices in a wet year and, in a dry one, **overstates** them early and **understates** the spike.

## 6.5 Geographic smoothing

Interconnection does more than arbitrage average price levels; it aggregates weather. Figure 6.7 shows the two Danish zones' onshore wind profiles and the duration curves of each zone and of their average.

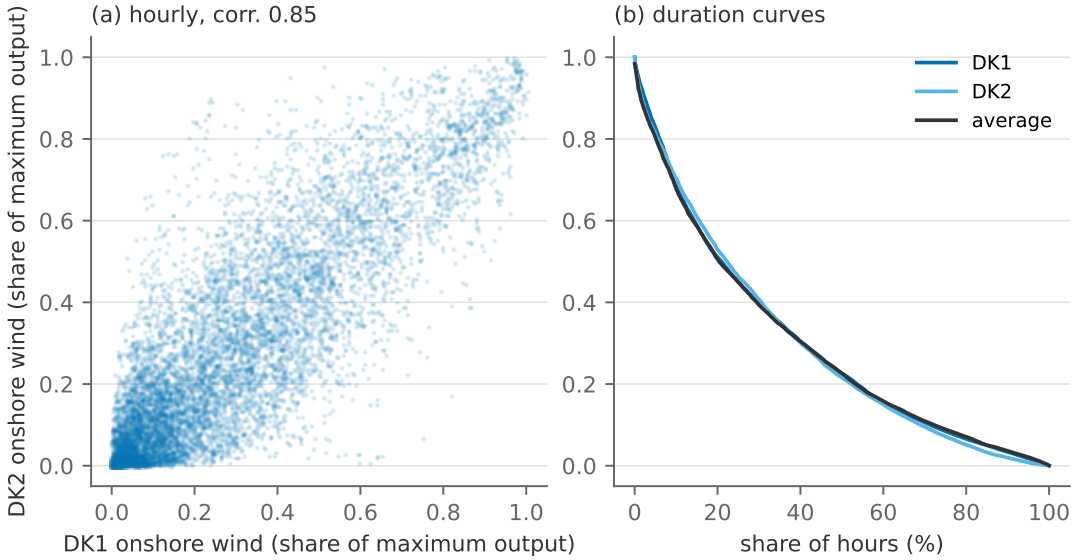



Figure 6.7: DK1 and DK2 onshore wind, hourly 2024: (a) the two profiles against each other, (b) the duration curves of each zone and of the averaged profile.

The figure shows a *negative* result: At a correlation of 0.85 (three hundred kilometres is simply not far in meteorological terms) averaging the two zones flattens the duration curve only barely. Denmark cannot diversify its wind against itself. The value of aggregation is real but it lives at *distance*: Correlations fall roughly exponentially with separation, Iberian and Nordic wind are close to uncorrelated, and the same figure drawn for DK1 against Spain would show the averaged curve visibly flatter at both ends. That is the statistical argument for a continental grid rather than national self-sufficiency. It is also the reason that the smoothing this network can offer runs through its *price* channel (Norwegian reservoirs absorbing Danish surpluses) rather than through wind diversification among near-identical neighbours.

## 6.6 The value of another gigawatt

Let us now ask the planner's question directly: What is a bigger cable worth? Figure 6.8 re-solves the year with the Skagerrak cables at every size from zero to three times today's 1.6 GW, and reports two things: the annual system cost saving relative to no cables, and the congestion rent the interconnector itself earns. Both are gross of what the cable costs to build. The figure is the benefit side of a cost-benefit analysis, to be set against the annualised investment of Table 3; the model of this section takes the grid as given and does not do that arithmetic.

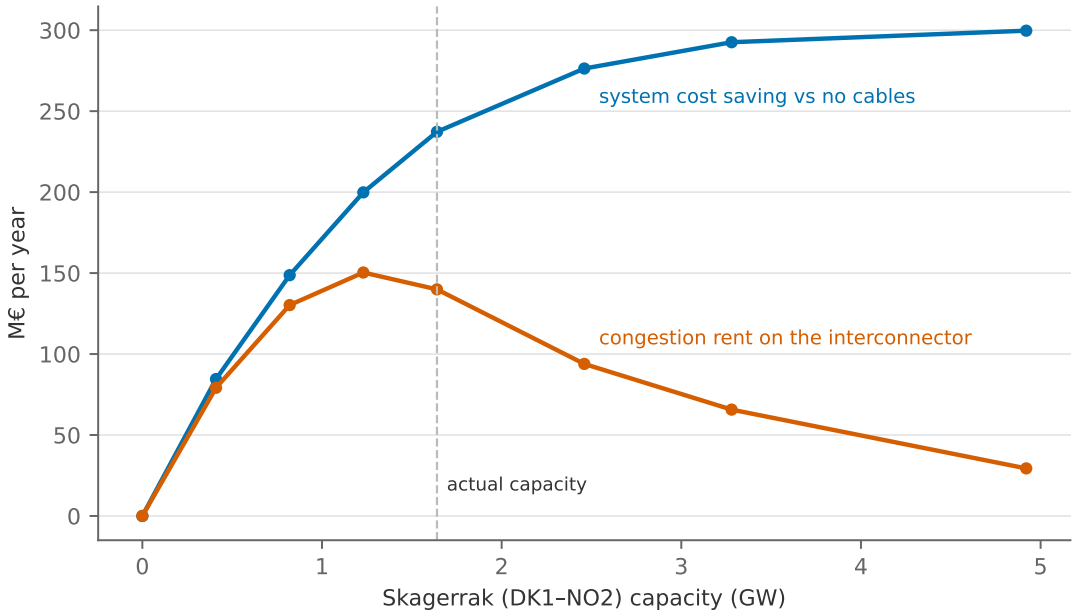



Figure 6.8: The Skagerrak interconnector swept from zero to three times its actual capacity: system cost saving relative to no cables, and the congestion rent the interconnector earns.

The figure shows the following:

i. The *social* value, system cost saved relative to no cables, rises concavely, from 237 M€ a year at actual capacity to 300 M€ at triple: Each gigawatt closes more of the spread the next would have monetised.

ii. The *merchant* value, the rent the interconnector itself collects, traces a hump. It peaks a little below today's size, stands at 140 M€ at the capacity actually built, and then collapses, to 29 M€ at triple capacity, because a big enough interconnector competes away the very spread it earns. It does not fall to zero at any size the figure shows, since reservoirs are finite and no cable can export water that has not rained. Where the peak sits is a property of this instance, its calibrated water and its fuel prices, not a finding.

iii. A merchant investor, paid only the rent, would stop at the hump, while every gigawatt beyond it still creates social value.

Beyond the peak, more capacity is strictly *against* the cable owner's interest even as it benefits the system. This is the standing argument for regulated rather than merchant transmission investment,

made formally by Joskow and Tirole (2005), and a live issue every time a TSO's bottleneck income depends on the bottleneck not being fixed. We leave the annuity arithmetic to Section 8, where the model builds interconnectors itself.

## 6.7 Who wins, who loses, and who pays

So far throughout this note, we have generally looked at system efficiency. However, a distinct and very important topic is how policies and infrastructure investments affect different stakeholders and consumers. This is particularly clear in a model with the transmission network, where we do not only distinguish between consumers and producers, but also between different countries/zones.

To understand this more carefully, let us again deploy the model and analyse who wins, loses, and who pays what. In general, connecting a low-price zone to a high-price zone generally means that prices tend to converge; Table 4 lists who gains and who loses from that.

Table 4: Who gains and who loses when a cable joins a low-price zone to a high-price zone.

| | **Low-price (exporting) zone** | **High-price (importing) zone** |
|---|---|---|
| Price moves | up | down |
| Consumers | **lose** | **win** |
| Producers | **win** | **lose** |
| Cable owner | collects the congestion rent (35) | |
| Net | positive: the sum is the system cost saving in Figure 6.8 | |

The total effect is unambiguously positive. That is the gains-from-trade result, and it is why the model builds the cable. But two of the four cells are negative, and they are concentrated. Figure 6.9 puts the model's own numbers on the cells, country by country, for two projects: the Skagerrak interconnector as built, against none at all, and a second interconnector of the same size on top of today's. Consumers gain what their bill falls by, since load is fixed; producers gain what their revenue net of variable cost rises by, hydro included; and each interconnector's rent is split equally between the two countries it joins, which is how the Nordic cables are in fact owned.

Panel (a) shows the following:

i. The cells of Table 4 are there, and they are large. In this wet 2024 Norway is the cheap side: Without the interconnector, southern Norway's water is trapped and its price collapses, so the interconnector lifts the Norwegian price toward the continent's. Norwegian consumers pay 1682 M€ more and Norwegian producers earn 1869 M€ more, and Sweden, wired to Norway by AC, moves with it, consumers paying 1228 M€ more and producers earning 1557 M€ more. These transfers are an order of magnitude larger than the system cost saving of 237 M€ that the planner sees.

ii. On the importing side Danish consumers gain 106 M€ and Danish producers, selling into a DK1 price that cheap water now holds down, lose 130 M€. Denmark as a country ends a small net loser, by 19 M€, while owning half the cable; Germany gains a little.

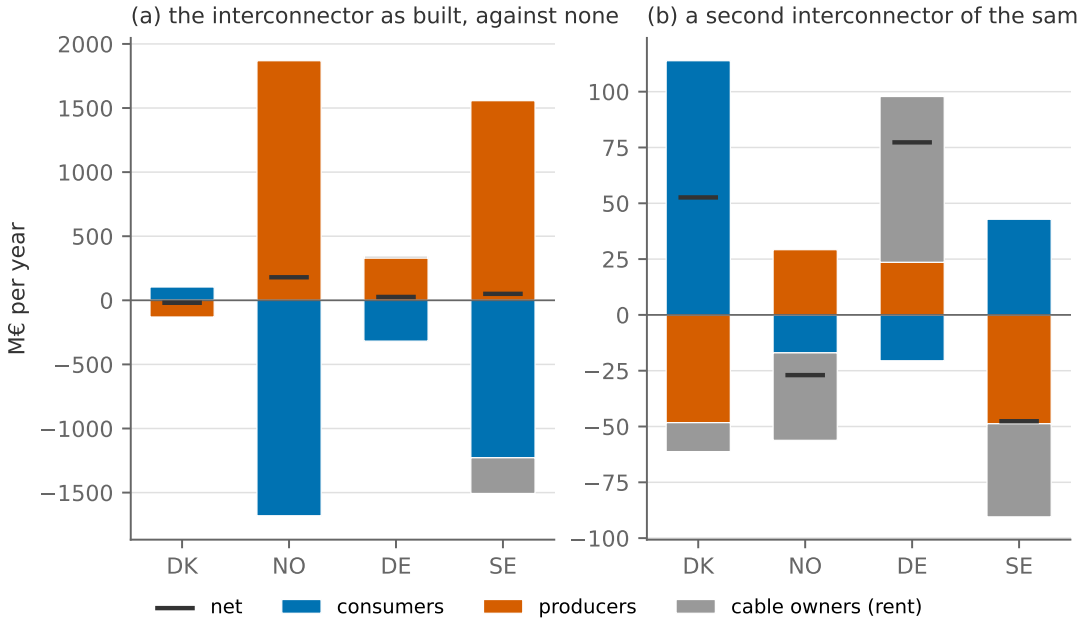



Figure 6.9: Who wins and who loses from the Skagerrak interconnector, by country and by group: the change in consumer surplus, producer profit and the country's half of the congestion rents, (a) for the interconnector as built against none, (b) for a second interconnector of the same size on top of today's. The dash marks each country's net.

iii. Norway takes most of the surplus, a net 180 M€, through its producers. Sweden, which owns none of the cable, still nets 50 M€, because its producers gain more than its cable owners lose: The new interconnector takes over export duty from Sweden's own interconnectors, whose owners lose 279 M€. A third country moves as much as the two parties to the deal.

Panel (b), a second interconnector of the same size, is the case that matters for the next investment decision, and it looks different. The system gains 55 M€. Germany gains 77 M€, most of it as income on the Jutland border, whose spread widens as cheap water lowers DK1; Denmark gains 53 M€, nearly all of it to consumers. Norway is now a net *loser*, by 27 M€, because its half of the rent falls on the far side of the hump in Figure 6.8 while its consumers and producers barely move, and Sweden loses 48 M€ the same way. The country that must host and half-own the cable would, on these numbers, have to be paid to build it, and the two countries that gain own none of it. Two reading notes. The rent shares here are what the cable owners actually collect, the value delivered at the receiving end minus the value taken at the sending end, so they are net of losses, unlike the flow-times-spread bars of Figure 6.5; with that convention the country nets add up exactly to the system cost saving. And the signs are those of one instance, a wet year in which the Nordics are the cheap side, as they were in 2021 and 2022 when the debate about the cables began.

**Sharing the bill across a border.** The figure states the problem a cross-border project has to solve before it is built. Its benefit, the system cost saving of Figure 6.8, is spread over consumers, producers and cable owners in two countries and in their neighbours, in shares the market decides; its cost, Table 3, has to be put on somebody's tariff. There are four ways this is done, and the model has something to say about each.

i. *Split the cost by ownership.* The Nordic convention, and the simplest: Skagerrak is owned half by Energinet and half by Statnett, NordLink half by Statnett and half by the German TSO, and each pays its half and keeps its half of the rent. Simple to agree, but unrelated to who benefits: In Figure 6.9 the two halves of the rent are small and alike while the two countries' nets differ by two hundred million euros.

ii. *Let the beneficiaries pay.* The principle written into both American and European rules. In the United States, Order 1000 of the Federal Energy Regulatory Commission requires the cost of a regional line to be allocated "roughly commensurate with" the benefits it brings (FERC 2011). In the European Union, a cross-border project designated a *project of common interest* goes through a **cross-border cost allocation**: The promoters compute each country's net benefit with the European network planners' scenarios and cost-benefit method, the regulators decide who pays what, a country whose net benefit is negative can be compensated, and if the regulators cannot agree, the EU Agency for the Cooperation of Energy Regulators decides (European Parliament and Council 2022; ACER 2023). The benefit that is allocated is exactly the sum of the bars in Figure 6.9, including the bars of the third countries, which is what makes the exercise contentious: Hogan (2018) explains why the benefits are hard to measure and easy to dispute.

iii. *Let the cable pay for itself.* A merchant line keeps its rent and nothing else. Figure 6.8 says what that buys: an interconnector sized to the hump, short of what the system would want, and Joskow and Tirole (2005) show that a merchant investor also has reasons to build the line in the wrong place and to keep it scarce. The European rules accordingly treat merchant interconnectors as an exemption to be applied for, not the default.

iv. *Bargain over the surplus.* Once the benefits are on the table, how to divide a joint surplus is a question cooperative game theory has answers to. Kristiansen et al. (2018) apply them to a North Sea offshore grid, and show that a split that keeps every country better off than it would be alone, such as the Shapley value, can differ substantially from splitting by ownership or by generation share.

Whatever the rule, the congestion rent is the money on the table. In Europe it may by regulation be used only for keeping the interconnector available or for more capacity, or, failing that, for lower network tariffs (European Parliament and Council 2019), so Figure 6.5 is also a map of where the next cable could be financed from. And the Norwegian case stands: Norwegian households facing higher bills because their hydro can now be sold to Germany is not a hypothetical grievance but a recurring political controversy, and the model says the grievance is *correct*. The consumers of an exporting country genuinely are worse off. What the model adds is that the country as a whole is better off, because its producers gain more than its consumers lose, which is an argument for compensating transfers, not for refusing to build.

**From duals to markets.**

| Constraint | Dual | What it measures | Market counterpart |
|---|---|---|---|
| zonal clearing (32b) | $\lambda_{z,h}$ | the zone's system marginal cost | the Nord Pool / EPEX area price |
| cross-zonal capacity (NTC) (32c) | $\overline{\omega}_{\ell,h} = \eta_\ell \lambda_{b,h} - \lambda_{a,h}$ | value of one more MW of interconnector | the TSOs' *bottleneck income* (35), net of losses |
| AC loop flow (32d) | $\zeta_{C,h}$ | the physics constraint on routing | loop flows; the redispatch a TSO must buy |
| reservoir energy (Section 4's $\theta_h$) | water value | value of a stored MWh of water | Nordic producers' water-value bids |

**The three questions.**

| **Adding interconnection to the system. . .** | |
|---|---|
| . . . does what to the two tails? | Compresses both, and by more than storage does: Imports cover the scarcity tail, exports drain the surplus tail. Note the asymmetry with storage: Trade moves energy in *space*, so it works on week-long weather systems that a 4-hour battery cannot touch, provided the neighbour's weather differs. |
| . . . does what to the value of wind? | Raises it substantially, but through the *price* channel (Norwegian reservoirs absorbing Danish surpluses) rather than through weather diversification: DK1 and DK2 wind correlate at 0.85, so Denmark cannot diversify against itself. |
| . . . does what to system cost? | Lowers it by 237 M€ a year on the Skagerrak interconnector alone, with concave returns. In Section 7 it turns out to be load-bearing: Cut off from trade, Denmark needs a much larger and more costly fleet to serve the same load. |

To sum up: Trade adds a subscript, not a new theory. Zonal prices, rents as capacity shadow prices net of losses, and decreasing returns to relaxing a constraint are by now familiar from every previous section. What is new is the physics, one linear constraint per AC loop, and the politics, a distribution the objective function cannot see. What this section cannot answer is what the fleet behind the prices should be. That is where the note turns next.

# 7 Investment

Every model so far has taken the fleet as given and asked how to run it. The transition's real question runs the other way: What should be *built*? This section makes capacity a choice variable. Each technology's installed MW becomes part of the optimisation, charged to the objective at an

annualised capital cost. The change to the model is small; the change to the economics is a whole layer: Prices must now cover not only fuel but capital, and the scarcity rents that Sections 2 to 6 treated as a distributional curiosity become the system's investment signal. It is also where the closing remark of Section 2 is finally answered: whether solar and wind are cost-effective once the whole system, and not only the marginal cost of running it, is priced.

We proceed in four steps. First, a short discussion on levelised costs and screening curves. Second, the model, and the result that ties this section to every previous one: long-run zero profit, and what it says about the value of different plant types. Third, we explain the results of optimising the model of the Northern European electricity markets that we have used since Section 6. Finally, we go through different experiments in the model and investigate how, e.g., carbon prices, trade and a Danish carbon budget change the system.

> **Reading alongside this section.** This section is where the note's threads tie together, so the reading is mostly backward. Result 6 below is the formal version of the cost-recovery question Borenstein (2012) discusses informally, and Joskow (2011) is the standing warning against the levelised-cost reasoning that screening curves invite. Butters et al. (2025) show what happens when battery investment is determined in market equilibrium rather than by a planner, the closest published analogue to what this section computes. Dechezleprêtre et al. (2023) is worth rereading here for how far the real EU ETS is from the frictionless carbon price of this section. The lecture note "Generation Technologies" (Berg 2026b) treats the levelised cost at greater length, technology by technology, and Appendix C lists what the model does not represent.

## 7.1 Annuities, levelised costs and screening curves

Most investments in the electricity system are long-lived assets that, once the investment is made, are operational for decades. A comprehensive cost-benefit analysis thus in principle requires identifying long time series predicting market outcomes and costs. To simplify the analysis, however, we often discard the dynamic perspective by rescaling investment costs appropriately using an annuity payment. Implicitly, this exercise asks what it would cost to run a system that repeats itself indefinitely year after year. This helps to represent everything in a simple way and aligns well with the **greenfield** optimisation framework that we present next.

**The annuity.** Let $I$ be the investment cost per MW of some project (e.g. a wind turbine, a battery, or a transmission line). Let $r$ denote the discount rate and $n$ be the technical lifetime of the investment. The annuity $a(r,n)I$ is the flow of investments that, when discounted and summed over the lifetime, equals the investment costs, that is $I = a(r,n)I \sum_{k=1}^{n}(1+r)^{-k}$, where $a(r,n)$ is the **annuity factor**

$$a(r,n) \;=\; \frac{r}{1-(1+r)^{-n}}. \tag{37}$$

An investment of $I$ euros per MW with lifetime $n$ is thus economically identical to a rent of $a(r,n)\,I$ euros per MW per year: a stream of capacity services purchased annually. Adding the

fixed operation and maintenance costs, usually quoted as a share $f$ of the investment, gives the **annual fixed cost** of a MW,

$$F \;=\; (a(r,n)+f)\,I. \tag{38}$$

We note that the discount rate $r$ is essential when determining the annuity cost. For very long-lived investments, we have that $a(r,n) \approx r$. For certain technologies like wind and solar the annuity is nearly the whole annual cost, so that cost rises and falls with $r$, while much of a gas plant's cost is fuel, which does not depend on $r$ at all; we return to this in Section 7.4.

At the note's $r = 7\%$, the annual fixed costs of the generators this section can build run from 29 k€/MW for solar to 168 k€/MW for offshore wind, and nuclear's is several times higher again, at 754 k€/MW. These are cost projections from an open technology-cost database for 2050, so they should be treated as relatively uncertain (Section 9.1 asks what happens under different cost projections).

**Levelised cost.** The annual fixed cost is a cost per MW, and the most common way of comparing technologies turns it into a cost per MWh: Divide it by the energy a 1 MW generator produces in a year, $8760 \cdot CF$ MWh with $CF$ the capacity factor of Section 3, and add the marginal cost $c$. The result is the familiar **levelised cost of electricity** (LCOE),

$$LCOE \;=\; \frac{(a(r,n)+f)\,I}{8760 \cdot CF} + c, \tag{39}$$

the constant price per MWh at which the plant would exactly break even. Computing it requires assumptions on the investment, the lifetime, the discount rate, costs, and the capacity factor. All but one of these are technical or financial assumptions, but the capacity factor is an equilibrium object that requires identifying how the electricity system balances throughout the year.

**Screening curves.** Screening curves plot the total annual cost of a technology type against assumed full-load hours, cf. Figure 7.1. At zero full-load hours (the intercept), the curves identify the annual fixed cost $F$ of the technology, and the slopes of the curves are the marginal costs of operation. For wind and solar, the curves are flat, whereas gas-fired generators (OCGT, CCGT) slope significantly upwards.

The levelised cost measure and screening curves are useful tools that condense a lot of information in a single number/curve: For a set of technology types, they reflect how costly a technology is and *roughly* how often it needs to be active for it to be efficient to include in the energy mix. There are, however, important limitations of a simple screening analysis. First, they are silent about the revenue channel that is driven by how well the generators' supply matches demand, and thus how prices form throughout the year; this is the cannibalisation and value factor discussed throughout this note. Second, other technologies do not fit squarely into this representation, for instance batteries, whose cost-effectiveness is determined primarily by price volatility.

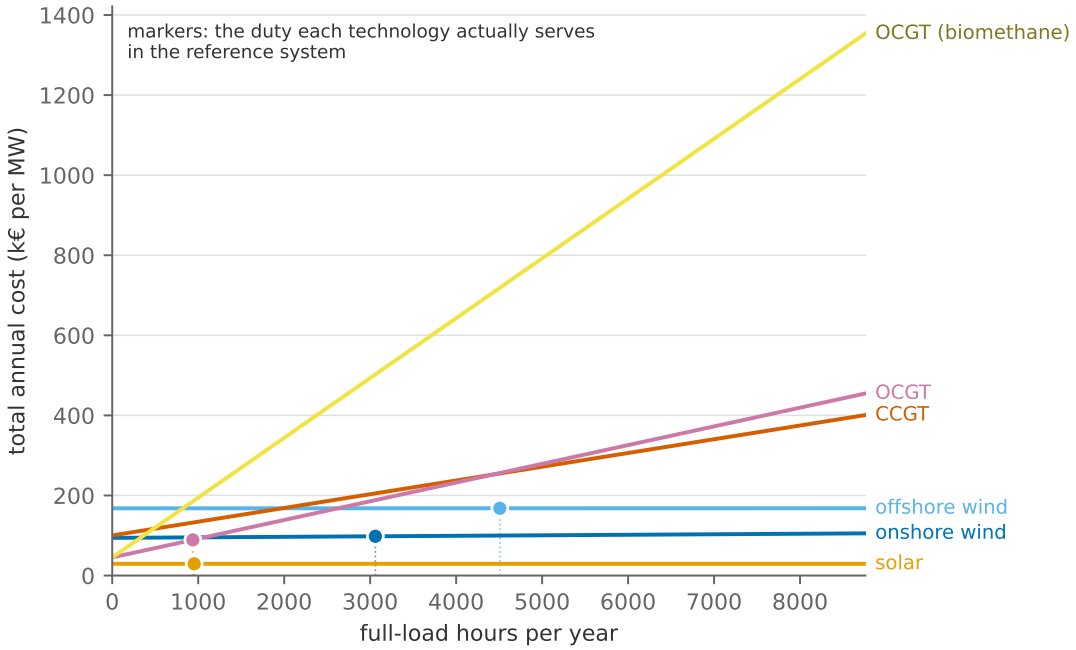



Figure 7.1: Screening curves: total annual cost per MW against running hours. Fixed cost is the intercept, marginal cost before the carbon price the slope. Nuclear is not drawn; its fixed cost alone is 754 k€/MW a year. The markers show where each technology ends up in the reference system of Section 7.3.

## 7.2 The greenfield problem

The greenfield problem is a type of investment problem that looks at how we would design an optimal system if we were to start from scratch, as opposed to a brownfield problem, where we take existing infrastructure and sunk investments into account. In this section, we consider a greenfield problem with a model with ingredients from all our previous sections: We include the variability in demand and generation introduced in Section 3, the storage from Section 4, the demand response from Sections 2.4 and 5 (albeit in a slightly different way, see below), and the network from Section 6.

Previously, we have seen two types of demand response. In Section 2, we discussed the *value of lost load*, and that a system planner could use load shedding as a last resort to balance the system; Section 5 discussed how certain demand types like heat pumps coupled with thermal storage result in another type of flexible electricity demand that enters in periods with relatively low prices. In this section, we rely on a more reduced form of flexible demand that simply breaks hourly demand up into segments with different willingness to pay. Formally, the load $D_{z,h}$ of each zone is split into blocks $j$, each a share $\alpha_j$ of the load with its own willingness to pay $v_j$.

**The model.** To summarise, the model now consists of generators, demand blocks, storage facilities, and interconnectors. The full presentation of this model can look daunting, but it is mostly a collection of all the components that we have collected throughout previous sections. What is new is that capacities that were data parameters before are now investment choices that come at the annual fixed cost $F_g$.

The planner chooses generation, trade flows, storage operation, *and* capacities to maximise a measure of welfare $W^{\text{inv}}$:

$$\max \quad W^{\text{inv}} \equiv \overbrace{\sum_h \sum_z \sum_j v_j \, d_{j,z,h}}^{\text{consumer value}} - \overbrace{\sum_h \sum_g (c_g + \tau e_g) \, q_{g,h}}^{\text{running costs, generators}} - \overbrace{\sum_h \sum_s c^s \left(g^c_{s,h} + g^d_{s,h}\right)}^{\text{running costs, storage}}$$
$$\underbrace{- \sum_{g \in \mathcal{G}^+} F_g \, \bar{q}_g - \sum_{s \in \mathcal{S}^+} F_s \, \bar{s}_s}_{\text{investment and fixed costs}}. \tag{40a}$$

This measures the consumers' value of the served load minus system costs that now include the investment and fixed costs of new capacity. To distinguish between new investments and existing capacity, we use $g \in \mathcal{G}^+$ and $s \in \mathcal{S}^+$ to indicate *new* generators and storage facilities; conversely, the investment costs of existing capacities do not enter here.

The balance constraint contains no new elements, but collects, for every hour and zone, all the ingredients from earlier; with shadow variable $\lambda_{z,h}$, the conditions read

$$\sum_{g \in \mathcal{G}_z} q_{g,h} + \sum_{s \in \mathcal{S}_z} g^d_{s,h} + \sum_{\ell:\, b_\ell = z} \eta_\ell \, f_{\ell,h} = \sum_j d_{j,z,h} + \sum_{s \in \mathcal{S}_z} g^c_{s,h} + \sum_{\ell:\, a_\ell = z} f_{\ell,h} \qquad \forall z, h. \tag{40b}$$

The boundary conditions for demand and generation are the usual ones, but we now add a similar condition to the investment choice $\bar{q}_g$ stating that it cannot go negative and that there may be an upper bound $\overline{Q}_g$ of *buildable potential*:

$$0 \le d_{j,z,h} \le \alpha_j \, D_{z,h} \qquad (\underline{\nu}_{j,z,h}, \overline{\nu}_{j,z,h}) \qquad \forall j, z, h \tag{40c}$$
$$0 \le q_{g,h} \le \gamma_{g,h} \, \bar{q}_g \qquad (\underline{\mu}_{g,h}, \overline{\mu}_{g,h}) \qquad \forall g, h \tag{40d}$$
$$0 \le \bar{q}_g \le \bar{Q}_g \qquad (\underline{\beta}_g, \overline{\beta}_g) \qquad \forall g \in \mathcal{G}^+. \tag{40e}$$

We think of these bounds as land or resource constraints, representing suitable locations for hydro reservoirs or politically feasible land available for solar and onshore wind.

The constraints on storage operation are the same as in section 4 with a similar cap of buildable potential here in terms of power capacity $\overline{s}$:[15]

$$soc_{s,h} = soc_{s,h-1} + \eta^c_s \, g^c_{s,h} - g^d_{s,h}/\eta^d_s + w_{s,h} \qquad (\theta_{s,h}) \qquad \forall s, h \tag{40f}$$
$$0 \le g^c_{s,h} \le \bar{s}_s \qquad (\underline{\phi}^c_{s,h}, \overline{\phi}^c_{s,h}) \qquad \forall s, h \tag{40g}$$
$$0 \le g^d_{s,h} \le \bar{s}_s \qquad (\underline{\phi}^d_{s,h}, \overline{\phi}^d_{s,h}) \qquad \forall s, h \tag{40h}$$
$$0 \le soc_{s,h} \le \bar{\delta}_s \, \bar{s}_s \qquad (\underline{\psi}_{s,h}, \overline{\psi}_{s,h}) \qquad \forall s, h \tag{40i}$$
$$0 \le \bar{s}_s \le \overline{S}_s \qquad (\underline{\beta}_s, \overline{\beta}_s) \qquad \forall s \in \mathcal{S}^+. \tag{40j}$$

[15] A small change compared to our battery-only model from section 4 is that we have added $w_{s,h}$ to the law of motion in eq. (40f); this represents an inflow of energy into the hydro reservoirs that comes for free (from rain and melting snow), and which is zero for a battery.

Finally, constraints related to the operation of transmission lines are the same as in Section 6:

$$0 \leq f_{\ell,h} \leq \bar{f}_\ell \qquad (\underline{\omega}_{\ell,h}, \overline{\omega}_{\ell,h}) \qquad \forall \ell, h \tag{40k}$$

$$\textstyle\sum_{\ell \in C} \varepsilon_{C\ell}\, x_\ell\, f_{\ell,h} = 0 \qquad (\zeta_{C,h}) \qquad \forall C, h \tag{40l}$$

Finally, as we will see in the simulations below, we also allow generators that rely on natural gas to switch to biomethane, which is a relatively expensive, but carbon-free substitute for natural gas. This will work as an important *backstop* mechanism that the model relies on in deep decarbonisation scenarios. To keep the model formulation manageable, this extension does not show up in eq. (40).

**Duality: the investment condition.** The Lagrangian of the model is one that accumulates all the terms from previous models; the recipe for doing so is the same, so we will skip the full presentation here and instead write up the specific parts that we are interested in. For instance, collecting all the terms related to the optimal operation and investment in a generator $g \in \mathcal{G}^+$ in zone $z$, a "generator Lagrangian" can be written as

$$\begin{aligned}\mathcal{L}_g^{\text{inv}} = &\sum_h \big(\lambda_{z,h} - c_g - \tau e_g\big)\, q_{g,h} + \sum_h \underline{\mu}_{g,h}\, q_{g,h} + \sum_h \overline{\mu}_{g,h}\big(\gamma_{g,h}\bar{q}_g - q_{g,h}\big) \\ &- F_g\, \bar{q}_g + \underline{\beta}_g\, \bar{q}_g + \overline{\beta}_g\big(\bar{Q}_g - \bar{q}_g\big).\end{aligned} \tag{41}$$

Besides the usual dispatch condition for $q_{g,h}$ (first line), the new condition for optimal investments in a generator capacity (second line) then reads

$$\begin{aligned}\lambda_{z,h} - c_g - \tau e_g + \underline{\mu}_{g,h} - \overline{\mu}_{g,h} &= 0 \qquad \forall h, \\ -F_g + \sum_h \gamma_{g,h}\, \overline{\mu}_{g,h} + \underline{\beta}_g - \overline{\beta}_g &= 0,\end{aligned} \tag{42}$$

alongside the usual complementary slackness conditions. We read it the usual way:

- Interior: built, below the cap. Both multipliers vanish, $\underline{\beta}_g = \overline{\beta}_g = 0$, and $\sum_h \gamma_{g,h}\, \overline{\mu}_{g,h} = F_g$: The rents equal the fixed cost.
- Lower bound: not built. Then $\underline{\beta}_g = F_g - \sum_h \gamma_{g,h}\, \overline{\mu}_{g,h} > 0$: The rents a MW would earn fall short of its fixed cost.
- Upper bound: at the potential cap. Then $\overline{\beta}_g = \sum_h \gamma_{g,h}\, \overline{\mu}_{g,h} - F_g > 0$ collects the excess of rents over fixed cost, the resource rent.

**Result 6** (Long-run zero profit)**.** *At the optimum of the capacity expansion problem, every technology that is built and not blocked by its potential cap earns exactly its annualised fixed cost in availability-weighted scarcity rents:* $\sum_h \gamma_{g,h}\, \overline{\mu}_{g,h} = F_g$ *for every* $g$ *with* $0 < \bar{q}_g^* < \bar{Q}_g$*. A technology at its cap earns its fixed cost* plus *the resource rent,* $\sum_h \gamma_{g,h}\, \overline{\mu}_{g,h} = F_g + \overline{\beta}_g$*; and no unbuilt technology could earn more than its fixed cost. Equivalently: In the price system the model generates, entry is exactly worthwhile for what is built and not worthwhile for what is not, and where entry*

*is rationed by land, the excess return is capitalised into the site.*

A battery's condition has the same form. Its power rating appears in the limits on charging, discharging and the energy it holds, so a battery that is built earns exactly its fixed cost in the rents on all three, $\sum_h \left(\overline{\phi}^c_{s,h} + \overline{\phi}^d_{s,h}\right) + \bar{\delta}_s \sum_h \overline{\psi}_{s,h} = F_s$.

This is the logic of Section 2.3 applied one level up. There, the *dispatch* condition priced energy; here, the *investment* condition prices capacity. Rents are no longer a transfer; they are the mechanism by which prices recover capital. The LP imposes the condition by optimality; a competitive market with free entry imposes it by arbitrage; and this equivalence is the welfare theorem extended to the capital stock. Every market design debate about whether "energy-only" prices can finance a capital-intensive system is a debate about whether the premises of Result 6, scarcity prices allowed to spike and investors able to collect them, hold in practice (Section 9 returns to this as *missing money*).

**What makes a backup plant worth building.** The investment condition answers a natural question: Why does a system full of cheap renewables still want to build gas turbines? Differentiate $\mathcal{L}^{\text{inv}}$ with respect to a technology's capacity $\bar{q}_g$, leaving out the bounds on $\bar{q}_g$ itself. Capacity enters the objective through its annualised fixed cost, and it enters the constraint set through the hourly capacity limits, whose multipliers we have been reading as scarcity rents since Section 2. So

$$\frac{\partial \mathcal{L}^{\text{inv}}}{\partial \bar{q}_g} \;=\; - \underbrace{F_g}_{\text{annual fixed cost}} + \underbrace{\sum_h \gamma_{g,h}\, \overline{\mu}_{g,h}}_{\text{scarcity rent}}, \qquad \overline{\mu}_{g,h} = \begin{cases} \lambda_{z,h} - c_g - \tau e_g, & q_{g,h} = \gamma_{g,h}\bar{q}_g \\ 0, & \text{otherwise.} \end{cases} \tag{43}$$

This is an envelope argument: The dispatch variables are already at their optimum, so their derivatives drop out and only the direct effects survive (Appendix A.4). Now read eq. (43) for a peaking plant that runs, say, 100 hours a year, and notice how each term behaves:

- *Marginal cost barely matters.* The indicator inside $\overline{\mu}_{g,h}$ is zero in *almost every hour*, because the plant is almost never running. A technology's fuel cost is weighted by the number of hours it is used, and for backup capacity that number is tiny. Two peakers differing by 20 €/MWh in fuel cost differ by almost nothing in annual value.
- *Fixed cost matters enormously.* It enters with weight one, every year, regardless of utilisation. For a plant that runs 100 hours, the fixed cost is essentially the entire cost.
- *The value comes from a handful of extreme hours.* The revenue term is $\sum_h(\lambda_{z,h} - c_g - \tau e_g)$ over the hours the plant is at capacity, precisely the scarcity hours, where Section 2.4 showed the price can reach a consumer's willingness to pay. A peaker is not selling energy; it is selling *availability*, and it is paid for it in a few hours a year at large margins.
- *Flexibility matters, in the same expression.* With ramping (Result 4), a plant that cannot get to full output quickly fails to be at capacity in exactly the hours when $\lambda_{z,h}$ spikes, so the

indicator switches off where the money is. Inflexibility does not show up as a cost; it shows up as forgone revenue.

So the ideal backup technology has low capital cost and high flexibility, while its fuel cost matters little. That is an open-cycle gas turbine, the gas plant with the *worst* efficiency and the *highest* marginal cost on Table 1. It is also, for the same reason read in reverse, the argument against conventional nuclear as a partner for variable renewables: High capital cost and low flexibility are precisely the wrong combination for a plant whose job is to fill residual gaps.

## 7.3 Greenfield investments in Denmark, 2050

**The instance.** We take the network of Section 6 as given and perform the greenfield experiment in the two Danish zones. We let the model rebuild the system from the following menu of generation candidates: solar, onshore wind, offshore wind, CCGT, OCGT, the same open-cycle machine fired with biomethane, nuclear, and batteries with durations $\bar{\delta}_s$ of 2, 4 and 8 hours (power and energy are priced separately, so each duration has its own fixed cost $F_s$, and the mix of durations is itself a choice). This menu allows the model to build a system based on variable renewables, fossil gas plants, other emission-free technologies (biomethane and nuclear), and to invest in batteries for flexibility.[16] Solar and both winds are capped by a published buildable potential, at 62.3, 12.5 and 24.3 GW across the two Danish zones: the most ambitious trajectory the transmission operators publish for each zone, never more than the land allows (Appendix C). Technology and fuel costs and demand levels are all projections for 2050.

The neighbours are not a part of the greenfield problem, but they are not kept completely frozen either. Each neighbour keeps the generation fleet it has and may add to it from the same candidate menu, up to the same kind of published potential. The question is what *Denmark* should build, given a neighbourhood that is itself adapting to the horizon's demand. Every emitter in every zone pays the same carbon price, representing the EU ETS.

In every zone, demand consists of the three blocks $j$ of eq. (40c): One representing "standard" electricity demand with a willingness to pay at 23.4 thousand €/MWh, a Danish survey-based VoLL averaged across sectors,[17] a second representing interruptible heavy industry, 10% of the load, with a willingness to pay at 3,000 €/MWh, and a third flexible block, 5% of the load, with a willingness to pay at 100 €/MWh. This provides some demand flexibility, but a more detailed demand curve with survey-based valuations is a natural extension; Brown, Neumann, et al. (2025) show in this type of model that even a little demand elasticity is important for a renewable-heavy system.

**Optimal generation.** Figure 7.2 shows the system the model builds at a carbon price of 85 €/t, the level of the carbon-tax experiment in Section 2.5: (a) the capacity of each candidate in the

[16] Biomethane is relatively costly at 149.4 €/MWh marginal cost.

[17] Surveyed VoLL measures the cost of an *unannounced* outage; as a bid price for anticipated, voluntary curtailment it overstates willingness to pay by a wide margin. The level matters little here, since the price never comes near it, but the caveat matters when such numbers are used as market price caps.

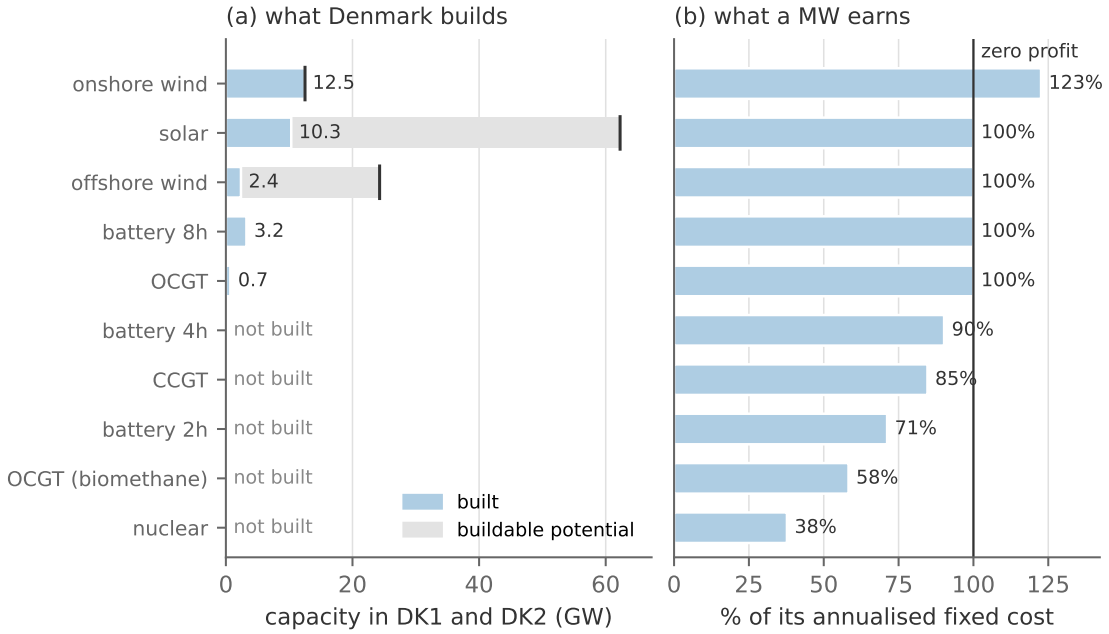



Figure 7.2: The reference system. (a) Capacity built in DK1 and DK2 together; the grey bars run to the buildable potential, with a tick at the cap. (b) What one MW of each candidate earns in the solved prices, as a share of its annualised fixed cost: for a technology the model builds, its Danish fleet, and for one it does not, the better of the two zones.

two Danish zones together, against its buildable potential where it has one, and (b) what a MW of each candidate earns in the prices of that system.

Panel (a) shows a system built on onshore wind and solar. Onshore wind is built up to its potential in both zones and is the largest item. Solar comes second, but stops well short of its potential in both zones. Offshore wind is built only on a small scale, almost all of it in DK2, far below its potential. Flexibility comes from batteries, all of them at the longest duration on the menu, from the interconnectors, and from a small fleet of open-cycle gas turbines in DK2. The model builds no combined-cycle plant, no nuclear, no biomethane turbine and no short battery, and Denmark ends up a net exporter, generating about a fifth more than it consumes (22%).

**Zero profit, in the numbers.** Panel (b) checks Result 6 on the solved system. For a generator, the marginal MW installed earns the scarcity rent from eq. (43). For a battery, the marginal MW installed earns its revenue from taking advantage of price volatility, discharging when prices are high and charging when prices are low. Consistent with Result 6, the model has three groups of technologies:

i. Solar, offshore wind, the 8-hour battery and the gas turbine are built below their caps and earn exactly their fixed cost, to the euro.

ii. Onshore wind is at its cap and earns more than its fixed cost. The excess is the resource rent $\overline{\beta}_g$: what a permit for one more MW would be worth (21 k€ per MW and year). It is the only technology that earns one.

iii. Nothing that is not built could pay for itself. Nuclear would recover little more than a third

Table 5: Levelised cost and capture price in the reference system, per built generator. The LCOE excludes the carbon price; for the gas turbine the carbon cost of 39 €/MWh has to be added to compare it with its capture price.

| Technology | Capacity GW | Full-load hours h/yr | LCOE €/MWh | Capture price €/MWh | Value factor | |
|---|---|---|---|---|---|---|
| solar | 10.3 | 953 | 30.8 | 30.8 | 0.58 | interior |
| onshore wind | 12.5 | 3,060 | 32.0 | 39.0 | 0.73 | at its cap |
| offshore wind | 2.4 | 4,508 | 37.3 | 37.3 | 0.70 | interior |
| OCGT | 0.7 | 938 | 95.0 | 134.2 | 2.51 | interior |

of its fixed cost (38%), and the biomethane turbine, the combined-cycle plant and the shorter batteries fall short too.

**Renewables and capture prices.** The first limitation of the screening curves in Section 7.1 was that they are silent about revenue. Table 5 puts the revenue side next to the cost side for every generator the model built: the levelised cost of eq. (39), and the capture price and value factor of Section 3.

For the three variable renewables, the table shows the following. First, solar and offshore wind, built below their caps in both zones, capture exactly their levelised cost. This is Result 6 restated per MWh: A technology is built until the average price it captures has fallen to its levelised cost. Second, for onshore wind, at its cap, the capture price exceeds the levelised cost by the resource rent per MWh. Third, all three value factors are well below one, and solar's is the lowest.

The third point follows from the first. For a technology built below its cap, the capture price equals the levelised cost, so the value factor is the levelised cost divided by the average price: The cheaper the technology, the lower the value factor at which the model stops building it. That is why solar, with the lowest levelised cost on the table, is built to only a fraction of its potential in either zone, since every MW added lowers the price solar sells at. It is the cannibalisation of Section 3 in its investment form, and the reason why ranking technologies by levelised cost alone, the practice Joskow (2011) warns against, gets the answer wrong. The full-load hours make the same point from the other side: Each variable technology runs as many hours as the Danish weather allows (the markers in Figure 7.1), so a screening curve says what a duty costs but not which duty a technology will get.

**Peakers and batteries.** The gas turbine is the odd row in Table 5. It runs for about a tenth of the year, and its levelised cost is well over twice that of wind, yet it is built. Add the carbon cost to its levelised cost and the sum equals its capture price, as zero profit requires: The turbine sells only in expensive hours, at 2.5 times the average price, and that is what pays for running so little. Comparing levelised costs across technologies that serve different duties therefore says little about which of them to build. Figure 7.3 shows how the turbines are paid: It sorts the hours of the year into bands of the price in DK2, where the whole fleet stands, and for each band sets the share of the year's hours against the share of the turbines' annual operating margin earned in it.

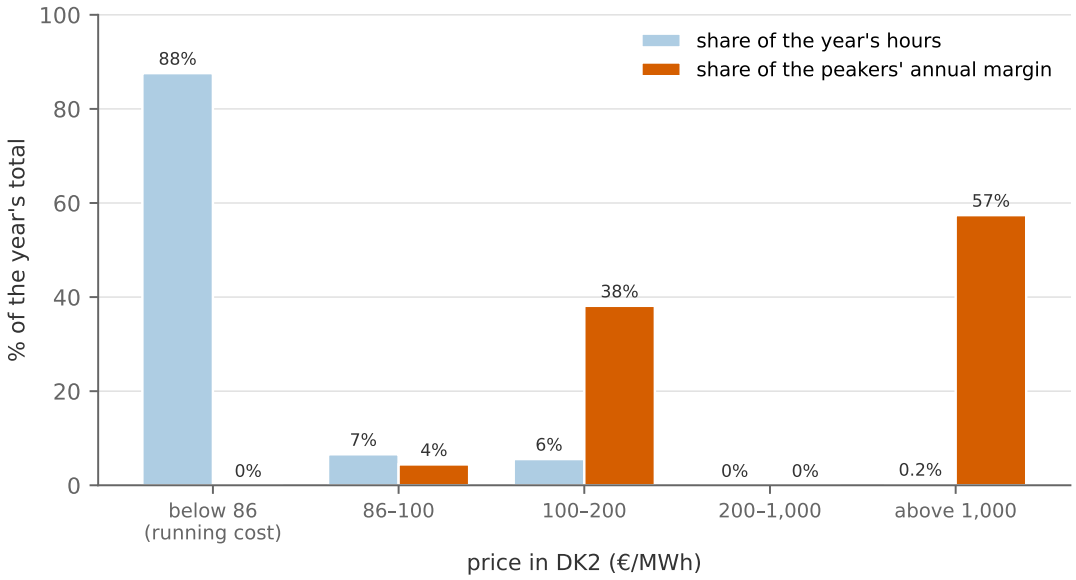



Figure 7.3: How the gas turbines are paid in the reference system: the hours of the year in bands of the DK2 price, and for each band the share of the year's hours against the share of the turbines' annual operating margin earned in it.

The figure shows that for most of the year (88%), prices are below the gas turbine's running cost, carbon included (86 €/MWh). The few hours in which prices are above 200 €/MWh (just 21 hours in this simulation) pay for more than half of the turbines' annual operating margin (57%), and most of the rest is earned in the several hundred hours priced just above the flexible block's willingness to pay.

The hours priced at or above the flexible block's willingness to pay do not arrive one at a time. They come in 10 spells with little wind or sun and high demand, lasting between 14 and 109 hours, so every one of them is longer than the 8-hour battery can carry at full output. Figure 7.4 follows the system through the longest of them.

The figure shows that throughout the spell, the system relies on a combination of imports, gas turbines, batteries and flexible demand to stay in balance, with imports by far the largest and prices just above the flexible block's willingness to pay. Batteries hold their charge until the price steps up, then spread it over the rest of the spell at a fraction of their power, refilling partly in a short lull and running empty just as the spell ends. The windy days after it refill them at prices near zero while Denmark exports. The price rises much further only in the short spell at the end of the window, when the neighbours are short as well and imports are no longer cheap: It peaks at 1,750 €/MWh, and the batteries sell there what they bought at near-zero prices days before. Those few hours matter far more to the turbines than to the batteries, which earn most of their margin from the everyday swings between low and high prices.

## 7.4 Experiments

We now do five experiments to see how the system responds to (i) carbon prices, (ii) removing trade (autarky), (iii) higher gas prices, (iv) lower capital costs and (v) a Danish carbon budget.

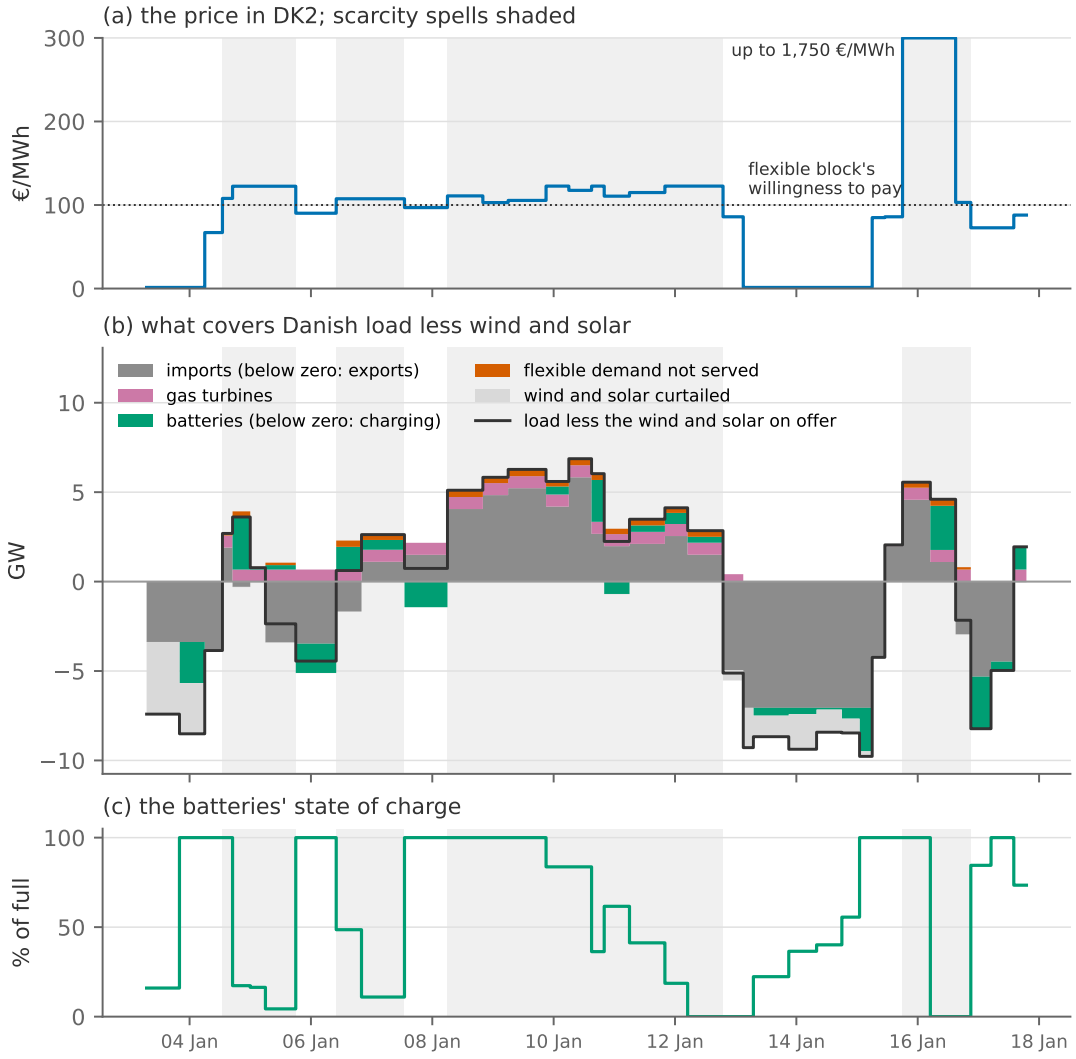



Figure 7.4: The reference system through its longest scarcity spell, with a few days on either side: (a) the price in DK2, with the spells shaded; (b) Danish load less the wind and solar on offer, and what covers it; (c) the batteries' state of charge.

Each experiment changes one price or one constraint and solves the model again. Since the model minimises cost, it adjusts along the cheapest margin first, and for each experiment we ask which margin that is, and why.

**Carbon prices.** Figure 7.5 sweeps the common carbon price from zero to 300 €/t and shows what Denmark builds and what everyone emits.

The figure shows that even without a carbon price, the Danish system is predominantly wind and solar, with onshore wind close to its potential. As carbon prices increase, especially from 0 to 100 €/t, the model builds out primarily solar and replaces the gas turbines, which go from producing energy at moderate prices to standing by for a few hours a year. At around 50 €/t and above, the mix of cheap renewables and costly gas generates a price spread that warrants investments in batteries as well.

Beyond the reference price, Danish emissions fall slowly. What is left is the backup generation from gas, in particular in the spells of Figure 7.4, in which more solar does little and the batteries

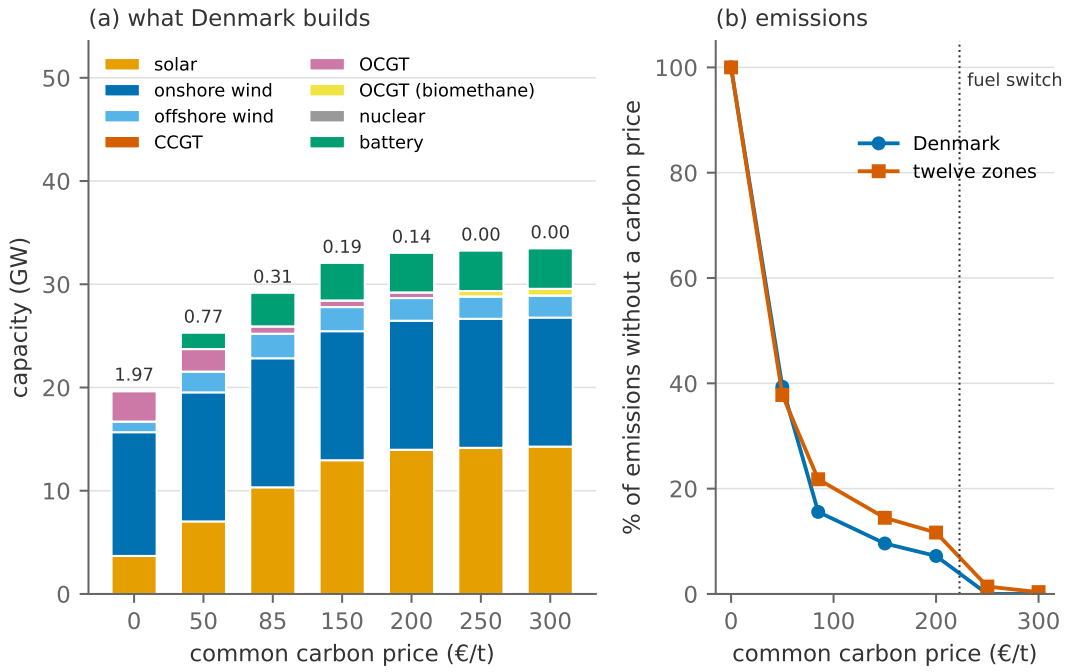



Figure 7.5: The common carbon price swept from zero to 300 €/t: (a) the optimal Danish capacity mix, with Danish territorial emissions in Mt above each bar; (b) Danish and twelve-zone emissions, each as a share of its level without a carbon price, with the fuel-switching price of the peakers marked.

run empty. Solar and batteries keep growing and the flexible demand block is curtailed more often, but each tonne costs more than the last. The last Danish emissions are abated when the carbon price is sufficiently high to make biomethane cheaper than natural gas in the turbines. The marginal abatement cost of this fuel switch is the fuel-price premium divided by the gas turbine's emission rate, $(149.4 - 46.8)/0.46 = 223$ €/t.

Panel (b) puts Danish and twelve-zone emissions on one scale, each as a share of its level without a carbon price. In proportion the two fall at almost the same pace, and fast: By the reference price both are down by about four fifths (84% and 78%), in the twelve zones mostly as German coal and lignite give way to solar and wind. At the fuel switch both drop to almost nothing, because the neighbours' gas plants can burn biomethane too: Above it, Danish emissions are zero and the twelve zones keep only a small remainder (1 Mt at 300 €/t). In tonnes, Denmark's share of the total is tiny throughout.

Figure 7.6 shows the other four experiments: (a) what Denmark builds in each, and (b) what moved, as the change in each technology's capacity against the experiment's baseline. The first three keep the reference carbon price and are compared with the reference system. The Danish budget replaces the price, so it is compared with the system that has no carbon price at all, set beside it in panel (a).

**Autarky.** Cutting Denmark off from its neighbours makes the system rebuild what the interconnectors were doing, and three things change. First, wind is cut back: Onshore wind falls well

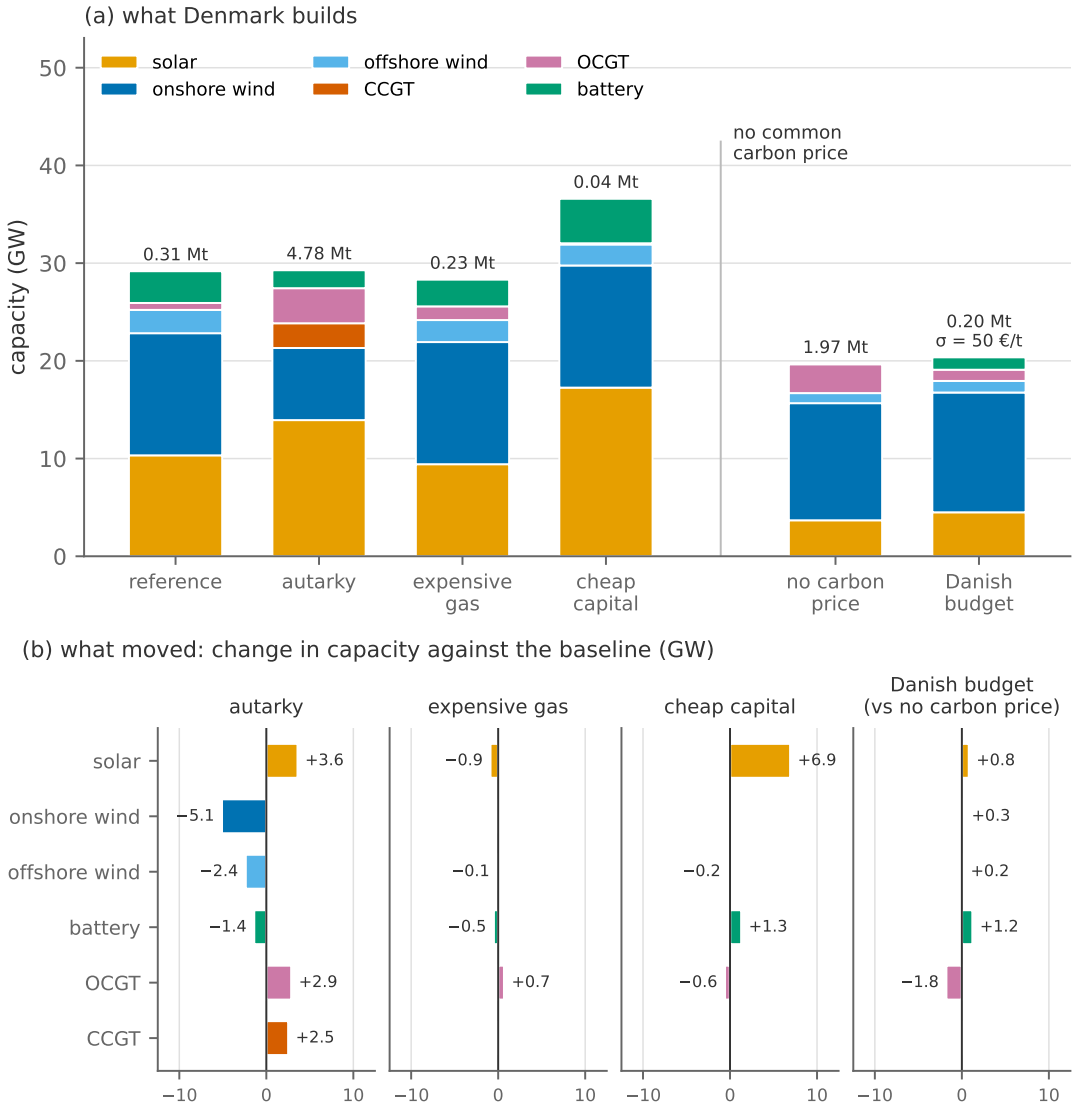



Figure 7.6: The other four experiments. (a) What Denmark builds under autarky (no interconnectors), a doubled gas price and capital at 2% instead of 7%, all at the reference carbon price, and under a Danish $CO_2$ budget 90% below its unpriced emissions, set beside the system with no carbon price at all; Danish emissions and the budget's shadow price above the bars. (b) The change in Danish capacity against each experiment's baseline.

below its potential, the largest move in panel (b), and the little offshore wind disappears. In the reference system, Denmark sells much of its wind abroad; with nobody to sell to, wind beyond what Denmark can use in the same hours is worth little, and the two winds blow at the same times. Solar, which produces at other times, is built instead. Second, gas returns as a source of energy and not only as backup capacity: The system builds combined-cycle plants that run for a large part of the year, the long duty for which the screening curves of Figure 7.1, with the carbon price added, pick the combined cycle, and a larger fleet of open-cycle turbines for the peaks. Third, Danish emissions rise by an order of magnitude, as gas turns out to be the cheapest way to make up for the missing imports.

Without the interconnectors, the cost of the Danish system increases by 12% to serve the same load. The comparison has to be made net of trade, since the reference fleet includes wind capacity

built partly for export, and comparing gross bills would compare an exporter's costs with a self-supplier's. What trade buys is mostly a matter of timing. Denmark exports when its own price is low and imports when it is high, so the interconnectors stand in for firm capacity that Denmark would otherwise have to build and fuel itself.

**Expensive gas.** Doubling the gas price in every zone separates the two jobs of a gas turbine. Its job of providing energy at medium prices disappears as costs increase. However, as high prices become more frequent, Denmark roughly doubles the turbine capacity, increasing production from gas in those hours. The net effect is that emissions fall, in Denmark and across the twelve zones, even though turbine capacity is built out.

The reason works through the neighbours, in particular Germany: In the reference they build combined-cycle plants that produce energy for much of the year and, as a by-product, stand at full output through the scarcity spells, which is what lets Denmark import in them. With gas this expensive, that energy comes from existing coal and lignite plants and from new wind and solar instead, the combined-cycle plants are not built, and the region meets the spells with less capacity. High prices in the spells become more frequent, so a turbine available in them would earn more than its fixed cost even with the higher fuel cost, and the model adds turbines until it no longer does.

**Cheap capital.** Lowering the discount rate from 7% to 2% lowers the annual cost of every technology in proportion to how much of its cost is capital. Solar and batteries, whose costs are almost all capital, gain the most: The optimal system holds much more of both and few gas turbines, and Danish emissions almost vanish with no change in the carbon price. Offshore wind, capital-heavy too, holds roughly steady, because the neighbours build far more wind on the same cheap capital and the market for Danish wind abroad does not grow.

The discount rate matters this much because almost the entire cost of the reference system is fixed, and most of it is capital: Annual fixed costs are 95% of the Danish system's cost, with running costs making up the rest, and only the gas turbine has an operating cost of the same order as its capital cost. At 2% the cost of the Danish system is about a third lower (31%), and the average Danish price falls by about as much. This moves the political economy of the sector from fuel prices to financing costs. It also makes the sector's costs fixed and its marginal costs close to zero, the world in which the missing-money questions of Section 9 bite hardest.

**A Danish carbon budget.** The last experiment removes the common carbon price everywhere, in the neighbouring countries too, and instead caps Danish territorial emissions 90% below what Denmark emits with no carbon price. With $\tau = 0$ in eq. (40a), the model gains the cap of Section 2.6 on Danish plants alone,

$$\sum_{h} \sum_{g \in \mathcal{G}_{\mathrm{DK1}} \cup \mathcal{G}_{\mathrm{DK2}}} e_g \, q_{g,h} \;\leq\; E^{\mathrm{DK}} \qquad (\sigma). \tag{44}$$

In the Lagrangian, $\sigma$ takes the place of $\tau$ in the groups of the Danish plants and nowhere else:

The budget is a carbon price on Danish emitters only. Absent any carbon regulation, the Danish system still relies heavily on renewables, but also holds a significant fleet of open-cycle gas turbines. With the budget in place, the system decarbonises using three margins to replace gas: a fleet of batteries, somewhat more solar and wind, and increased imports. A share of the gas turbines are kept as backup, at less than half their unpriced capacity. The budget's shadow price comes out at $\sigma = 50$ €/t.

Two remarks on this price. First, it is low: A common carbon price has to reach about 150 €/t before Danish emissions fall as far. The difference is the neighbours: Under a common price, the power Denmark imports in its scarce hours is priced too, so importing does not avoid the carbon price; under a territorial budget the neighbours' turbines pay nothing for carbon, and their power counts as carbon-free in the Danish accounts. Second, part of the cut moves rather than disappears: Summed over all twelve zones, emissions fall by less than Denmark's do, and 31% of the Danish cut reappears abroad as **carbon leakage**.

The leakage is modest but real, and it comes from imports being one of the budget's margins: Power Denmark no longer generates from gas in its scarce hours is in part generated from gas abroad instead. The lesson about the price is general: A territorial budget in an open economy prices the cheapest way to cut Danish emissions, imports included, and that can lie far below the cost of cutting emissions. Section 8 puts the budget over the whole neighbourhood, where there is nowhere to leak to.

> **Which way does it bias the answer?** *Perfect foresight over the investment horizon.*
>
> The model chooses capacity knowing every hour of weather and every price it will face. This **understates** the cost of the transition, because a real investor hedges against a future they cannot see and therefore builds a portfolio more diversified, and more expensive, than the optimum for any single realised future. And it **overstates** how confidently the mix can be specified: With the future known, the LP picks a corner (Appendix A.1), so it never builds a diversified portfolio for diversification's own sake, even though diversification is exactly what uncertainty rewards. Section 9.1 measures the second effect, and Section 9.2 discusses the first.

**From duals to markets.**

| Constraint | Dual | What it measures | Market counterpart |
|---|---|---|---|
| zonal clearing (40b) | $\lambda_{z,h}$ | marginal cost incl. scarcity | energy-only market prices |
| investment optimality (42) | — | rents = annualised fixed cost | free entry; cost recovery without subsidy |
| buildable potential (40e) | $\overline{\beta}_g$ | resource rent: value of one more permitted MW | the price of a site permit or seabed lease |
| a Danish $CO_2$ budget (44) | $\sigma$ | the cost of the last Danish tonne, imports included | a national target's implicit carbon price |

To sum up: Making capacity a variable adds one optimality condition per technology, and that condition is the long-run zero-profit result that ties the rents of every previous section to cost recovery. The reference system verifies this and explains both why a cheap technology is built only until its own capture price has fallen to its cost and why an expensive turbine is built at all for a few hours a year. The experiments then show the system adjusting along its cheapest margin first: gas energy before gas capacity under a carbon price, firm gas capacity in place of imports under autarky, turbines for availability when gas is expensive, capital-heavy technologies when capital is cheap, and Danish wind and solar together with unpriced imports when only Danish tonnes count. All of it was computed for one year of weather and one set of cost projections. Section 8 widens the boundary to the whole neighbourhood and puts one budget over all twelve zones; Section 9 then asks how much of this section's precision survives other weather years and other cost projections.

# 8 Expansion at European scale

Section 7 was our first capacity expansion model exploring how to decarbonise the energy system. In this section, we extend our exploration in several ways: First, we take the *transition* part of the green transition more seriously by looking into optimal investments *given* that investments in the current system are already sunk. Next, we look more carefully into what shapes the cost of decarbonising the energy system, in particular looking into flexible demand from electric vehicles (EVs) and the role of a sizeable hydrogen sector.

> **Reading alongside this section.** This section tests claims from two papers. Victoria et al. (2019) looks at storage technologies in a decarbonising European system, batteries for the daily cycle and hydrogen for the multi-day one. Neumann, Zeyen, et al. (2023) is the hydrogen half of the same research programme and the source of the extension this section most obviously points at. The note's models resemble those of the papers, albeit simpler in several ways. Appendix D elaborates on where the note's models differ from the literature and what this means.

## 8.1 The scenarios

This section takes our first greenfield experiment from Section 7 and explores how four perspectives shape the green transition of the energy system. Before we explore this using our numerical model, we will briefly go through each and explain how it changes our model.

**Baseline: Brownfield with interconnector investments.** The first departure from Section 7 is that we change our focus from a Danish greenfield experiment to a full "European" **brownfield** experiment. Section 7 looked into the decarbonisation of the Danish system from a greenfield perspective that ignored the current fleet of power plants. However, as power plants and electricity system infrastructure are generally long-lived durables that exist for decades, this "sunk capital"

can be important when we decide how to invest in new technologies even for a longer time horizon such as 2040–2050. This capacity expansion approach is called a **brownfield** experiment.

The second departure is that we allow for investments in new transmission lines. Again, we do this as a brownfield experiment: We take the existing line capacities as given and ask whether, and by how much, it is worth expanding them.

**Flexible demand from EVs.** Electric vehicles (EVs) are important not only for the decarbonisation of emissions from transportation; their role in the electricity system has long been discussed because of the big battery fleet that comes naturally with EV adoption. Expectations as to what a big fleet of EVs will do for an electricity system range from adding a lot of new demand, to adding a lot of new *flexible demand*, to even adding a huge battery fleet that may not only change its demand in response to prices (*smart charging*), but even supply power back to the grid when prices are high (*vehicle-to-grid*). We test both smart charging and vehicle-to-grid and explore how this competes with the usual "stationary" batteries that we have looked into previously.

**The hydrogen sector.** Another important feature for the energy system is expectations for a future hydrogen sector. Hydrogen is mainly discussed as a fuel for the uses that are hard to electrify directly: industrial processes that use it as a feedstock, such as fertiliser, refining and steel, and fuels for heavy transport, shipping and aviation. How large this sector will become is very uncertain. Policy ambitions are big, but for many of the proposed uses of hydrogen the trend points the other way: Direct electrification keeps getting cheaper and reaches more uses over time, which shrinks the set of uses where hydrogen is the only option.

For the electricity system, a hydrogen sector matters for three reasons. First, it is potentially a huge block of new demand, since every MWh of hydrogen made by electrolysis takes more than a MWh of electricity. Second, that demand can be operated flexibly: An electrolyser can run when power is cheap and stand idle when it is expensive, as long as something absorbs the difference between when the hydrogen is made and when it is used. Third, hydrogen can be stored in large quantities and burnt back to electricity, which makes it a candidate for the long-duration storage that Section 4 found batteries cannot provide.

## 8.2 The baseline

**The model.** The model is almost identical to that of Section 7 in eqs. (40); the only change is that we allow for the planner to invest in new interconnectors. We model this in a way that mirrors the generator investments: Let $k$ be a candidate for a new transmission line that can transport electricity from zone $a_k$ to $b_k$ and compute the total annual fixed cost of this cable as we did for investments in generators (section 7.1) with a cost of $F_k$ per MW of capacity.[18] To simplify things, we assume that all the new candidates are controllable DC cables.

[18] In the simulations below, all candidates are actually implemented as a combination of two cables running in opposite directions; this makes the formal presentation slightly more complicated, but the general idea is the same.

With this as the only change, the planner now maximizes

$$\max \quad W^{\text{exp}} \equiv W^{\text{inv}} - \sum_k F_k \, \bar{f}_k, \tag{45a}$$

subject to all the same constraints as in Section 7 *and* a new buildable potential constraint on the size of the new cables $\overline{f}_k^{\max}$:

$$0 \leq \bar{f}_k \leq \bar{f}_k^{\max} \qquad (\underline{\beta}_k, \overline{\beta}_k) \qquad \forall k. \tag{45b}$$

**Duality: the value of an interconnector.** As in Section 7, it is a long cumbersome task to write up the full Lagrangian for this model. Instead, we collect all the relevant terms for a candidate $k$ in an "interconnector Lagrangian" that can be written as:

$$\begin{aligned} \mathcal{L}_k^{\text{int}} = \sum_h \Big[ f_{k,h} \left(\eta_k \lambda_{b_k,h} - \lambda_{a_k,h}\right) + \underline{\omega}_{k,h} f_{k,h} + \overline{\omega}_{k,h} \left(\overline{f}_k - f_{k,h}\right) \Big] \\ - F_k \, \bar{f}_k + \underline{\beta}_k \, \bar{f}_k + \overline{\beta}_k \big(\bar{f}_k^{\max} - \bar{f}_k\big), \end{aligned} \tag{46}$$

The optimality conditions for optimal flow (first line) and the new condition for optimal investments in new capacity (second line) then read:

$$\begin{aligned} \eta_k \lambda_{b_k,h} - \lambda_{a_k,h} + \underline{\omega}_{k,h} - \overline{\omega}_{k,h} &= 0, \qquad \forall h \\ - F_k + \sum_h \overline{\omega}_{k,h} + \underline{\beta}_k - \overline{\beta}_k &= 0. \end{aligned} \tag{47}$$

This is equivalent to the generator condition with the scarcity rent of the interconnector coming through a congested line, which occurs when the price spread net of losses $\eta_k \lambda_{b_k,h} - \lambda_{a_k,h} > 0$ persists even as the cable is fully utilized, i.e. when $f_{k,h} = \overline{f}_k$. We read the investment condition the usual way:

- Interior: built, below the ceiling. Then $\underline{\beta}_k = \overline{\beta}_k = 0$ and the price spreads net of losses exactly cover the annual fixed cost in congestion rents $\sum_h \overline{\omega}_{k,h} = F_k$.
- Lower bound: not built. Then $\underline{\beta}_k = F_k - \sum_h \overline{\omega}_{k,h} > 0$ and congestion rents do not cover the first MW of installed capacity.
- Upper bound: at the ceiling. Then $\overline{\beta}_k = \sum_h \overline{\omega}_{k,h} - F_k > 0$ collects the excess of rents over fixed costs, which is the transmission line's equivalent of a resource rent.

Section 6.6 measured both sides of this condition for the Skagerrak cables. The system cost a bigger cable saves rose concavely with its size, while the rent the cable itself collected first rose and then fell, because a large enough cable competes away the very spread it earns on. Equation (47) says where the planner stops: where the rent on the *last* MW, which is also the system cost the last MW saves, has fallen to its fixed cost.

**Result 7** (The value of a wire)**.** *At the optimum, every new cable built below its ceiling earns exactly its annual fixed cost in congestion rent per MW: the price difference between its two ends net of losses, $\eta_k \lambda_{b_k,h} - \lambda_{a_k,h}$, summed over the hours in which that difference is positive, which are the hours the cable runs full. That rent is also the system cost one more MW would save. A cable that is not built would earn less than its cost; one at its ceiling earns more, and the excess $\overline{\beta}_k$ is the value of permitting more. An owner paid only the rent, who knows that new capacity lowers the spread on the old, builds less than the planner.*

**The instance.** Our baseline model is the one from Section 7 with the few adjustments outlined above. The model describes a 2050 scenario with existing generators and transmission lines taken as given; the optimisation then builds out new generators and transmission lines within their buildable potentials. As always, the full set of data and a replication package is available in the online repository. Compared to Section 7 that focused mostly on Denmark, we focus on the entire energy system here.

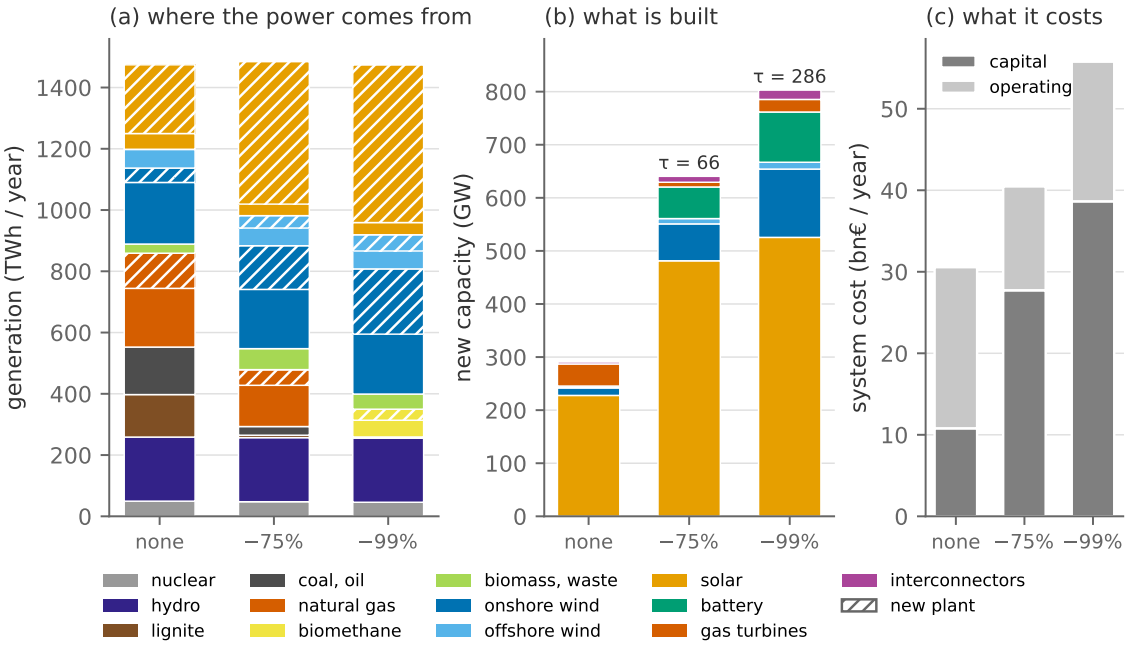



Figure 8.1: The baseline without a budget and at cuts of 75% and 99%: (a) generation by source, existing plants solid and new plants hatched; (b) new capacity by technology, interconnectors included, with the carbon tax $\tau$ in €/t above each bar; (c) the annual system cost, capital and operating.

**The carbon budget.** Figure 8.1 shows model outcomes with different emission budgets: panel (a) shows where electricity comes from, panel (b) new capacity, and (c) the total costs of the system.

The figure shows that even without a carbon tax, wind and solar already supply two fifths of the energy, and the system builds hundreds of GW of solar. Imposing a carbon tax of 66 €/t eliminates 75% of emissions, mainly by replacing coal and lignite in the generation mix by building

out wind and solar alongside adding 8-hour batteries and using existing biomass plants more often to help balance the system. The system cost increases by about a third, driven by higher capital costs. In order to (almost) fully get rid of emissions, the carbon tax has to increase significantly to 286 €/t: Every gas plant now burns biomethane, and a fleet of new gas turbines are built only to run a few winter weeks where the combination of variable renewables and batteries cannot balance the system. These last emission reductions are expensive both in terms of capital and operating costs from the expensive biomethane.

**Prices.** Figure 8.2 shows (a) the price duration curve of the whole system, each hour's zonal prices weighted by the zones' demand, and (b) what sets the price, at the three budgets.

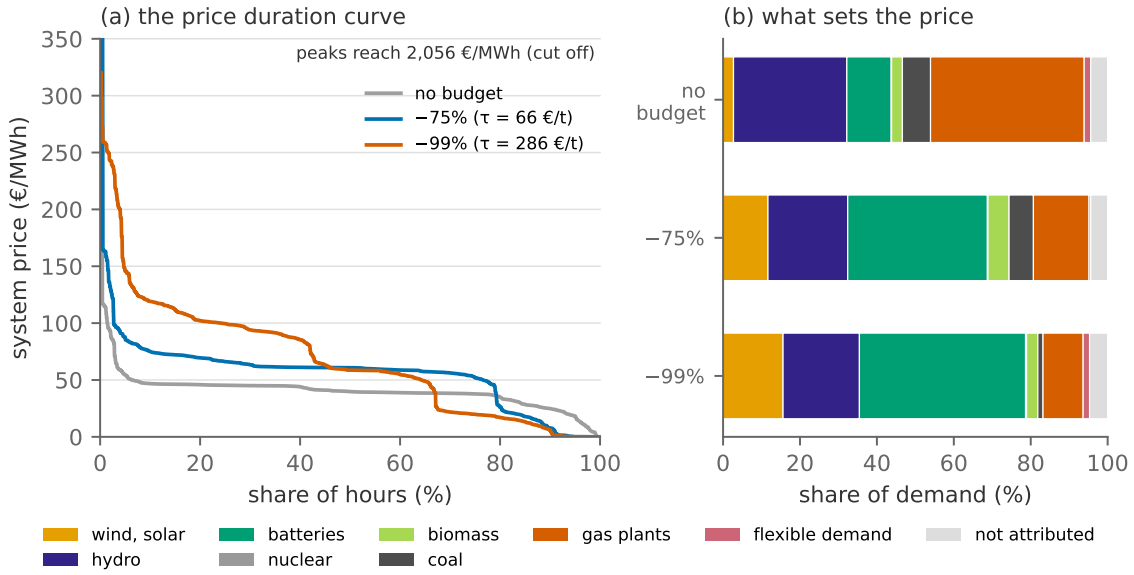



Figure 8.2: The system's price duration curve and decomposition of what sets the price as shares of the year's demand. "Not attributed" is demand whose price no single plant, storage facility or block of demand sets, mostly where the AC grid's loop flows tie the Nordic zones' prices to each other (Section 6.3).

The figure shows that imposing a carbon tax that eliminates (almost) all of the emissions raises the average system price from 44 to 68 €/MWh. The decarbonised system's price distribution, however, also features "fatter tails", i.e. more hours where the price is near zero and wind and solar are curtailed (grows from 155 to 868 hours), but also more hours where the prices are relatively high above the flexible demand block's willingness to pay (grows from about a hundred to about two thousand hours). The highest scarcity prices disappear as interruptible industry is no longer required to balance the system.

The figure's panel (b) also shows that in the decarbonised system, the price goes from being primarily set by thermal plants (i.e. plants burning fuels) and hydro, to much more often being determined by wind, solar, and batteries.

**What covers the last quarter of emission reductions.** As mentioned above, the first three quarters of emission reductions come relatively cheaply through a combination of wind, solar, and

batteries. For the last quarter where emissions primarily come from gas plants, this becomes increasingly costly. The system essentially has four margins that compete to balance the system: storage, flexible demand block, backup capacity, and the interconnectors that let one zone's surplus serve another's demand. Figure 8.3 looks at those hours from two sides: (a) what covers the tenth of the year with the highest residual demand, and (b) what one more MW of storage would earn against its annual cost, for durations from eight hours to a month.

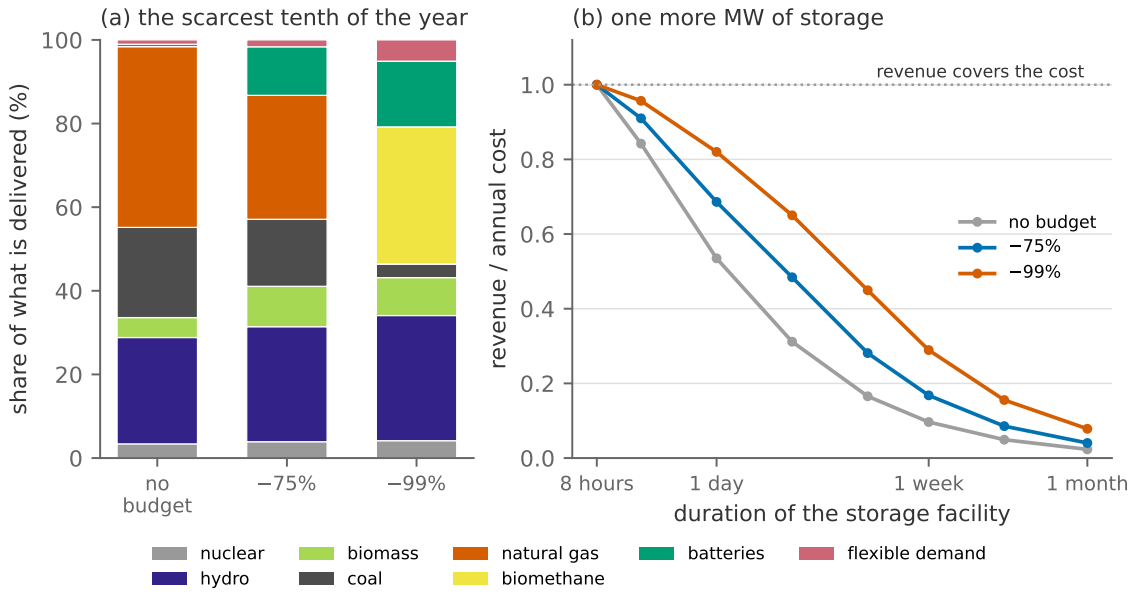



Figure 8.3: The baseline without a budget and at cuts of 75% and 99%: (a) what covers the tenth of the year with the highest residual demand, demand less the wind, solar and run-of-river available, as shares of what is delivered in it; (b) what one more MW of storage of each duration would earn trading against the solved prices, over its annual cost on the battery menu, in the zone where it earns the most.

The figure shows that in the scarcest hours, the gas plants burning biomethane, batteries, the existing biomass plants and flexible demand that gives way take over from natural gas and coal, while hydro covers about the same share with and without a carbon tax. Panel (b) shows the revenue that an extra MW of battery storage would earn relative to its investment costs as a function of its duration. As the systems are optimized, they all start at one at the 8-hour duration that is actually built (this is the zero-profit case in Result 6), but it also shows how the value of longer-duration storage changes with the carbon budget. A higher carbon tax raises the value of long storage, because it creates long stretches of high prices, but a facility that holds a week cycles only about twenty times a year, so each cycle has to earn a twentieth of its annual cost. It would pay only if its energy capacity cost less than a third of a battery's.

The other margins run into limits of their own. The flexible demand blocks are simple in the sense that they can always reduce demand by 5% if prices exceed 100 €/MWh, but the next consumer willing to lower demand is the block of "interruptible industry" demand, which requires prices to exceed 3,000 €/MWh. Thus, flexible demand becomes a relatively cheap margin at the deepest emission cuts, but this relatively cheap option is limited. In the decarbonised system, the

interruptible industry demand is never used, because the other margin, backup capacity switching from natural gas to biomethane, is always cheaper.

## 8.3 Flexible demand from EVs

**What the extension adds.** A significant share of demand for electricity by 2050 comes from EVs. In the baseline scenario, this was simply a part of the standard demand block. Now, we consider what happens first when EVs optimize when they charge (*smart charging*) and subsequently what happens if the EVs are allowed to sell electricity back to the grid when they are plugged in for charging, e.g. overnight (*vehicle-to-grid*). Smart charging and vehicle-to-grid are types of flexible demand and battery storage, but they are still different from what we have seen throughout this note in the important sense that they only interact with the grid when they are plugged in. This means that the profit maximization from the stationary batteries in Section 4 is only a secondary motive compared to the transportation service that they provide. We leave the formal presentation for Appendix E.1.

**What the cars do to prices and margins.** Figures 8.4 and 8.5 repeat figs. 8.2 and 8.3 for the three ways of charging in the decarbonised system.

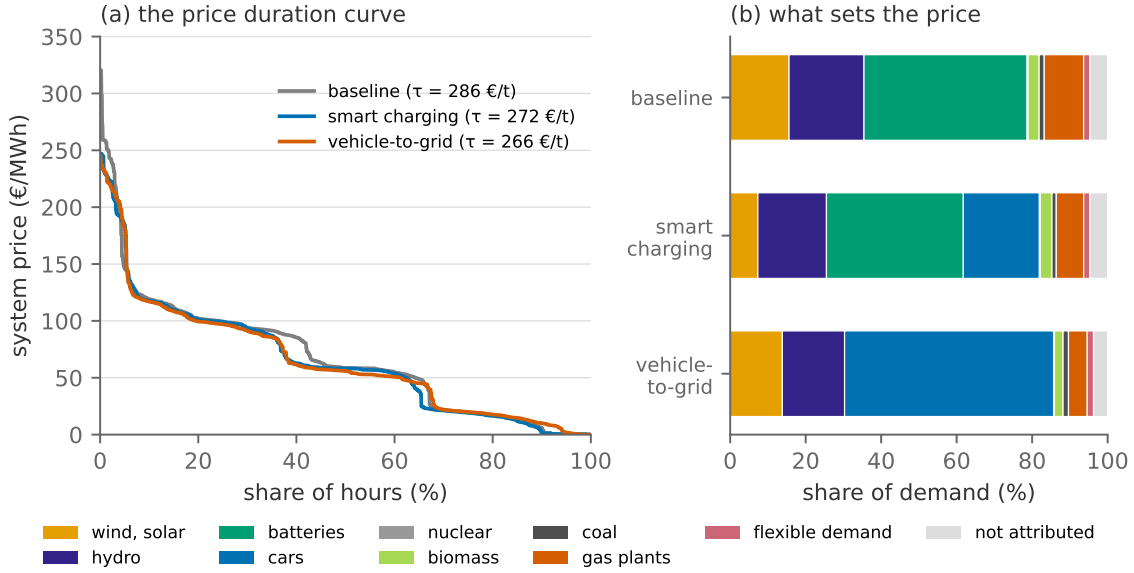



Figure 8.4: The baseline, smart charging and vehicle-to-grid at a cut of 99%, read as fig. 8.2: (a) the system's price duration curve; (b) what sets the price, as shares of the year's demand.

Figure 8.4(a) shows that both smart charging and vehicle-to-grid have roughly the same effect on prices for the higher end of the price distribution: For the half of the year with highest prices, either will generally lower prices almost identically. For the other half, in particular the hours with lowest prices, vehicle-to-grid further raises the prices compared to only smart charging. Panel (b) shows that both steps crowd out the stationary batteries, with vehicle-to-grid removing them entirely.

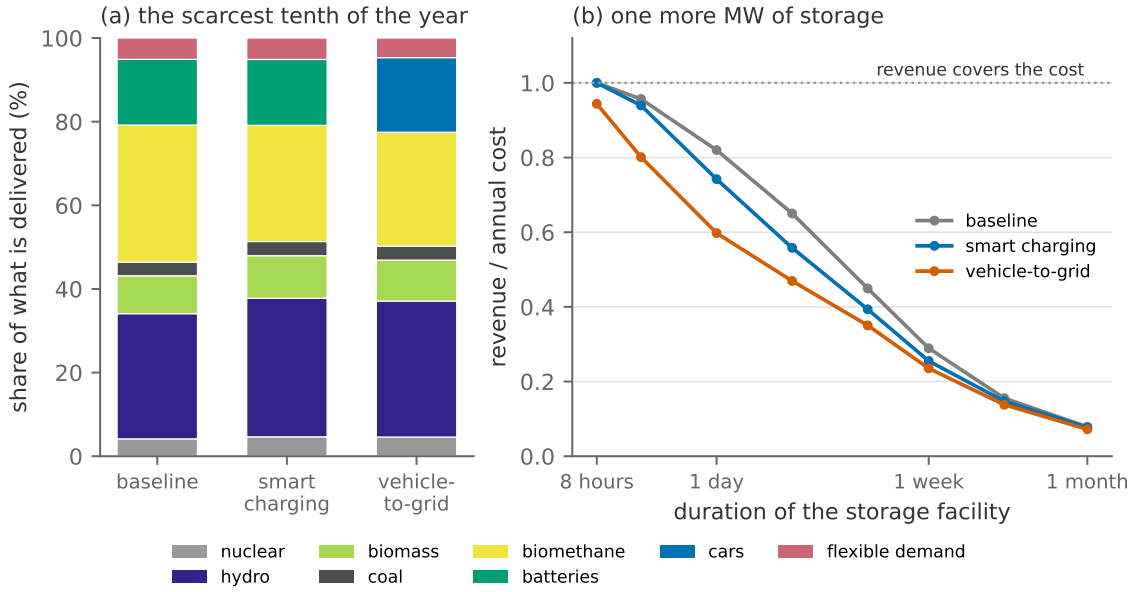



Figure 8.5: The baseline, smart charging and vehicle-to-grid at a cut of 99%, read as fig. 8.3: (a) what covers the tenth of the year with the highest residual demand; (b) what one more MW of storage of each duration would earn over its annual cost.

Figure 8.5 shows that the cars take over from the batteries. With vehicle-to-grid, they set the price more often than anything else, cover a sixth of the scarcest hours, and take so much of the batteries' revenue that one more MW of eight-hour storage no longer pays. Week-long storage earns almost as little as in the baseline, because the cars, like the batteries they replace, move energy within days, not across weeks.

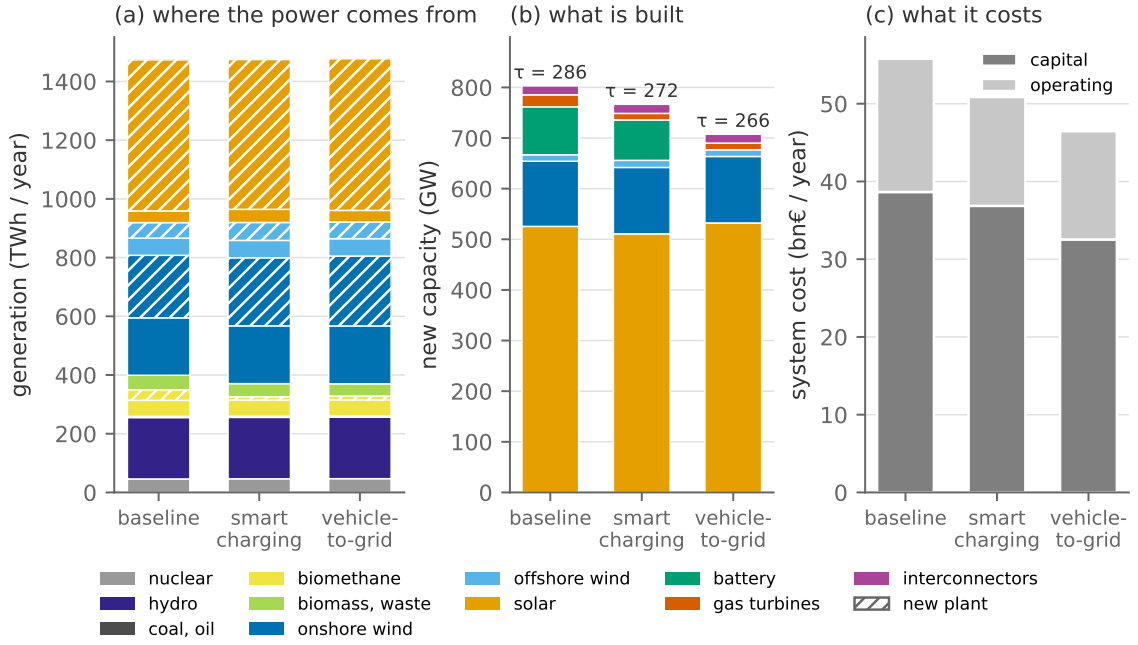



Figure 8.6: The baseline, smart charging and vehicle-to-grid at a cut of 99%, read as fig. 8.1: (a) generation by source, existing plants solid and new plants hatched; (b) new capacity by technology, with the carbon tax $\tau$ in €/t above each bar; (c) the annual system cost. The cars' batteries and chargers are bought for driving and are not in (b).

**What flexibility is worth.** Figure 8.6 repeats fig. 8.1 for the three ways of charging in the decarbonised system. The figure shows that beyond replacing stationary batteries, the generation mix and built capacity are only marginally affected by smart charging or vehicle-to-grid. The system cost falls, however, with vehicle-to-grid mainly because the system no longer includes the cost of the stationary batteries, and with smart charging mainly because the gas plants burn less biomethane. The cost of the EVs is not included in this measure.

## 8.4 The hydrogen sector

**What the extension adds.** The hydrogen scenario keeps the smart-charging cars and adds a market for hydrogen in every zone with a hydrogen demand. The model can supply hydrogen in three ways: so-called *reformers* that make hydrogen from natural gas or biomethane, *imports* from outside the geographic areas covered in our model, and *electrolysers* that turn electricity into hydrogen.[19] Hydrogen can be stored over time in salt caverns where the geology allows and in tanks everywhere, and a *hydrogen turbine* can convert it back to electricity.

This way of modelling the hydrogen market is important, because the combination of hydrogen storage and hydrogen-fuelled turbines means that this new hydrogen market represents a new flexible demand that can even feed power back into the electricity system, potentially providing seasonal storage.

The hydrogen system works in a similar way to the electricity system: The price of hydrogen is the cost of whichever route is at the margin. For the reformers, the marginal cost is the cost of its fuel, including the carbon tax. The electrolyser runs flat out whenever electricity is sufficiently cheap relative to the hydrogen it makes, and a turbine runs in the opposite case. We leave the formal presentation for Appendix E.2.

**Where the hydrogen comes from.** Figure 8.7 shows how the hydrogen demand is met at three budgets, and month by month in the decarbonised system.

The figure shows that without a carbon tax, nearly all the hydrogen is reformed from natural gas, as it is today. At a cut of 75%, imports take a quarter of the market, and the dearest import step in use sets the price of hydrogen and with it the carbon tax: $\tau$ is the tax at which a reformer costs as much as that step (eq. (61)), the plateau of Section 7.4 again. In the decarbonised system, all but the dearest import step run at full capacity, electrolysis makes a fifth of the hydrogen, and the reformers make the rest from biomethane. Almost no hydrogen storage is built: The reformers do the seasonal balancing, running hardest from autumn to early spring, when electrolysis makes the least.

**What the hydrogen sector builds.** Figure 8.8 repeats fig. 8.1 for the hydrogen scenario and smart charging, the scenario it adds to, in the decarbonised system. The two are not like for like, since their budgets differ in tonnes.

[19] The imports come into Germany, as a step supply curve: a quantity at a price for each import route.

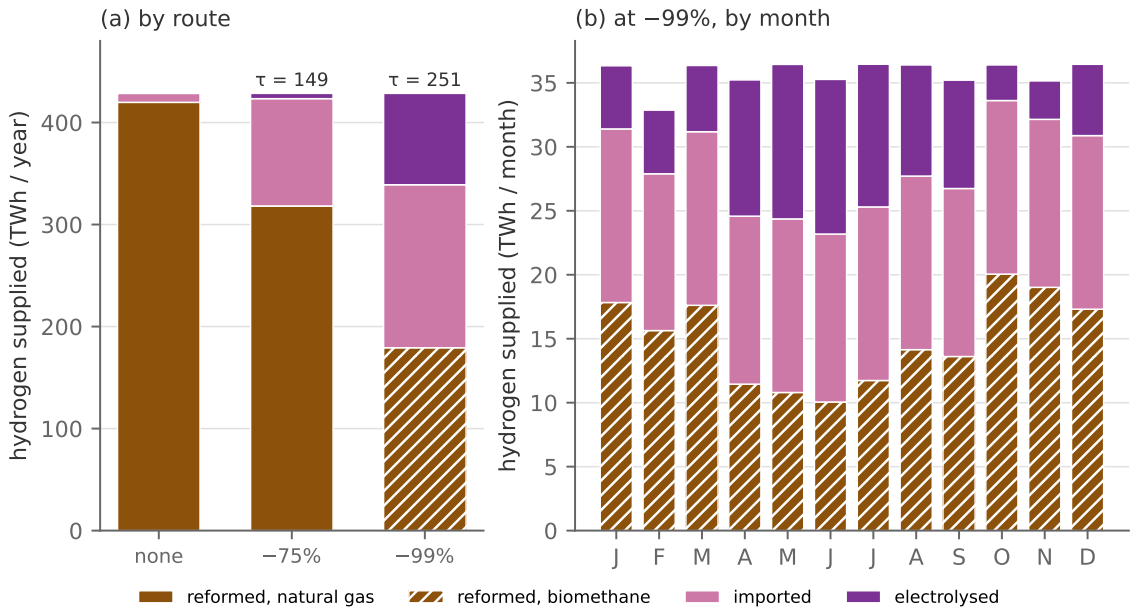



Figure 8.7: (a) Hydrogen supply by route (reformed from natural gas or from biomethane, imported, electrolysed) without a budget and at cuts of 75% and 99%, with the carbon tax $\tau$ in €/t above each bar. (b) The same at a cut of 99%, month by month, summed over the zones.

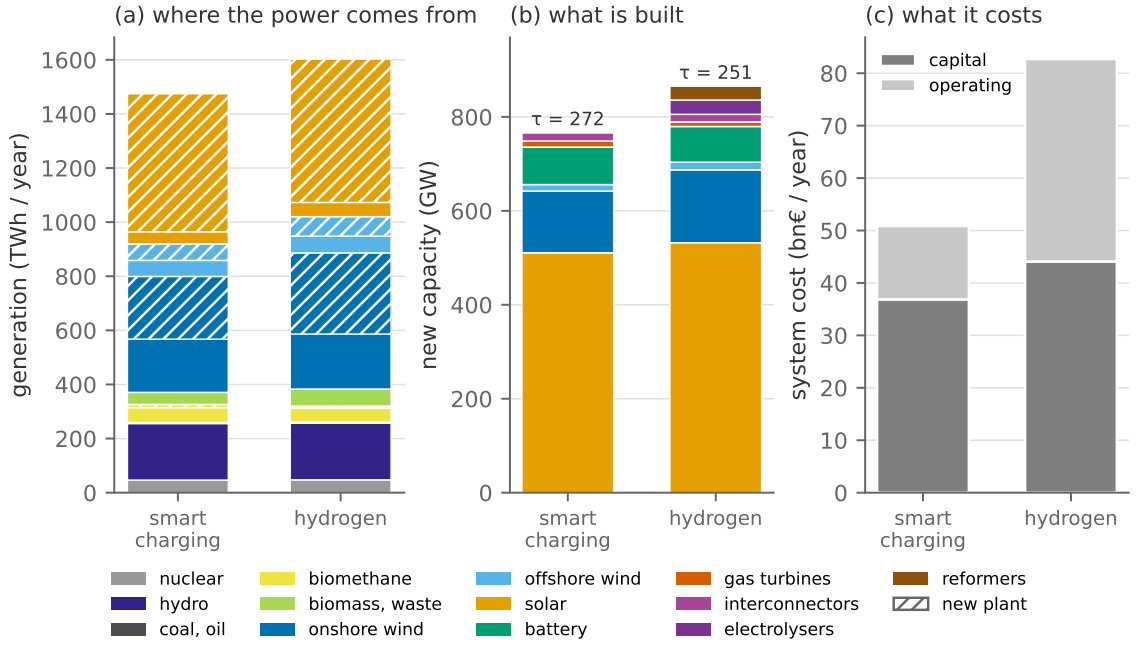



Figure 8.8: Smart charging and the hydrogen scenario at a cut of 99%, read as fig. 8.1: (a) generation by source; (b) new capacity by technology, the reformers in GW of hydrogen and everything else in GW of electricity; (c) the annual system cost.

The figure shows that the electrolysers add about a tenth to the demand for electricity, and that the system meets it mostly with more onshore wind. Electrolysers are built at scale only from a cut of 90%, mostly in Germany, where the demand is; they run about 4,200 hours a year and earn exactly their fixed cost (Result 6). The system cost rises by more than half, because the system now serves a whole sector more.

**Electrolysers as a flexible load.** Figure 8.9 compares the prices of the two scenarios.

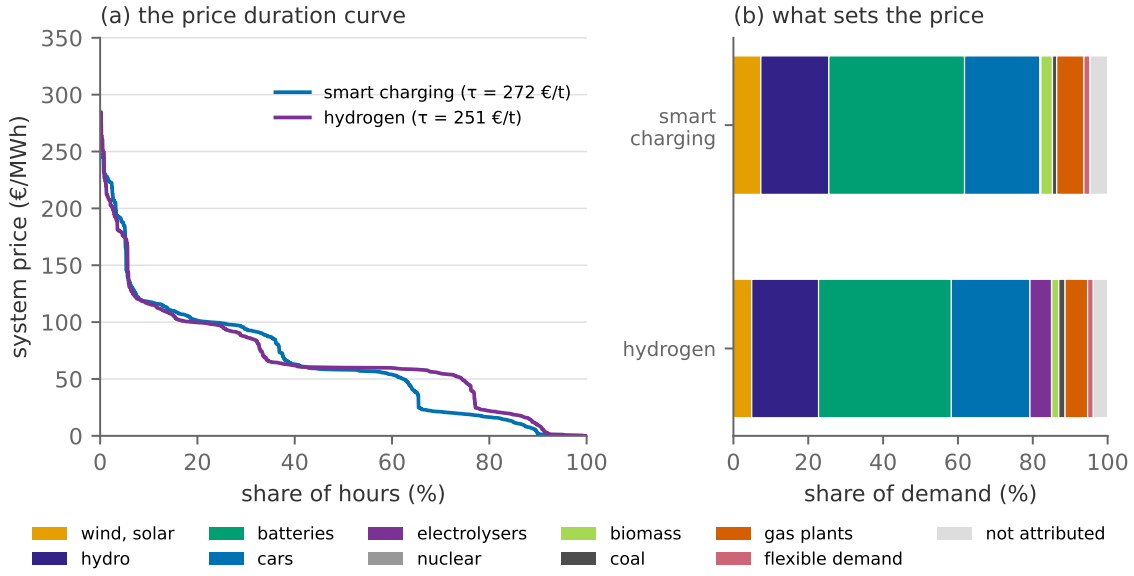



Figure 8.9: Smart charging and the hydrogen scenario at a cut of 99%, read as fig. 8.2: (a) the system's price duration curve; (b) what sets the price, as shares of the year's demand.

The figure shows that the electrolysers fill the bottom of the price duration curve: There are fewer hours near zero, and the share of wind and solar output that is curtailed halves. In the hours between cheap and dear, the electrolysers are the marginal consumer and set the electricity price at the value of the hydrogen they make, which shows in the curve as a long step at that value, running on well after the price without them has fallen away. That is why the average price rises even as the cheap hours fill: A new load, however flexible, fills the cheap tail of the price distribution, but it does not empty the expensive one (the qualification of Section 5.2 at scale).

**Hydrogen as electricity storage: no.** No solve builds a hydrogen turbine, at any budget. The hydrogen turbine is the same machine as the open-cycle gas turbine and costs the same to build, so the choice between them is a choice of fuel: A hydrogen turbine is worth building only if hydrogen is the cheaper fuel in the hours when power is dear enough to run a turbine (Results 1 and 6), and in no zone and at no budget is it. Hydrogen is worth its price to industry. Where the reformers set that price, as in Germany, it is the cost of the gas they burn grossed up for the losses of reforming, so a gas turbine burns the same gas more cheaply itself. Where electrolysers supply the hydrogen, as in Denmark, it is cheap only when power is cheap, and then no turbine runs. Without the industrial customer, the electrolyser would run only to fill the storage for the turbine's

few hours, and its capital would be spread over the few MWh the turbine sells. Figure 8.10 repeats fig. 8.3 for the two scenarios.

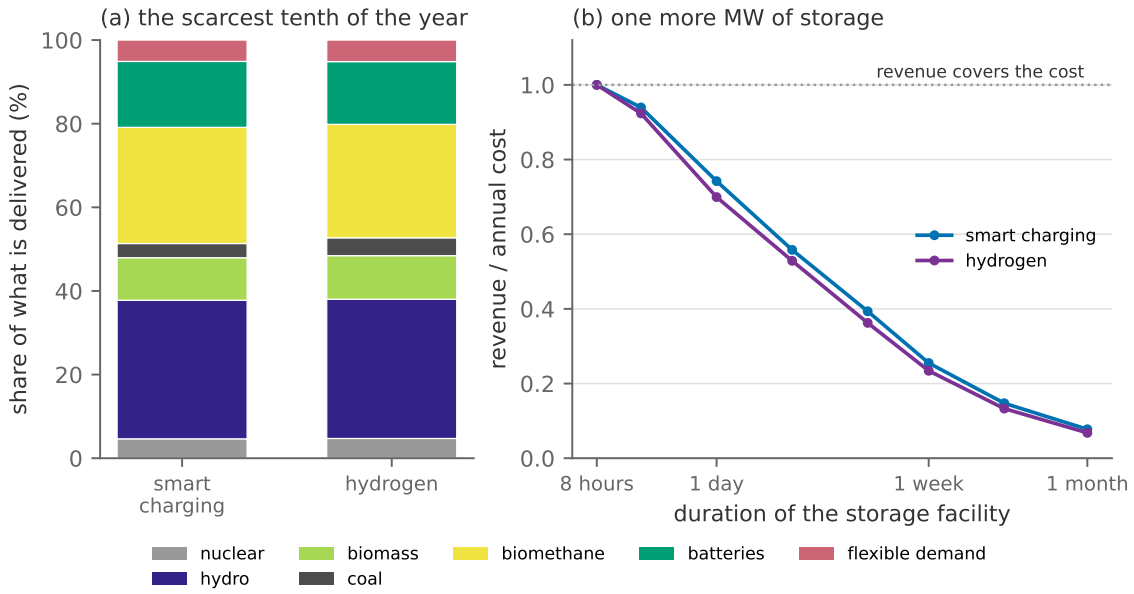



Figure 8.10: Smart charging and the hydrogen scenario at a cut of 99%, read as fig. 8.3: (a) what covers the tenth of the year with the highest residual demand; (b) what one more MW of storage of each duration would earn over its annual cost.

The figure shows that no hydrogen is burnt in the scarcest hours, which are covered as with smart charging alone; the electrolysers step aside instead, running at a small fraction of their average. Nor does a hydrogen sector make long storage pay, because the electrolysers buy in the cheap hours that storage would have bought in.

Victoria et al. (2019) divide the work of storage between batteries, which follow solar's daily cycle, and hydrogen, which smooths the variability of wind over days and weeks. We find the batteries' half of that division but not the hydrogen half: The seasonal work is done by reformers burning biomethane, a fuel that can be bought in any month at the same price and so is a seasonal store that costs nothing to hold. Nor does the scenario have the hydrogen network that Neumann, Zeyen, et al. (2023) find worth a few per cent of the system cost.

## 8.5 The four scenarios side by side

Figure 8.11 sets the four scenarios side by side at every budget.

The figure shows that every scenario passes the same three margins. The first cuts are switches between existing plants and cost a few tens of euros per tonne. Then $\tau$ climbs to the switch price and stays there while the gas plants switch to biomethane. Beyond this plateau, $\tau$ is set by the new gas turbines that burn biomethane in a few winter weeks to push out the last coal, and that must earn their capital in those weeks. Flexibility lowers the system cost at every budget, and the carbon tax wherever the marginal tonne is not a fuel switch. The hydrogen scenario costs more at every budget, and on the way down its carbon tax is set by a step of hydrogen imports.

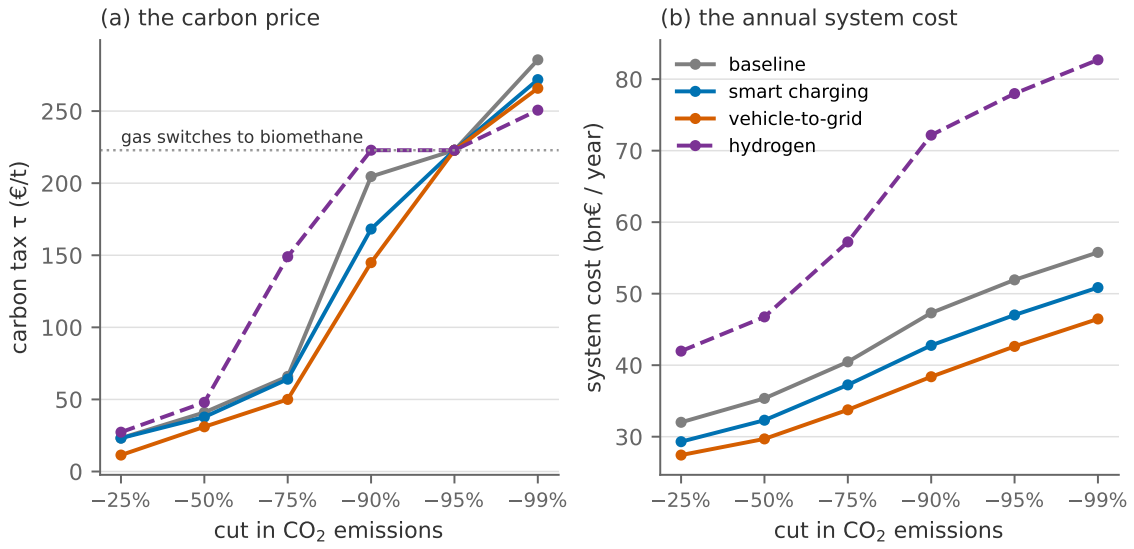



Figure 8.11: The four scenarios at six budgets, as cuts against each scenario's unconstrained emissions: (a) the carbon tax $\tau$, with the price at which a gas plant switches to biomethane dotted; (b) the annual system cost. The hydrogen scenario is dashed: It is a larger system with a budget of its own.

**Which way does it bias the answer?** *Today's fleet at no cost, biomethane without scarcity, a cheap demand block everywhere, no hydrogen network, cars that do not wear, and a year in segments.*

- *Today's fleet, kept at no cost.* **Understates** what must be built and what it costs.
- *Biomethane at one price all year, its ceiling never binding.* A free seasonal store that caps the cost of abating gas at the switch price: **understates** the deep-end carbon price and the value of hydrogen storage.
- *A cheap demand block in every zone, with no limit on how often it gives way.* **Overstates** how long demand can step back, and so **understates** the deep-end carbon price.
- *No hydrogen network, and imports free of the carbon tax.* **Understates** the electrolytic route and the cost of decarbonising hydrogen.
- *Half the cars' batteries on offer, a low morning floor, no wear.* **Overstates** the fleet as a battery.
- *The year in segments.* Segments average away swings that batteries earn on: **understates** the stationary battery and, with vehicle-to-grid, **overstates** how much of the storing the cars do.

**From duals to markets.**

| Constraint | Dual | What it measures | Market counterpart |
|---|---|---|---|
| interconnector investment optimality (47) | — | congestion rent = annual fixed cost of a wire | a merchant cable's break-even; regulated cost recovery |
| interconnector ceiling (45b) | $\overline{\beta}_k$ | value of one more MW of *permitted* transmission | the option value of an interconnector not yet permitted |
| car batteries' law of motion (53b) | $\theta^{EV}_{z,h}$ | value of energy in the cars' batteries | what a smart-charging tariff or an aggregator can pay for flexibility |
| hydrogen market clearing (57a) | $\lambda^{H_2}_{z,h}$ | marginal cost of a MWh of hydrogen | a hydrogen contract price, absent a hydrogen market |

To sum up: In a European brownfield, the first three quarters of the cut in emissions come cheaply, with wind, solar, 8-hour batteries and the existing biomass plants, and the last quarter is expensive. It is about the hours with little wind and sun, and every margin that can cover them runs into a limit of its own: Storage pays only as a daily device, the cheap demand block is small, and the rest falls to biomethane in the gas plants and to turbines that run a few winter weeks, whose capital the carbon tax pays. Electric vehicles take over much of the batteries' daily work and make every target cheaper, but they do not carry energy across weeks. A hydrogen sector adds a large flexible load that fills the cheap hours, and it never burns hydrogen back, because in the hours when power is dear, industry values the hydrogen above the biomethane the same turbine could burn. In every case the carbon price could be traced to one margin: a switch between two plants or two fuels, a step in a supply curve, or the capital of a machine that runs a few hours a year. Section 9 re-solves the model of Section 7 in other weather years and under other cost projections, and discusses the limits of both sections.

# 9 Uncertainty and the limits of the model

The results of Sections 7 and 8 come from a planner who knows the future and from markets that deliver the planner's prices. This section first shows how much those results vary with what the planner is assumed to know, and then discusses the limits of the model.

> **Reading alongside this section.** This section requires no supplementary reading. Gøtske et al. (2024) repeat the weather-year experiment for a net-zero European system over six decades of weather, and Reichenberg et al. (2023) treat the same variability as an investor's revenue risk. Bushnell and Smith (2025) discuss how to represent uncertainty when energy models evaluate real climate policy, and Joskow (2008) sets out the missing-money problem. O'Neill et al. (2005) show how prices can support an efficient schedule despite start-up costs, and Neumann and Brown (2021) map the many

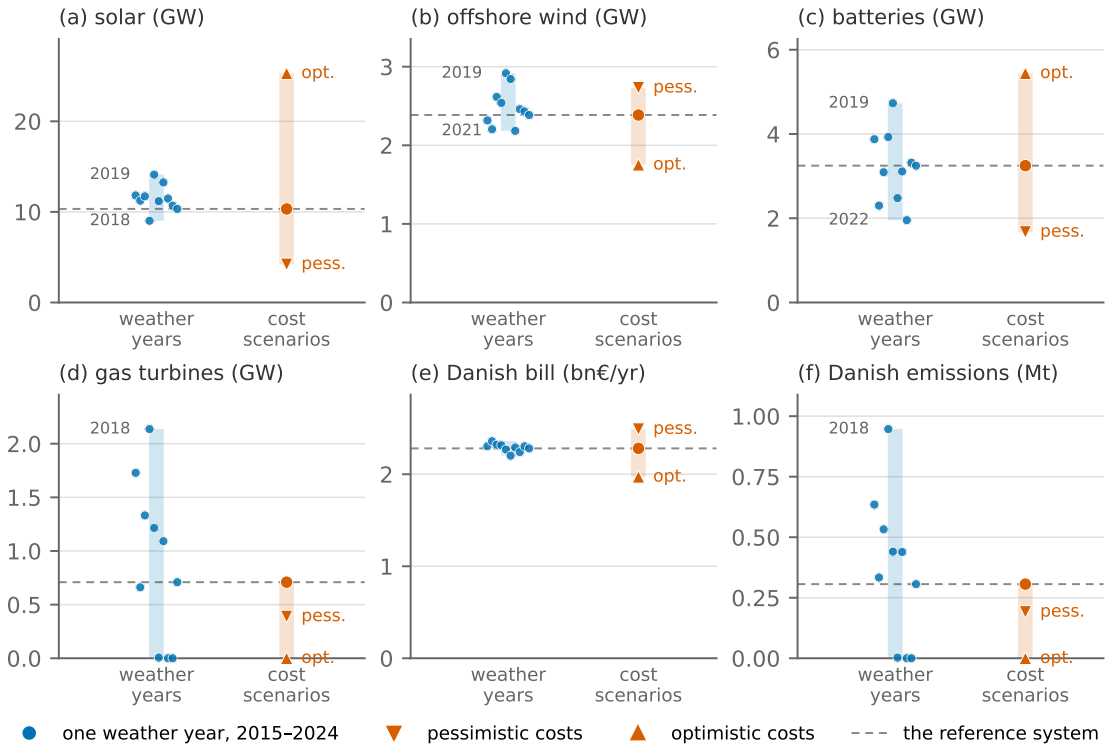



Figure 9.1: Six results of the model of Section 7 across the weather years (one dot per year) and the cost scenarios. The dashed line is the reference system; the Danish bill is net of trade.

> European systems that cost almost as much as the optimum. Appendix C lists what the models leave out, with the direction of each omission's error.

## 9.1 Variability

Two inputs to the results of Sections 7 and 8 are highly uncertain: the weather, since both sections built their systems for the wind and sun of a single year, 2024, and the technology costs, which are projections for 2050, decades ahead.

We explore what this means by solving the model of Section 7 again for the 10 weather years from 2015 to 2024 (Appendix B.6) and for two cost scenarios, with everything else as in the reference case. In the *pessimistic* scenario, investment costs fall 50% less from 2025 to 2050 than the database projects, and in the *optimistic* one 50% more. This is not uncertainty, but **variability**: Every run is optimised knowing the weather and the costs it will face.

Figure 9.1 shows the range of six results across the weather years and the cost scenarios:

i. Much of the system does not move. Every run builds onshore wind up to its potential (not drawn) as well as solar and batteries, and no run builds a combined-cycle plant, nuclear or a biomethane turbine.

ii. The cost scenarios move solar most, far more than the weather does, and the batteries with it, since the database projects steep falls in their costs and modest ones in wind's. Cheaper solar and batteries replace offshore wind.

iii. The weather moves the gas turbines most: Some years need none and 2018 the most, since the turbines are paid in scarcity spells (Figure 7.4) whose length and depth depend on the year. Danish emissions follow, the caveat of Result 2: A carbon tax fixes the price of a tonne and leaves the number of tonnes to the weather.

iv. The Danish bill moves much less than the fleets (by 6.9% across the weather years and 23.1% across the cost scenarios): Very different fleets are built for different years at almost the same cost.

A single run therefore reports more precision than the model has: The broad answer, a Danish system of wind, solar and batteries with gas only as a small fleet of turbines, holds in every run, but the details are ranges.

> **Which way does it bias the answer?** *Two constructed cost scenarios, and ten weather years that differ only in timing.*
>
> - *Cost scenarios, not a distribution.* Their width is a choice, and past projections have repeatedly overestimated what solar, wind and batteries would cost (Way et al. 2022): **understates** cost uncertainty, towards cheaper machines.
> - *The years differ only in timing.* Annual wind and solar output, reservoir inflow and demand are the same in every year, so no year is still, dry or cold throughout: **understates** the weather's range.
> - *Ten years.* Too few to contain the rare spell a system must also survive: **understates** the backup a robust system needs.

## 9.2 Discussion and limits

**Decisions under uncertainty.** Every system in Figure 9.1 was optimised with perfect foresight, and we never evaluated what happens when expectations are wrong. Yet capacity is built years before anyone knows the weather it will meet, and the two kinds of error cost very different things: A system built for a mild year that meets a year like 2018 is short of turbines in its calm spells and must import or ration demand, while a system built for 2018 that meets a mild year has only paid for idle turbines. The costs, too, are revealed only over time, and an investment that cannot be undone can be worth delaying until more is known; a model that already knows the costs puts the value of that option at zero (Dixit and Pindyck 1994). The operation has foresight as well, since every storage facility knows the year's prices in advance. This matters little for the battery of Section 4, which loses only about a sixth of what it saves the system when operated one day at a time (8.20 of 51.2 M€ a year), but more for a reservoir that stores water for months.

Formally, let $W^{\text{inv}}(\bar{q}, \omega)$ be the welfare of eq. (40a) that a fleet $\bar{q}$ achieves in weather year $\omega$. The sweep solved $\max_{\bar{q}} W^{\text{inv}}(\bar{q}, \omega)$ for each year, while a planner who must build before the year

is known maximises expected welfare, and

$$\mathbb{E}_\omega \left[ \max_{\bar{q}} W^{\mathrm{inv}}(\bar{q}, \omega) \right] \;\geq\; \max_{\bar{q}} \mathbb{E}_\omega \left[ W^{\mathrm{inv}}(\bar{q}, \omega) \right],$$

since on the left every year gets its own fleet. The sweep is therefore optimistic on average, by the value of knowing the weather in advance, and the fleet that solves the right-hand side is in general none of the fleets it found. Its investment condition is Result 6 in expectation, $\mathbb{E}_\omega \left[ \sum_h \gamma_{g,h}\, \overline{\mu}_{g,h} \right] = F_g$: A gas turbine is paid in the bad years and stands idle in the good ones. Gøtske et al. (2024) run this evaluation for a net-zero European system and find that the systems built for years with compound extremes are the robust ones.

**Market and policy imperfections.** The model's prices are market prices only if every firm takes them as given, the premise of Result 1 and of the free entry behind Result 6. The premise is weakest in the hours that pay for capacity: Demand responds little, almost every plant is running, and a firm that owns several plants can raise the price on all of them by holding one back. Borenstein et al. (2002) attribute more than half of the rise in California's wholesale power costs in the summer of 2000 to such market power, and a hydro producer can do the same by holding water back from the expensive hours. The planner is also indifferent to risk, while an investor bears the variability of Section 9.1 as revenue risk and, unable to insure it fully, demands a higher return (Reichenberg et al. 2023).

The gas turbines of Section 7 earn more than half of their annual margin in only 21 hours (Figure 7.3), the hours in which prices spike and a well-intended price cap bites. Capped below the scarcity level, the turbines no longer recover their fixed cost, fewer are built than the optimum, and in the next calm spell the shortfall is rationed by the system operator rather than by price: the **missing money** problem (Joskow 2008). Investors who expect intervention whenever prices spike discount those rents even without a formal cap, and with reason: From December 2022 to June 2023, the EU capped the market revenues of wind, solar, nuclear and other inframarginal generators (Council of the European Union 2022). The problem grows as the system decarbonises, since the more of its cost is capital, the more must be recovered in such rents. Capacity mechanisms, which pay for availability directly, are the usual response; the model contains none.

**Non-convexities.** Every cost in the model is linear, but a real thermal plant also pays to start and, once started, cannot run below a minimum load. These costs make the problem non-convex and break the second half of Result 1: A plant that is marginal in a two-hour peak is paid its marginal cost but not its start-up cost, so there may be no price at which every plant chooses the efficient schedule. American markets therefore add side payments, called **uplift** (O'Neill et al. 2005), and European exchanges let a plant bid a block of hours that is accepted only as a whole. The model's prices are too smooth as a result: Real plants run through cheap hours to avoid a restart and need more than their marginal cost to start, so real prices fall lower in the troughs and rise higher in short peaks. Since storage and flexible demand earn that spread, the model

understates their value. The system of Section 7 is less exposed than today's: Its only thermal plants are open-cycle turbines that start within minutes, and with much wind and sun, how the year is represented matters more than how plants operate (Poncelet et al. 2016).

**Congestion inside a zone.** Section 6 showed that zonal prices ignore congestion inside a zone, which redispatch pays for after the market has cleared. In the investment models the same omission also decides where capacity goes: Each zone is a **copper plate**, so the planner builds wind and solar wherever they produce most and pays nothing to carry their power within the zone. The assumption is weakest for Germany, whose wind is in the north and most of whose consumption is in the south. The model therefore understates the cost of the grid and redispatch the system needs, and overstates the value of wind behind the bottleneck, Danish wind included.

Operators can also move such congestion to the border by offering the market less interconnector capacity. The Swedish operator did so until a Commission case, opened on a Danish complaint, led to Sweden being split into four bidding zones in 2011 (European Commission 2010), and the German operator set the capacity for imports from western Denmark to zero when German wind output was high, until it committed to a guaranteed minimum in 2018 (European Commission 2018). EU rules now require at least 70% of transmission capacity to be offered for trade between zones (European Parliament and Council 2019, Art. 16); the model's interconnectors always offer all of it.

**Many systems at almost the same cost.** The model reports the cheapest system, but the cost changes little as the fleet moves away from it: At the optimum, every technology built below its cap earns exactly its fixed cost (Result 6), so a little more of one and a little less of another leaves the cost unchanged to a first approximation. This is why the fleets of Figure 9.1 move so much more than the bill. Neumann and Brown (2021) map the systems near the optimum of a fully renewable European model of the same family, using modelling to generate alternatives (DeCarolis 2011): Within half a per cent of the optimal cost the investments already range widely, and only wind, onshore or offshore, some hydrogen storage and a stronger grid are shared by every system within ten per cent. The model therefore says much about what such systems share and little about which of them to build, and the choice can rest on what it does not price, such as land use, local acceptance or who pays, at little cost. Real transitions do leave the optimum: The electricity system the United Kingdom built from 1990 to 2014 cost 9–23% more than the cheapest path a model finds for the same years (Trutnevyte 2016).

To sum up: Other weather years and cost scenarios leave the broad answer of Section 7 standing and turn its details into ranges, with the costs deciding how much solar and storage to build and the weather how many turbines to hold. Without perfect foresight and competitive markets, the capacity that pays only in bad years is the hardest to choose and to finance. Linear costs and copper-plate zones understate the value of flexibility and the cost of moving power, and since many systems cost almost as much as the cheapest, the model says most about what they share and least about the exact mix.

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# Appendices

## A Convex optimisation, duality and KKT: a refresher

The main text assumes you can set up a Lagrangian, write down first-order conditions, and interpret multipliers, including Karush-Kuhn-Tucker conditions for inequality constraints. This appendix provides a refresher on the techniques including a short cookbook for how to approach it. This is not a strictly rigorous presentation with fancy proofs; we focus on the basics and on giving you a tool that can be used. Note that to arrive at one simple cookbook for how to use it, we always write up the Lagrangian for a *maximization problem*. When a model is naturally written as a cost *minimisation*, we maximise minus the costs instead.

### A.1 Convex problems and linearity

In this note, all models belong to the class of so-called *linear programs* (LP). While I will spare you a presentation of the different types of optimisation problems, the LP deserves a short mention. First, LPs belong to a greater family of optimisation models called *convex problems* that are famous for being (relatively) easy to solve (both theoretically and numerically). A convex problem maximises a concave function (or minimises a convex one) over a convex set. The convex set roughly means that the space of possible solutions we search over is easy to navigate and the concave function means that gradients/derivatives will tell you in what direction the optimum lies.

The linear models in this note are special cases of the convex problems that are linear in objective, equality, and inequality constraints. A linear function is both concave and convex, so an LP is the friendliest corner of the convex world, and three properties are used repeatedly:

i. *Local is global.* A local optimum of a convex problem is a global optimum. When a solver reports an optimum, there is no better dispatch hiding elsewhere (up to ties).

ii. *First-order conditions are exact.* For a convex problem the KKT conditions below are not merely necessary at an optimum; they are also sufficient.

iii. *Solutions sit on corners.* A linear objective on a polyhedron attains its optimum at a vertex (roughly meaning a point where as many constraints bind as there are decision variables). This is why optimal dispatch is bang-bang (plants at capacity or at zero, one marginal plant in between) and why, in Section 7, an LP never builds a diversified portfolio for diversification's own sake.

## A.2 The standard form, and a recipe for the Lagrangian

You will typically meet optimisation problems written in a *standard form* that looks like this:

$$\begin{aligned}
\max_{x_1,\,x_2} \quad & f(x_1, x_2) && && \text{(48a)}\\
\text{s.t.} \quad & g(x_1, x_2) \le k_g && (\lambda_g) && \text{(48b)}\\
& h(x_1, x_2) = k_h && (\lambda_h) && \text{(48c)}\\
& x_1 \in [\underline{x}_1, \overline{x}_1] && (\underline{\theta}_1,\ \overline{\theta}_1) && \text{(48d)}\\
& x_2 \in [\underline{x}_2, \overline{x}_2] && (\underline{\theta}_2,\ \overline{\theta}_2) && \text{(48e)}
\end{aligned}$$

where $x_1, x_2$ are the decision variables, $f$ is the *objective*, $g$ is an *inequality constraint*, $h$ is an *equality constraint*, and $(\underline{x}_i, \overline{x}_i)$ are *domain constraints* (also inequality constraints, really, but so common that the recipe treats them separately). As in the main text, each constraint's multiplier is named in parentheses next to it.

The *Lagrangian* replaces each constraint by a linear penalty term:

$$\begin{aligned}
\mathcal{L} = \underbrace{f(x_1, x_2)}_{\text{objective}} &+ \underbrace{\lambda_h\Big(k_h - h(x_1, x_2)\Big)}_{\text{equality constraint}} + \underbrace{\lambda_g\Big(k_g - g(x_1, x_2)\Big)}_{\text{inequality constraint}}\\
&+ \underbrace{\underline{\theta}_1\Big(x_1 - \underline{x}_1\Big) + \overline{\theta}_1\Big(\overline{x}_1 - x_1\Big)}_{\text{domain constraint I}} + \underbrace{\underline{\theta}_2\Big(x_2 - \underline{x}_2\Big) + \overline{\theta}_2\Big(\overline{x}_2 - x_2\Big)}_{\text{domain constraint II}},
\end{aligned} \tag{49}$$

where $\lambda_h, \lambda_g, \underline{\theta}_1, \overline{\theta}_1, \underline{\theta}_2, \overline{\theta}_2$ are called *shadow variables* (equivalently: dual variables, or Lagrange multipliers); we return to their interpretation below. If there were more constraints of any type, the Lagrangian would simply extend additively, exactly as it does for the two domain constraints.

Three remarks on getting the terms right, one per constraint type. This is where sign mistakes are made, so the rules are worth committing to memory:

- *Equality constraints:* $\lambda_h$ can be positive or negative, and the constraint may be added either way round:

$$\text{Do this: } \lambda_h \left(k_h - h(x_1, x_2)\right), \qquad \text{or this: } \tilde{\lambda}_h \left(h(x_1, x_2) - k_h\right).$$

  Both give the same solution except for the sign of the multiplier: $\lambda_h = -\tilde{\lambda}_h$.

- *Inequality constraints:* write the term so that it **is non-negative when the constraint holds**. With the constraint $g(x_1, x_2) \leq k_g$:

$$\text{Do this: } \lambda_g \underbrace{\left(k_g - g(x_1, x_2)\right)}_{\geq 0}, \qquad \text{and } \underline{\text{not}}\text{: } \tilde{\lambda}_g \underbrace{\left(g(x_1, x_2) - k_g\right)}_{\leq 0}.$$

  The reason the direction matters is that the sign of $\lambda_g$ is itself a condition for optimality: the KKT conditions below require $\lambda_g \geq 0$, and that requirement only means what it should if the term is written this way round. This changes if the problem is a *minimisation* instead of *maximisation* problem.

- *Domain constraints:* $x_1 \in [\underline{x}_1, \overline{x}_1]$ is just the pair of inequality constraints $x_1 \geq \underline{x}_1$ and $x_1 \leq \overline{x}_1$, so the same rule applies: We add both terms such that they are non-negative:

$$\begin{aligned} \text{Do this:} \quad & \underline{\theta}_1 \overbrace{(x_1 - \underline{x}_1)}^{\geq 0} + \overline{\theta}_1 \overbrace{(\overline{x}_1 - x_1)}^{\geq 0} \\ \text{and } \underline{\text{not}}\text{:} \quad & \tilde{\underline{\theta}}_1 \underbrace{(\underline{x}_1 - x_1)}_{\leq 0} + \tilde{\overline{\theta}}_1 \underbrace{(x_1 - \overline{x}_1)}_{\leq 0} \end{aligned}$$

## A.3 Optimality conditions

Now that we have set up the Lagrangian, identifying an optimum $(x_1^*, x_2^*)$ requires two sets of conditions:

**(1) First-order conditions for optimality.** All derivatives of $\mathcal{L}$ with respect to the decision variables equal zero at the optimum:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x_1} = \quad & \frac{\partial f}{\partial x_1} - \lambda_h \frac{\partial h}{\partial x_1} - \lambda_g \frac{\partial g}{\partial x_1} + \underline{\theta}_1 - \overline{\theta}_1 \quad =: 0 \\ \frac{\partial \mathcal{L}}{\partial x_2} = \quad & \frac{\partial f}{\partial x_2} - \lambda_h \frac{\partial h}{\partial x_2} - \lambda_g \frac{\partial g}{\partial x_2} + \underline{\theta}_2 - \overline{\theta}_2 \quad =: 0 \end{aligned}$$

**(2) Complementary slackness conditions.** For every inequality constraint, including the domain constraints, three things hold at once:

$$\begin{aligned} \lambda_g \Big(k_g - g(x_1, x_2)\Big) &= 0, & k_g - g(x_1, x_2) &\geq 0, & \lambda_g &\geq 0 \\ \underline{\theta}_i \Big(x_i - \underline{x}_i\Big) &= 0, & x_i - \underline{x}_i &\geq 0, & \underline{\theta}_i &\geq 0 \end{aligned}$$

$$\overline{\theta}_i\Big(\overline{x}_i - x_i\Big) = 0, \qquad \overline{x}_i - x_i \geq 0, \qquad \overline{\theta}_i \geq 0,$$

for $i = 1, 2$. Each row is one inequality constraint, and the three columns are, in order, the complementary-slackness condition proper, *primal feasibility* (the constraint itself holds) and *dual feasibility* (its multiplier is non-negative).

The interpretation of these conditions, and of the shadow variables in them, is generally really important for understanding the models. We read the set as follows:

- For every inequality constraint, at least one of the following two must hold: (i) the shadow value of the constraint is zero (e.g. $\lambda_g = 0$), or (ii) the constraint holds with equality, e.g. $k_g - g(x_1, x_2) = 0$. Both may hold at once; neither may fail.
- If a constraint holds with equality, $k_g - g(x_1, x_2) = 0$, we say that the constraint is *active* (or *binding*).
- If it holds strictly, $k_g - g(x_1, x_2) > 0$, we say that the constraint is *inactive* (or *slack*), and its shadow value is then zero.

Complementary slackness is the condition that carries the most economics per symbol: *only active constraints can have positive prices.* A capacity limit that does not bind is worth nothing at the margin ($\lambda_g = 0$); a capacity limit with a positive price must be exhausted. Scarcity and value are the same fact seen from the primal and the dual side. A useful corollary is that the lower- and upper-bound multipliers of the same variable can never both be positive, since the two bounds naturally cannot both bind at once (unless $\underline{x}_i = \overline{x}_i$).

One caveat worth a sentence: for general nonlinear problems the KKT conditions require a regularity condition on the constraints (a "constraint qualification") to be necessary. For linear constraints (every model in this note) no such condition is needed.

## A.4 Interpreting the multipliers: derivatives of the value function

Let $f^*$ be the optimal value of (48). This is, naturally, a function of the parameters that define how *tight* each constraint is: $(k_h, k_g, \underline{x}_i, \overline{x}_i)$. This is useful to interpret what the multipliers measure: **the change in the optimised objective from a marginal relaxation of its own constraint**. Wherever $f^*$ is differentiable, it is straightforward to show that[20]

$$\lambda_h = \frac{\partial f^*}{\partial k_h} \lessgtr 0, \qquad \lambda_g = \frac{\partial f^*}{\partial k_g} \geq 0, \qquad -\underline{\theta}_i = \frac{\partial f^*}{\partial \underline{x}_i} \leq 0, \qquad \overline{\theta}_i = \frac{\partial f^*}{\partial \overline{x}_i} \geq 0. \tag{50}$$

This envelope property is the licence for every sentence of the form "the dual of market clearing is the marginal cost of serving one more MWh". Taking the constraint types in turn:

- *Equality constraints:* $\lambda_h$ is the gain in the objective from increasing $k_h$ marginally, and it can go either way. If $f$ is a utility function and $k_h$ represents income, we'd have $\lambda_h > 0$

[20] You can show this by differentiating $\mathcal{L}$ with respect to the relevant parameter, using the fact that at the optimum the Lagrangian coincides with the objective, $\mathcal{L} = f^*$.

measuring specifically the marginal utility of income.[21]

- *Inequality constraints:* the two cases of complementary slackness are exactly the two cases of the interpretation. If the constraint is inactive, a marginal relaxation changes nothing that was binding, so $\lambda_g = 0$. If it is active, relaxing $k_g$ enlarges the feasible set in a direction the optimiser actually wants, and the objective rises at rate $\lambda_g > 0$.
- *Domain constraints:* raising $\overline{x}_1$ hands the optimiser more options, which can never hurt ($\overline{\theta}_1 \geq 0$) and helps only if the bound was the thing stopping it ($\overline{\theta}_1 > 0$ exactly when active). Raising $\underline{x}_1$ removes options, which can never help ($-\underline{\theta}_1 \leq 0$) and hurts only if the optimiser is pushed off its preferred choice.

Two further points matter specifically for the LPs of this note.

**Solvers may report the other sign.** Once again, whether or not we report the problem as minimisation or maximisation problem matters. If the numerical solvers work on the minimisation form and report duals as derivatives of that objective instead, the signs flip: the dual of a $\leq$-constraint in a cost minimisation comes back non-positive (relaxing a constraint cannot raise minimised cost). The course code flips the sign once, so that what you see matches the non-negative multipliers of the Lagrangian as written here. If you ever wire a solver up yourself and the rents come out negative, it is important to check this convention.

**At a kink, the multiplier is not unique.** The value function of an LP is piecewise linear in the constraint levels, and at a kink where the optimal basis changes, i.e. where the identity of the marginal plant or the active abatement margin switches, the derivative in eq. (50) does not exist: left and right derivatives differ, and any multiplier between them is valid. This is not a technical nuisance to be waved off; it is visible in the results. Figure 2.4's step function is the derivative of a piecewise-linear value function, flats and jumps included, and on any of its vertical segments the permit price is indeterminate between the segment's endpoints. When a solver reports one number at such a point, it has picked an endpoint of the valid interval, not discovered a unique price.

## A.5 Duality, and the dual as the price-setting problem

Every optimisation problem comes with a twin. The problem we started from, (48), is called the *primal*: it chooses quantities $(x_1, x_2)$ and takes the constraint levels ($k_g$, $k_h$ and the bounds) as given. Its *dual* is a problem in the multipliers instead: it chooses a price for each constraint and takes the quantities as given. The two are built from the same ingredients and they swap roles completely, so that each constraint of the primal supplies one variable of the dual, and each variable of the primal supplies one constraint of the dual. For the models in this note the twins also have the same solution, and that is what lets us move between a planner choosing quantities and a market setting prices without changing the model at all.

[21] Note that which way round the term was written (Appendix A.2) matters: with the $\tilde{\lambda}_h\,(h - k_h)$ version, the interpretation flips to the gain from *decreasing* $k_h$.

Where does the dual come from? From asking how good the primal optimum $f^*$ could possibly be, *without* solving for it. A lower bound is easy: any feasible $x$ we can think of provides one, since the optimum is at least as good as anything feasible. An upper bound is the hard direction, because it has to hold for every feasible point at once, including all the ones we have not looked at. This is what the Lagrangian gives us. Fix the multipliers, and let the planner choose freely, paying the penalties instead of respecting the constraints:

$$D(\lambda, \theta) = \max_{x} \; \mathcal{L}(x, \lambda, \theta). \tag{51}$$

For any $x$ that is actually feasible, every penalty term in $\mathcal{L}$ is non-negative, so $\mathcal{L}(x) \geq f(x)$: the penalised planner can always claim at least what the constrained one achieves. And since $D$ maximises over *all* $x$, feasible or not, it is at least as large again. Hence $D(\lambda, \theta)$ is an *upper* bound on the optimum of (48) for any sign-feasible multipliers: each set of prices certifies a ceiling on what the constrained planner can achieve, and it does so without anybody having solved (48). This is *weak duality*. Some of those ceilings are useless (set every multiplier to zero, and $D(0, 0)$ is just the value of the problem with all constraints thrown away), so the *dual problem* is the search for the tightest one:

$$\min_{\lambda,\, \theta} \; D(\lambda, \theta), \qquad \lambda_g,\, \underline{\theta}_i,\, \overline{\theta}_i \geq 0. \tag{52}$$

For linear programs the bound closes completely: primal and dual optima are *equal* (*strong duality*), provided the primal is feasible and bounded. Carrying out the inner maximisation for the dispatch LP of Section 2 is a one-line computation, because $\mathcal{L}$ is linear in $q$: unless the coefficient on every $q_g$ is exactly zero, the maximum is infinite, so the dual's constraints are precisely those zero-coefficient conditions — the first-order conditions, reappearing as the constraints of the price-setting problem. That computation turns (52) into the dual program (6a)–(6b) quoted there. Note also that the prices solving the dual are exactly the multipliers of Appendix A.3: the dual is not a second computation to carry out, but the one we have already done, seen from the other side.

The economic reading of the dual pair: the primal chooses *quantities*, with prices emerging as multipliers; the dual chooses *prices*, subject to a no-arbitrage condition on quantities (no generator can be made to earn more than its cost structure allows), with quantities emerging as multipliers. Strong duality is the statement that the quantity-planner and the price-setter agree on the value of the system — the LP restatement of the welfare theorems that Section 2.3 uses to move between planner and market without changing the model.

## A.6 The technique, end to end

Every "derive and interpret" exercise in this course has the same shape, and it is the shape the main text follows silently each time:

i. Write the objective and constraints in the standard form (48) — as a maximisation — and name a multiplier per constraint.

ii. Form the Lagrangian, writing every inequality term so that it is non-negative when its constraint holds (Appendix A.2).

iii. Differentiate with respect to each decision variable and set the derivatives to zero — the first-order conditions for optimality — remembering the multiplier of a non-negativity constraint if the variable has one.

iv. List complementary slackness for every inequality.

v. Solve the resulting case distinction — which constraints bind is the economic content, and there are usually only a few coherent patterns.

vi. Interpret each multiplier as the price of its constraint, using eq. (50).

One clarification from experience: steps 1–4 are mechanical, but step 5 is not, and it is where the training lies. With inequality constraints there is no single formula — solving usually means *guessing* which constraints are active, solving the easy system that remains, and checking the guess against feasibility and the sign conditions. A failed check is not failure; it is information about which guess to try next, and the example below runs exactly that loop. Sometimes the case distinction is too rich to solve analytically at all; the conditions are still worth deriving, because interpreting them — or handing the problem to a numerical solver — is the payoff. Applied to (3a)–(3c), the recipe yields the merit order in half a page, as Section 2.3 shows. The exam expects the technique at exactly this level: set up, derive, interpret — not reprove strong duality.

## A.7 A worked example: boots and Teslas

With the technical material in place, let us run the recipe once on a small consumer problem where every number can be checked by hand.
The problem: As an example, consider the "simple" consumer utility maximization problem:

- $x_1, x_2$ are two consumer goods, e.g. boots and Teslas. Personally, I enjoy all boots just the same - no matter the number, but the more Teslas I get, the less it means to me. So, my utility function is:

$$f(x_1, x_2) = x_1 + 3 \cdot \ln(x_2)\,.$$

- I have 3€ available to buy boots at 1€ a piece and Teslas at 1/2€ (good price on Teslas, yes). I also have a horrendous credit score, so I cannot establish any debt. So, I have a budget:

$$\underbrace{x_1 + 0.5 \cdot x_2}_{h(x_1,x_2)} = \underbrace{3}_{k_h}$$

- I have promised Marie Kondo never to own more than 5 items, so:

$$\underbrace{x_1 + x_2}_{g(x_1,x_2)} \leq \underbrace{5}_{k_g}.$$

- It is silly to have "negative" amounts of boots, and I only have two feet, so I also face the domain constraint:

$$x_1 \in [0, 2].$$

Figure A.1 draws the three constraints and the set of bundles that satisfies all of them at once. Note what the budget does: because it must be spent *exactly*, the choice is confined to the edge of that set where the budget line runs — the thick stretch between the item cap and the two feet. That is worth holding on to, because it is what makes the answer a corner.

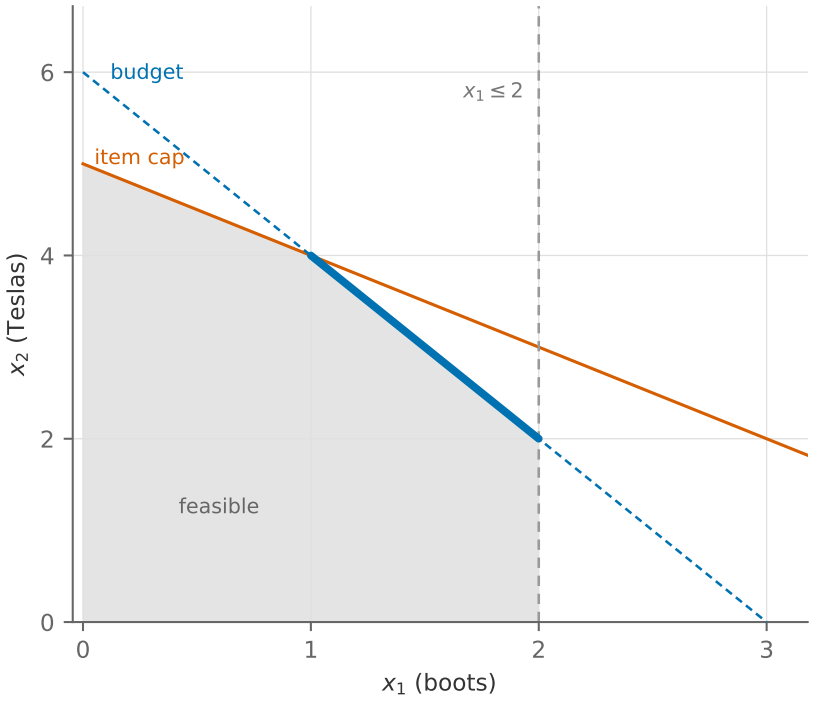



Figure A.1: The constraints of the example, and the feasible set they leave. Shaded is every bundle that is affordable and allowed: at most five items, at most two pairs of boots, nothing negative. Its upper-right edge is built from the two constraints that bind somewhere — the item cap from $(0, 5)$ to $(1, 4)$, then the budget from $(1, 4)$ to $(2, 2)$, where the two feet stop it. Since the budget holds with equality, the bundles actually on offer are that thick stretch of budget line.

The Lagrangian: following the recipe:

$$\mathcal{L} = \underbrace{x_1 + 3 \cdot \ln(x_2)}_{\equiv f(x_1,x_2)} + \lambda_h \Big( \underbrace{3 - (x_1 + 0.5 \cdot x_2)}_{\equiv k_h - h(x_1,x_2)} \Big) + \lambda_g \Big( \underbrace{5 - (x_1 + x_2)}_{\equiv k_g - g(x_1,x_2)} \Big) + \underline{\theta}_1 \Big( \underbrace{x_1 - 0}_{\equiv x_1 - \underline{x}_1} \Big) + \overline{\theta}_1 \Big( \underbrace{2 - x_1}_{\equiv \overline{x}_1 - x_1} \Big).$$

The KKT conditions: the first-order conditions for optimality,

$$\frac{\partial \mathcal{L}}{\partial x_1} = 1 - \lambda_h - \lambda_g + \underline{\theta}_1 - \overline{\theta}_1 := 0$$
$$\frac{\partial \mathcal{L}}{\partial x_2} = \frac{3}{x_2} - 0.5 \cdot \lambda_h - \lambda_g := 0,$$

and the complementary slackness conditions:

$$\begin{aligned} \lambda_g \left(5 - (x_1 + x_2)\right) &= 0, & 5 - (x_1 + x_2) &\geq 0, & \lambda_g &\geq 0 \\ \underline{\theta}_1 x_1 &= 0, & x_1 &\geq 0, & \underline{\theta}_1 &\geq 0 \\ \overline{\theta}_1 \left(2 - x_1\right) &= 0, & 2 - x_1 &\geq 0, & \overline{\theta}_1 &\geq 0. \end{aligned}$$

Solve and interpret: this is the guess-and-check loop promised in Appendix A.6.

*First guess: no constraint binds except the budget.* Set $\lambda_g = \underline{\theta}_1 = \overline{\theta}_1 = 0$. The remaining system,

$$\begin{aligned} 1 - \lambda_h &= 0 \\ \frac{3}{x_2} - 0.5\lambda_h &= 0 \\ x_1 + 0.5 \cdot x_2 &= 3, \end{aligned}$$

solves to $\lambda_h = 1$ and $(x_1, x_2) = (0, 6)$: spend everything on Teslas. But check the guess: $x_1 + x_2 = 6 > 5$ violates the Marie Kondo constraint, so this is *not* the optimum — and the failed check tells us which constraint to activate next.

*Second guess: the item cap is active.* Keep $\underline{\theta}_1 = \overline{\theta}_1 = 0$ but let $\lambda_g > 0$, so complementary slackness forces $x_1 + x_2 = 5$:

$$\begin{aligned} 1 - \lambda_h - \lambda_g &= 0 \\ \frac{3}{x_2} - 0.5\lambda_h - \lambda_g &= 0 \\ x_1 + 0.5 \cdot x_2 &= 3 \\ x_1 + x_2 &= 5. \end{aligned}$$

You may verify that the solution is $\lambda_h = \lambda_g = 0.5$ and $(x_1, x_2) = (1, 4)$. Now every check passes: $x_1 = 1$ lies strictly inside $[0, 2]$ (so the domain multipliers are indeed zero), and both remaining multipliers are non-negative. By sufficiency (Appendix A.1), this *is* the optimum. Figure A.2 puts both candidates on the picture, and the geometry is worth a look, because it is the same geometry every time. Utility rises along the feasible segment in the direction of more Teslas, so the consumer walks up it until something stops her, and what stops her is the item cap. The optimum is therefore the corner where budget meets cap, with both constraints active. The two feet cut the segment off at the other end, but she never wants to go that way, which is exactly why their

multipliers are zero.

The multipliers now carry the economics, via eq. (50). $\lambda_g = 0.5$: at the margin, loosening Marie Kondo's rule is worth 0.5 of utility per item. $\lambda_h = 0.5$: a marginal euro is worth 0.5 in utility. Note that the euro was worth 1 in the first guess, before the item cap entered: with the cap binding, extra money is less useful, because the cheap way to spend it — more Teslas — is exactly what the cap forbids. Constraints interact through the multipliers, and reading those interactions is what the last step of the recipe is for.

One caution to close on, and it is Appendix A.4's: a multiplier is a *derivative*. Allowing a sixth item does not raise utility by 0.5, but by $3\ln(3/2) - 1 \approx 0.22$, because the value of loosening the cap falls as the cap loosens: at six items the cap no longer binds at all, so $\lambda_g$ has reached zero by the time we get there. In the rest of this note the value function is piecewise *linear* rather than curved, so a shadow price does stay valid over a range of the constraint — but only up to the next kink.

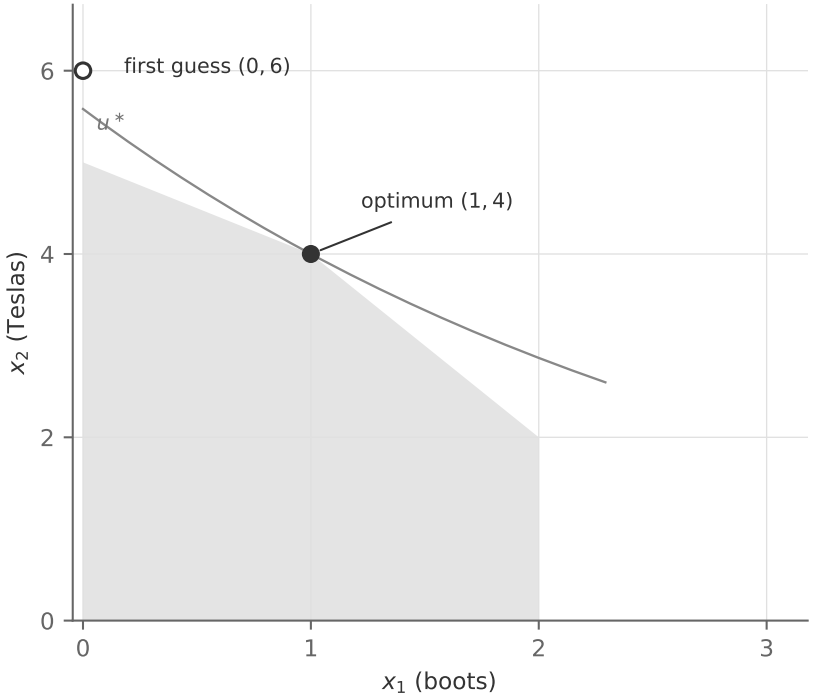



Figure A.2: The two candidates against the same feasible set, with the constraint lines dropped — the shape of the region carries them. The first guess ignores the item cap and lands outside it, which is what the check catches. The optimum is the feasible bundle on the highest indifference curve $u^*$: the whole feasible set lies below that curve, and everything the consumer would prefer to $(1, 4)$ lies outside it. Note that $u^*$ passes through the corner with a slope between those of the two constraints that meet there — the geometry behind $\lambda_h = \lambda_g = 0.5$.

# B Provenance: data and the network

Every number in this note is meant to be traceable to a source and reproducible. This appendix records where the inputs come from, and how the bidding-zone network of Sections 6 to 9 was derived.

The models themselves are built in PyPSA (Brown, Hörsch, et al. 2018), an open framework for power system analysis. Every model in this note *can* be solved with a free and open-source linear programming solver, and at the instance sizes the course repository ships by default that is what we recommend: nothing here requires a licence. The largest instances behind the note's own figures — the capacity expansion problems of Section 7 to Section 8, which run to around half a million variables — were solved with a commercial barrier solver under a free academic licence, purely for speed. The formulation is identical either way, and so is the answer. The course repository holds every model, dataset and script, together with the instructions for reproducing each figure and table here.

## B.1 Data sources

- **Danish market data** — Energi Data Service (Energinet), dataset `ProductionConsumptionSettlement`: hourly gross consumption and production by category for DK1 and DK2, 2019–2024. CC-BY 4.0, no API key. Availability profiles are production divided by its annual maximum — a classroom proxy that ignores within-year capacity growth. The same request also keeps production in *levels* by category (central thermal, local CHP, commercial) together with the cross-border exchange flows, which is what Figure 3.8 holds the modelled week against. All six of DK1's wires are read — Norway, Sweden, Germany, the Netherlands, Great Britain and DK2 — so that production plus net imports equals consumption hour by hour; omitting any of them silently breaks that identity.

- **Temperature** — Open-Meteo historical archive (derived from ERA5, a global reconstruction of past weather), hourly 2m temperature, Aarhus, 2024. CC-BY 4.0.

- **Zonal weather years** — the same archive, hourly wind speed at 100 m and surface radiation at each of the twelve zones' coordinates, 2015–2024. CC-BY 4.0, no account. The basis of Section 9's sweep; construction and caveats below.

- **Technology costs** — PyPSA's `technology-data` (PyPSA, Danish Energy Agency, et al. 2025) *fetched at a pinned tag* (`v0.13.2`) and reshaped, for each of the cost projection vintages 2025 to 2050 in five-year steps. It compiles the Danish Energy Agency's catalogue (Danish Energy Agency and Energinet 2024) together with several other published sources and converts everything to a common currency year. The small table of Table 1 is *generated* from the 2025 vintage; Sections 7 to 9 price investment on the 2050 vintage, and the cost scenarios of Section 9.1 are built from the fall between the 2025 and 2050 vintages. The compiled files carry no single licence, so nothing from them is redistributed by the course repository — students re-fetch byte-identically.

- **Battery costs** — the one place where the note overrides the compilation's investment costs, and deliberately so: the compilation's battery trajectory has fallen well behind the world market, its 2040 vintage costing roughly what turnkey systems sold for in 2025. Every vintage's

four-hour system cost is therefore capped at **165 €/kWh**, BloombergNEF's 2025 European average turnkey price (177 $/kWh in the *Energy Storage System Cost Survey 2025*, December 2025, press-release figures; converted at ∼1.07 USD/EUR; the global average was 117 $/kWh and the Chinese 73 $/kWh). The split between inverter (€/kW) and cells (€/kWh) keeps the compilation's own ratio, and vintages already below the cap are left alone: The cap binds for the 2025 and 2030 vintages only. It leaves the 2050 costs of Sections 7 to 9 untouched, but it shortens the projected fall from which Section 9.1 builds its cost scenarios, and with it the batteries' range across them. The currency conversion, and reading a nominal-2025 dollar price against the compilation's EUR-2020 basis, are part of the assumption; the processed data records the compilation's value beside the one used, in the same visible-override format as the fuel file.

All downloads land outside the published tree and every transformation is scripted, so the processed inputs are reproducible from the original sources.

## B.2 One set of plant characteristics

The note's data arrives in three tiers: the small table of Sections 2 to 5, the shipped network of Sections 6 to 9, and the investment costs of Sections 7 and 8. Each comes from a different source, and sources of this kind routinely disagree on fuel prices, so that the same combined-cycle plant would cost one amount to run in one section and another in the next. Three rules keep the tiers on one basis.

- **One file holds the fuel prices and carbon contents**, and all three tiers read it. It records the compilation's own published value beside the one this note uses, so every override is visible rather than buried; the note's prices are stylised 2025 forward levels, because the compilation's are inherited from a study over a decade old and sit well below what European plants have paid recently.
- **The small table is generated, not hand-written.** Every efficiency and operating cost in Table 1 is read from `technology-data` at the pinned tag, so the machines named there are the same machines, with the same numbers, as in the course's companion note on generation technologies — which derives its catalogue from the same source at the same pin.
- **The shipped network is re-priced when it is loaded.** Each thermal generator keeps the efficiency PyPSA-Eur gave it — that heterogeneity is real, and worth having — and takes this note's fuel price and operating cost. Carrier emission factors are set from the same file at the same time.

Two carbon entries are accounting choices rather than measurements. Biogenic carbon is counted as zero. Municipal waste carries only its fossil, largely plastic, fraction at 0.15 t per $\mathrm{MWh_{th}}$ — not the zero it is often waved through with, and enough, divided by a low electrical efficiency, to make waste one of the more carbon-intensive entries in the table.

Sections 7 to 9 read the 2050 vintage for *investment* costs, since they ask what it costs to build for 2050, and every other forward input they read, from fuel prices to demand, is at that same horizon; Section 9.1 varies that projection. The 2050 vintage's gas turbines are a few per cent more efficient than those of the small table, and the pipeline writes the difference out rather than leaving it implicit.

## B.3 The bidding-zone network

The network of Sections 6 to 9 was derived from **PyPSA-Eur** (Hörsch et al. 2018)[22] at release **v2026.08.0**, run once by us — the workflow's machinery (snakemake, atlite, a ∼7 GB weather cutout) is never a student dependency. The full configuration override is a few short blocks; everything else is the release default. The choices that matter:

- **Countries: DK, DE, SE, NO** — Denmark and its synchronous neighbours. The Dutch (COBRAcable) and British (Viking Link) interconnectors are thereby cut from the model.
- **Clustering: administrative, at bidding-zone level** — the transmission grid (built by PyPSA-Eur from OpenStreetMap data) is aggregated to the twelve actual market price areas rather than to synthetic clusters, so every zonal price in Section 6 has a direct Nord Pool/EPEX counterpart.
- **Weather and demand year: 2024** — the note's data year throughout, matched to the prebuilt `europe-2024-sarah3-era5` ERA5/SARAH-3 cutout, which is downloadable without a Copernicus account (2024 is a leap year; the workflow drops 29 February, leaving 8760 hours). Renewable profiles come from atlite on that cutout; loads are the realised 2024 hourly demand from an archived snapshot of the Transparency Platform of ENTSO-E, the association of Europe's transmission system operators; fleets from powerplantmatching and Open Power System Data (today's capacities). Reservoir hydro carries the 2024 runoff, with one caveat: the annual statistics used to normalise inflow levels end at 2023, so the 2024 level — and with it the Nordic water values of Section 6 — rests on a regression of those statistics on annual runoff rather than on a published figure. The resulting Norwegian inflow, 116 TWh, is within the historical range.
- **No carbon price** enters the marginal costs, and the exported network is *unsolved*: all optimisation, and any policy, happens in the note's own pipeline.

The workflow output (the clustered, electricity-only network) is then trimmed by a short export script: the sector-coupling placeholders PyPSA-Eur attaches for its downstream stages (zero-capacity hydrogen and battery components) are dropped, every capacity is fixed at its existing value, the zone codes of ENTSO-E's Ten-Year Network Development Plan (TYNDP) are renamed

[22] https://github.com/PyPSA/pypsa-eur; code MIT, data CC-BY-4.0/CC0 under REUSE annotations.

to the market names (DK1, DK2, DE, SE1–SE4, NO1–NO5), branches are renamed by interconnector, and time series are stored in single precision. The published artefact is ∼2 MB and loads with one line of PyPSA.

One adjustment and one honest limitation carry into Sections 6 to 9. The adjustment: PyPSA-Eur's AC capacities are thermal ratings, but a zonal market trades on commercial NTCs, which are much smaller on the borders — with the physical ∼10 GW on the Jutland–Germany border, DK1 would never decouple from the German price, which is not how the market behaves. The export therefore derates the two cross-border AC interconnectors (DK1–DE to 2.5 GW, DK2–SE4 to 1.7 GW) to the maximum NTCs published for 2024; intra-country AC and the DC cables keep their capacities. One wrinkle: the Øresund NTC is directionally asymmetric — 1.7 GW toward DK2 but 1.3 GW toward SE4 — and a symmetric AC line cannot carry that, so the export keeps the larger direction. The limitation is now a smaller one, and it is described in the next subsection.

## B.4 Calibrating the network to its year

Two properties of the shipped network needed attention before any section solved it, and Section 6, which holds the network against the 2024 market, needed one more.

**The hydro level.** The regression above lands Norwegian reservoir inflow at 124 TWh together with run-of-river, about ten percent below the 137.6 TWh that the Norwegian Water Resources and Energy Directorate (NVE) states as Norway's normal annual hydro production (reference period 1991–2020) and well below the 140 TWh Norway's plants produced in 2024 (Statistics Norway 2025, a record year, with reservoirs ending above their median, both figures are Statistics Norway's and NVE's). Every section's network is therefore put on the normal-year level when it is loaded: reservoir inflow is scaled so that inflow plus run-of-river over the year equals the normal production. That is the right level for the 2050 reference cases of Sections 7 to 9, which should not inherit either a dry regression or a record year. Solved at the shipped level, the model turns Norway into a net importer where the real Norway exported 18 TWh, and the whole Nordic plateau couples to the German thermal price.

**The internal Nordic grid.** The cross-border AC interconnectors were derated to commercial capacities on export, but the interconnectors inside Sweden and Norway and between them kept their thermal ratings, three to five times the transfer capacities the market trades on, so the model moves water south far more freely than the market can. Section 6 derates every interconnector whose capacity exceeds TYNDP's reference-grid transfer capacity to that figure, never uprating (10 interconnectors change, all of them Nordic).

**The year itself.** Section 6 scales Norwegian inflow once more, by a single factor found by a few secant steps, such that the solved hydro output equals the 2024 production (the factor on top of the normal level is 1.13; inflow and output differ because a full reservoir spills). Table 6 shows what

each step does, starting from the network at its normal-year level. Both steps are calibrations to an outcome, and the note says so where it uses them; they are confined to Section 6.

Table 6: What the two calibrations do to the Nordic price level, at the 65 €/t of Section 6, against the 2024 market.

| | NO2 | SE3 | SE4 | NO hydro | NO net export |
|---|---|---|---|---|---|
| | mean price, €/MWh | | | TWh | TWh |
| network at the normal-year hydro level | 61 | 59 | 59 | 125 | -1 |
| Norwegian inflow scaled | 58 | 58 | 58 | 140 | 14 |
| Nordic corridors at reference NTCs | 67 | 60 | 60 | 125 | -1 |
| both (the section's instance) | 59 | 58 | 58 | 140 | 14 |
| the 2024 market | 50 | 36 | 50 | 140 | 18 |

## B.5 Demand at the horizon

Sections 7 to 9 solve for 2050 and need its demand. Each zone's annual total comes from the TYNDP 2024 scenarios (ENTSO-E and ENTSOG 2024, Distributed Energy at 2050, averaged over the three climate years the workbooks carry), with the demand that rooftop generation serves behind the meter counted in; leaving it out would understate Danish demand by about a third. Only the annual total is used: the hourly shape stays the network's own 2024 profile, scaled to the new level.

Electric vehicles come on top, because TYNDP counts them outside its demand profiles. For every zone but Norway their charging energy is TYNDP's own projection, summed over cars, vans, buses and trucks. Norway is not in that dataset. For it the note takes the estimate of Statnett, the Norwegian transmission operator, that transport could draw about 26 TWh by 2050 (as quoted by Hvidsten et al. 2025), and counts 14 TWh of it as road vehicles, in line with the road-fleet estimates of DNV and NVE (12–15 TWh). The other 12 TWh (ferries, shore power, rail) is ordinary inflexible load. Section 7 adds all 26 TWh to fixed demand; Section 8 adds the 12 TWh and carries the 14 TWh in its vehicle fleet.

That fleet comes from the same scenarios' modelling inputs for electric vehicles: for each zone and vehicle type, the number of vehicles, their battery, home charger and charging efficiency, and the energy they draw from the grid in a year, with hourly profiles of driving, of the share of cars plugged in at home and of the minimum charge the batteries must keep. Only the passenger cars are flexible, as in TYNDP's own modelling; when the cars charge on arrival, a third of each hour's driving is charged in that hour and a third in each of the next two. Norway's fleet is Sweden's per head of population, each vehicle drawing more than a Swedish one so that the fleet draws the 14 TWh above, which gives about as many vehicles as stand behind DNV's estimate.

## B.6 The weather of the expansion sections

The network arrives with one year of availability profiles, computed on the full reanalysis grid with real power curves and land-use weighting. Section 6 solves on them, because it is held against the 2024 market and wants the best profiles available. The expansion sections need something the network cannot supply: Section 9's sweep re-solves the capacity expansion problem on one weather year at a time, and needs every zone to see the *same* year. So Sections 7 to 9 all solve on a second source: hourly wind speed at 100 m and surface solar radiation from the **Open-Meteo historical archive** (ERA5-derived, CC-BY 4.0, no account required — the same source as the temperature series of Section 5), for 10 years, at each zone's own coordinate in the network. Wind speed is converted to availability by the textbook power curve (nothing below 3 m/s, cubic to a rated 12 m/s, flat to a 25 m/s cut-out); radiation is scaled by 1000 W/m$^2$. Section 7 and Section 8 solve on the archive's 2024, so the reference case of Section 7 is one of the runs of Section 9's sweep, not a separate construction of the same year.

Two things about this construction matter more than the details.

**Levels come from the network, shapes from the archive.** Each swapped profile is rescaled so that its annual mean equals the mean of the profile the network already carried. The capacity factors therefore remain the carefully derived ones — computed with real power curves and land-use weighting over a grid — while the archive supplies only the timing and, crucially, the correlation between zones. A single point and a stylised power curve is a good way to know when a calm week happened and where; it is not a good way to know a capacity factor, and it is not asked to be.

**What it still simplifies.** One coordinate stands for a whole bidding zone. For Denmark that is defensible; for Germany it is not really, since a country that wide smooths its own wind across its geography in a way a single point cannot. The derived German profile is therefore more volatile than the German fleet actually is, and the cross-border correlations in the sweep are, if anything, a little too low. Demand is held at the network's values throughout, so the sweep does not capture the way a cold still week raises load at the same time as it removes wind — a correlation that would make the tails worse, not better. Both effects point the same way: the spread reported in Section 9 is a lower bound on the risk.

## B.7 How the year is aggregated

A year has 8760 hours, and a capacity expansion problem over twelve zones at full resolution runs to several million variables. The published instances therefore reduce the year to a smaller number of snapshots, and how this is done is an input like any other. Two schemes are implemented in the course repository, and the note uses one of them.

**Every $k$-th hour.** Keep one hour in $k$ and weight it by $k$. This is simple and unbiased for annual averages, but the kept hours are $k$ hours apart, so nothing shorter than $k$ hours can be

represented. At 1095 snapshots (every eighth hour) a day keeps three of its hours, and most of the daily cycle that batteries earn on is lost: the hydrogen scenario of Section 8 at its reference budget builds far fewer batteries than in the full year (44 against 98 GW) and more electrolysers (36 against 30 GW).

**Chronological segments.** Cut the year into 1095 contiguous segments of unequal length, between 2 and 26 hours (median 8), short where load and weather move and long where they do not, each entering the problem as one period at its hourly means and weighted by its duration. Chronology is intact, so storage of any duration sees the real sequence of the year; what is lost is the variation *within* a segment. This is the scheme every network section uses, Sections 6 to 9, at 1095 segments for the note's figures and 336 for the instances the repository ships by default.

**The check.** Both expansion sections were re-solved at all 8760 hours to grade the aggregation. The reference case of Section 7 builds 4.6 GW of battery in the full year against 3.2 on segments and 2.2 GW of offshore wind against 2.4, at a Danish bill net of trade of 2.27 against 2.28 bn€ and Danish emissions of 0.30 against 0.31 Mt. The bill, the emissions, offshore wind and the gas turbine agree; the battery does not, because the full year keeps the swings within each segment that segments average away, and those make a shorter battery worth building beside the 8-hour one (1.0 GW of 4-hour storage). In Section 8 the checks are the hydrogen and the vehicle-to-grid scenarios at the reference budget. In both, the carbon price and the system cost agree to within a few per cent (257 against 251 €/t with hydrogen, 270 against 266 €/t with vehicle-to-grid), as do the hydrogen scenario's electrolysers, and in both the battery again does not. The hydrogen scenario builds 98 GW of battery in the full year against 76 on segments. With vehicle-to-grid the segments build none, because the cars do all of the storing; the full year builds 38.5 GW beside them, and the cars give back less (113 against 137 TWh): The swings within each segment are worth a stationary battery even where the cars can store. Every number the note quotes from the network sections is the segmented one; the full-year figures are quoted where a section's headline depends on them. One consequence of segments is that every mean and every duration curve in Sections 6 to 9 weights each snapshot by the hours it stands for: a calm week that became one long segment counts for a week, not for one observation.

# C What is actually inside the models

Every model in this note is a simplification, and by now you have met the individual simplifications one at a time, in the "which way does it bias the answer?" boxes. This appendix collects them. It answers, in one place, the question you should ask of *any* energy system model somebody puts in front of you: What is in it, what is not, and which way does each omission push the result? Read it as a checklist rather than as prose.

## C.1 From the written models to the solved ones

From Section 6 onwards the simulations are assembled in a modelling framework from standard building blocks (a market, a generator, a storage unit, a line, a conversion between two markets) rather than written out equation by equation. Nothing is hidden by this: Every block expands into constraints of the kind written in the text, and every shadow variable the note reads is the dual of one of those constraints. The course repository documents the correspondence block by block; the note does not repeat it.

## C.2 What the models contain

The network behind Sections 6 to 9 is derived from PyPSA-Eur (Hörsch et al. 2018); Appendix B documents the derivation. What survives into our models is listed in Table 7.

## C.3 What the simulations add

The models written in the text are kept minimal, because every extra constraint is a page of algebra that buys no new economics. The *simulations* behind the figures of Sections 6 to 9 carry three further constraints, applied identically in every solve. Each is listed here with its parameters and its known limitations. None of the three changes the structure of any argument in the text, and their combined effect on the headline numbers is modest. One feature that is deliberately *not* here is an operational reserve requirement: Reserve and commitment detail belong to dedicated operational studies, and the capacity expansion literature this note builds toward omits them in comparable solves.

- **Ramping limits.** The slow thermal carriers get PyPSA-Eur's own hourly limits per unit of capacity: coal 0.9, lignite 0.6, nuclear 0.3 per hour. Biomass, which PyPSA-Eur does not rate, takes coal's 0.9 (the Danish units are converted coal boilers), and waste 0.6, from the Danish Energy Agency catalogue's note that the boiler regulates about 1% per minute; gas turbines traverse their whole range within an hour and get none. One wrinkle: Most figures solve on segments of the year several hours long (Appendix B.7), and there is no exact way to state an hourly ramp limit on a multi-hour step. Scaling the limit up by the step's length would say a coal plant can traverse its range between snapshots, but the aggregated series is already a smoothed version of the hourly path, so that reading compounds one optimism with another and every limit goes vacuous. The simulations instead impose the true hourly rates per step, deliberately conservative. Limitation: These are average-fleet numbers, and there are no start-up costs or minimum loads behind them (that would need integer variables, and the identity between dual and price would fail).

- **Forced outages.** Thermal availability is derated: 0.95 for gas and oil plants, 0.90 for coal, lignite, nuclear, biomass and waste. These are generic reliability assumptions, not statistics for these particular fleets. Limitation: Derating shaves every hour a little instead of removing

Table 7: What survives of the network into the models.

| Ingredient | As represented here |
|---|---|
| Geography | 12 real bidding zones, DK1, DK2, DE, SE1–SE4 and NO1–NO5, each with its own hourly price. |
| Generation fleet | 67 generators covering CCGT, OCGT, coal, lignite, nuclear, oil, biomass, waste, geothermal, run-of-river, onshore and offshore wind, and solar, at today's capacities. |
| Weather | Hourly availability profiles $\gamma_{g,h}$ per zone and technology: the network's own reanalysis profiles in Section 6, and the zonal weather archive in Sections 7 to 9, all twelve zones on one year at a time and correlated across the map as the atmosphere correlates them (Appendix B.6). |
| Transmission, DC | 9 named DC cables (Skagerrak, NordLink, Kontek, Konti-Skan, the Great Belt, Baltic Cable and others), each with a capacity limit and *controllable* flow in both directions. |
| Transmission, AC | 17 AC circuits with reactances, so flows obey Kirchhoff's voltage law: Loop flow is represented, and power cannot be routed freely within the synchronous area. Cross-border AC capacities are derated to commercial NTCs, and in Section 6 the internal Nordic interconnectors as well (Appendix B.4). |
| Hydro | 14 storage units, Nordic reservoirs and pumped hydro, with historical inflow, round-trip efficiencies, and durations from 19 hours up to several thousand hours. This is what makes the Nordic price duration curves flat, and the water value an equilibrium object the model computes rather than a number we assume. |
| Storage state | A cyclic condition: The state of charge must end the horizon where it began, so no energy is conjured by emptying the tank on the last day. |
| Curtailment | Represented, free, and unlimited (see below). |
| Sector coupling | Section 5: a heat market, a gas boiler, and a heat pump with an hourly, temperature-dependent COP. Section 8: the cars' batteries as storage between the grid and the driving, charged smartly or also discharged to the grid, and an industrial hydrogen demand served by reforming, imports, electrolysis and storage, with a turbine that may burn hydrogen back. |
| Demand curtailment | Sections 7 and 8: a step demand curve in every zone, with a last-resort block at the value of lost load and two cheaper blocks (Section 7.3). |
| Carbon policy | Section 7: a common carbon price on every zone, swept; and, as one experiment, a territorial $CO_2$ budget on Danish generation alone. Section 8: one $CO_2$ budget over all twelve zones, whose dual is the carbon price. |
| Investment | Section 7: the Danish fleet is wiped and rebuilt from continuous capacity variables priced at annualised capital cost, under published potentials; the neighbours keep their fleets and may add to them under the same kind of potential. Section 8: every zone builds beside its existing fleet, interconnectors may be bought. |

whole plants for whole weeks, so the model never experiences the correlated loss of several units at once, which is the tail event reserve margins exist for.

- **Transmission losses.** Each DC crossing loses a flat 3% (converter pair plus cable, independent of length); AC flows lose energy by a piecewise-linear approximation of the quadratic resistive loss. Limitation: The linearised loss enters as an inequality, so in a zero-price hour the optimiser may "burn" surplus energy in a line. Harmless here, where curtailment is free, but a known artefact of the formulation.

## C.4 What the models do not contain

Each row of Table 8 names an ingredient that is absent, says where it would enter, and states the *direction* in which its absence bends the answer.

Table 8: What the models do not contain, where it would enter, and the direction of the error its absence makes.

| Absent | What it would be | Direction of the error |
|---|---|---|
| **Start-up costs, minimum load, unit commitment** | Integer on/off decisions per plant per hour | Makes the problem non-convex, so the identity between dual and price would fail. Their absence understates the cost of cycling plants and overstates how cleanly the market can be priced. |
| **Value of lost load, in the dispatch models** | In Sections 3 to 6 load is a fixed number that must be served, effectively $v = \infty$ (Section 2.4). The expansion models of Sections 7 and 8 price it, as a three-block step demand curve | Removes the top of the price distribution wherever it is absent. Understates average prices, peaker revenue, storage value and demand-response value in the dispatch sections. In the expansion sections the remaining coarseness (three blocks rather than a demand curve) overstates the carbon price needed to hit a target. |
| **Uncertainty itself: forecast error and reserves** | Stochastic availability and outages; a reserve requirement per hour, holding headroom against them | The simulations derate for outages (Appendix C.3), but that is a deterministic proxy: The optimiser still sees every realisation in advance, and no reserve is ever held. The model sees *variability*, never *uncertainty*; it understates system cost and overstates how well a renewables-heavy system can be operated. |

*Table 8, continued*

| Absent | What it would be | Direction of the error |
|---|---|---|
| **Intra-zonal congestion** | A nodal network inside each zone (Section 6.1) | All congestion rents reported are cross-border only. Understates the true cost of moving power, and hides redispatch entirely. |
| **Reactive power, voltage, frequency** | The full AC power flow | Ancillary services are outside the model, as is most of what a system operator actually worries about minute to minute. |
| **Demand elasticity** | A downward-sloping demand curve per hour | Only the heat pump of Section 5 and the three blocks of Section 7 respond to price, and as step functions. Overstates the carbon price needed to hit a target, because almost nobody is ever asked to consume less. |
| **Must-run generation and negative bids** | Subsidised or inflexible plants bidding below zero | The model's price floor is zero; the market's is not. This is one of the three tail mismatches in Section 3.6. |
| **Market power** | Strategic withholding by pivotal suppliers | Model prices are a competitive *benchmark*: a lower bound on what a strategic player could extract. |
| **Lumpy investment, lead times, vintages** | Integer capacity, build delays, existing-plant ages | The expansion sections build fractional gigawatts instantly. They cannot speak to the *pace* of transition, nor to stranded assets. |
| **Technology learning** | Costs falling with cumulative deployment | Costs are fixed at a catalogue year; the model cannot represent the feedback from deployment to cost that has driven the actual solar and wind cost decline. |
| **CHP** | A fixed electricity-to-heat ratio (Section 5.4) | Misses the technology that historically defined Danish district heating, and the one that *decorrelates* heat and power prices. |

*Table 8, continued*

| Absent | What it would be | Direction of the error |
|---|---|---|
| **A hydrogen network** | Pipelines letting one zone sell hydrogen to another (Section 8.4) | Each zone's hydrogen sector is self-contained, so a zone with cheap power and no domestic hydrogen demand (Norway, here) gets no hydrogen sector at all and cannot export. Understates the electrolytic route, by the value Neumann, Zeyen, et al. (2023) measure. |
| **Everything outside four countries** | Zones beyond DK, DE, SE and NO | The footprint contains one very large hydro reservoir, so calm spells are easier to ride out than they are for Europe as a whole. Understates long-duration storage specifically. |
| **Offshore wind where none stands today** | Offshore candidates in zones with no existing offshore fleet | A zone with no plant of a kind today has no availability profile to build a candidate on, so six of the twelve zones cannot build offshore wind at all. Understates offshore, and pushes the build onto onshore wind and solar. |

**One that is not in the table, because it is not an omission.** Section 7 and Section 8 cap every zone, Denmark included, at the same published potentials, and both cap the biomethane all zones may burn at the same resource. What separates them is what each adds: Section 7 wipes the Danish fleet and keeps the wires fixed, while Section 8 builds beside every existing fleet and may buy wires. The two sections are therefore not two runs of one model, and a quantity carried from one to the other is a comparison of systems, not of scenarios.

## C.5 Three questions worth answering explicitly

Three entries above need a paragraph each.

**Is curtailment in the model?** Yes, and it is free. A renewable generator's constraint is an inequality, $q_{g,h} \leq \gamma_{g,h}\bar{q}_g$, so the planner may always use less than the weather offers, at no cost and with no limit. When it does, the spilled energy is curtailment (Section 3.3). Two consequences follow. First, this is why the model's price floor is zero rather than negative: The marginal generator in a surplus hour is a wind turbine that is indifferent, not one paying to stay on. Second, free curtailment is optimistic. Real curtailment is contested, sometimes compensated, occasionally

physically constrained, and in some markets a subsidised generator would rather bid negative than be curtailed at all.

**Are ramping or flexibility costs in the model?** In the written models, no: Every generator can move from zero to full output between two consecutive hours, at no cost. Section 4.3 derives what changes when it cannot (the merit order becomes time-dependent and plants enter it at an effective marginal cost $c_g + \xi_{g,h} - \xi_{g,h+1}$), precisely so that you can reason about the size and sign of the omission. The simulations do impose ramp limits on coal, lignite, nuclear, biomass and waste, held at their hourly values even on the multi-hour segments they solve on, and Figure 4.3 sweeps the rate on DK1's own slow plant, but ramping carries no *cost* anywhere. The note's models remain systematically optimistic about how easily a renewables-heavy system can be operated, just a little less so than the written versions alone would be.

**Do the storage plants have perfect foresight?** Yes, completely. The optimisation is a *single* problem covering the whole horizon, so every battery and every Norwegian reservoir chooses today's charging knowing every price for the rest of the year. This matters more the longer the storage; Section 9.2 prices it for the battery of Section 4 and says what to expect for a reservoir.

## C.6 The bias index

Finally, Table 9 collects the note's "which way does it bias the answer?" boxes. The habit they represent: Name the assumption, then work out the sign of its error before you argue about the number.

None of this is an argument against using the model. Every alternative to a transparent optimisation model is a less transparent one, and a model whose omissions are catalogued, with signs, is far more useful than one whose author has not thought about them. The point of the catalogue is that it converts "the model is a simplification" from an excuse into an instrument: Given a result you are unsure about, you can now find the row that threatens it and say which way it moves.

Table 9: The bias index: every bias box of the note, collected.

| Assumption | Where | Sign of the error |
|---|---|---|
| Load always served ($v = \infty$) | Section 2.4 | Understates prices, peaker revenue, storage and demand-response value |
| One zone, one country, one direction of trade | Section 3.5 | Understates the value of Danish wind (no exports); overstates the value of the wires; understates the network cost of VRE |
| Frictionless ramping, perfect foresight for storage | Section 4.3 | Overstates fleet flexibility; understates the value of everything that supplies flexibility; overstates storage value |
| Constant COP (had we used one); a fixed fleet | Section 5.4 | Would overstate heat pump value and understate the peak load electrification adds; overstates electrification's emissions |
| Zonal prices, fixed NTCs, perfect foresight over inflow | Section 6.4 | Understates the cost of moving power and the Nordic cable rents; overstates Nordic prices in a wet year |
| Perfect foresight in investment | Section 7.4 | Understates transition cost; overstates confidence in the mix |
| No hydrogen network; a small footprint; a biogenic backstop | Section 8.4 | All three understate the case for hydrogen as electricity storage |
| One point per zone; demand held fixed across weather years | Appendix B.6 | Understates cross-zone correlation and the cold-still-week coincidence of high load with low wind; the weather spread of Section 9 is a lower bound |
| A cost projection read as a probability | Section 9.1 | Understates cost uncertainty: The database publishes one number per technology per year, not a distribution, and its past errors on solar and wind have run in one direction |

# D What the literature finds, and where this note sits

Section 8 tests two published claims and reports the result. This appendix does the wider job: It collects what a handful of recent studies of the European system actually find, says which of their conclusions this note's models are equipped to reproduce, and says which ones they are *not* equipped to reproduce, and in which direction that bends our answers.

Two warnings about how to read it. First, every study below is between five and fifteen times the size of the model of Section 8, and most of them cover all of Europe and every sector; where we disagree with them, the prior should be that they are right and we are small. Second, this is a short list chosen for what it teaches, not a systematic review. It is enough to place this note honestly and not enough to settle anything. Everything cited here is on the course reading list, with an open-access link wherever the authors have provided one. You are not expected to have read any of it before this appendix; you are expected, afterwards, to be able to open one of these papers and work out which ingredient produced its headline result.

## D.1 Four strands

**(i) The division of labour between storage technologies.** Victoria et al. (2019) is the canonical statement, and it is the paper Section 8 is written against. Its findings, in its own terms: Electric batteries act as short-term storage counterbalancing solar, while hydrogen storage smooths wind's longer fluctuations; significant storage appears only once $CO_2$ is cut by more than about 80% of 1990 levels; and at a 95% cut the optimal system holds battery energy of about 1.4× and hydrogen storage of about 19.4× average hourly electricity demand. It also finds that flexibility from electric-vehicle batteries *avoids the need for additional stationary batteries* and reduces the use of pumped hydro.

**(ii) Offshore wind is an infrastructure question.** Once land binds, how much offshore wind a system can absorb depends on what it is allowed to build to bring the energy ashore. Glaum et al. (2024) put numbers on this for the North Sea: With point-to-point connections of the kind built today, 310 GW can be integrated; allow meshed offshore grids and offshore hydrogen and the figure rises to 420 GW, with savings up to 15 bn€ a year. Floating wind, about 75 GW, appears *only* when offshore hydrogen production is available, and the bulk of the offshore energy comes ashore as hydrogen rather than as electricity, by about two to one.

Lüth et al. (2023) ask the same question at Danish scale, for the planned energy islands, and find that distance decides: Close-to-shore farms serve the power system best through cables, while electrolysis earns its keep far from shore because it avoids expensive long-distance cable. Their sharpest result for our purposes is a different one: **Electrolyser investment is sensitive to the hydrogen price and much less so to the carbon price.**

**(iii) Flexibility is a taxonomy, not a quantity.** Chyong et al. (2024) classify what a 2050 European system needs and who supplies it: Hydrogen storage and production carry the inter-seasonal margin; a combination of electrical storage, hydrogen-based storage and hybrid heat pumps carries the intraday one; and *spatial* flexibility, national networks and cross-border interconnection, is a source in its own right rather than a way of delivering the others. Their headline comparative claim is that in deep-decarbonisation scenarios cross-border *electricity* transmission capacity comes to exceed that of natural gas, while the need for gas networks and storage falls substantially.

Odenweller et al. (2026) approach flexibility through prices, by coupling a long-horizon integrated assessment model to an hourly power system model. Their result is an asymmetry worth remembering: Flexible loads (electrolysers, smart-charging vehicles) pay *below*-average electricity prices, while heat pumps pay almost *twice* the average, because their demand peaks in winter when the system is tight. Remove demand-side flexibility and prices rise for every end-user.

**(iv) Hydrogen wants a network.** Neumann, Zeyen, et al. (2023) price the thing this note most conspicuously lacks: A European hydrogen network is worth up to 26 bn€ a year, about 3.4% of system cost, and between 64% and 69% of it could be repurposed natural-gas pipeline rather than new build. Expanding the electricity and hydrogen grids together saves the most, around 10%.

## D.2 Which conclusion becomes which ingredient

The table below is the design record for Section 7 and Section 8: For each conclusion above, the model ingredient it implies, and whether this note carries it. It is worth reading as an argument about *modelling* rather than about Europe. A conclusion in this literature is almost always the visible end of one ingredient, and the way to read any such study is to ask which ingredient produced it.

Table 10: Conclusions of the expansion literature, the model ingredient each implies, and whether this note carries it.

| Conclusion | Ingredient it implies | In? | Where, or why not |
|---|---|---|---|
| Batteries do the diurnal work | Storage with a duration choice | ✓ | Section 4; a duration menu in Section 7 |
| Hydrogen does the seasonal work | Electrolyser, storage, turbine | ✓ | Section 8.4: the electrolyser built, the turbine declined, and the seasons balanced by reformers burning biomethane |
| EV batteries displace stationary ones | The cars' batteries as storage, with smart charging and vehicle-to-grid | ✓ | Section 8.3 |
| A biogenic fuel puts a plateau into the abatement curve | Biomethane as a drop-in fuel | ✓ | Section 7.4: the switching price, and the plateau every scenario of Section 8 crosses |
| Spatial flexibility is its own category | Buyable interconnection | ✓ | Section 8: bought only once the budget binds |
| Demand elasticity shapes price formation | A step demand curve | ✓ | The three blocks of Section 7, in every zone |
| Firm low-carbon capacity competes for the same role | Nuclear candidate | ✓ | On the menu in both sections; never built |
| Hydrogen wants a network | Hydrogen transport between zones | — | **Absent.** The largest single omission; see Appendix C |
| Offshore integration is an infrastructure choice | Offshore hubs, meshed grid, offshore electrolysis | — | **Absent.** Our offshore wind lands directly in its zone |
| Costs fall with deployment | Endogenous learning | — | **Absent.** Costs are a catalogue year (Section 9.1 varies them exogenously) |
| Sector coupling deepens the storage threshold | Heat, transport, industry demand | partly | Heat in Section 5, transport and industrial hydrogen in Section 8; no aviation, shipping or industrial process heat |

## D.3 Where Section 8 agrees, and where it does not

Now the comparison, in Table 11. Our numbers are the ones Section 8 reports; theirs carry citations.

Down the right-hand column, the pattern is this. Where this note carries the ingredient, it broadly reproduces the literature: the storage threshold, the EV substitution, the value of spatial

Table 11: Published claims of the expansion literature against what Section 8 finds.

| Published claim | What we find | Reading |
|---|---|---|
| Storage matters only below ∼20% of 1990 emissions (Victoria et al. 2019) | Battery build goes 2.9 GW unconstrained → 59.5 at −75% → 94.7 at −99% | **Agrees.** Storage is a deep-end phenomenon in our model too |
| EV flexibility avoids the need for *additional* stationary batteries (Victoria et al. 2019; Hvidsten et al. 2025) | Smart charging displaces 48% of the battery at −75% and 16% at −99%; vehicle-to-grid builds none at any budget, though the full year keeps 38.5 GW at −99% | **Agrees, and quantifies.** Partial with smart charging; with vehicle-to-grid complete on segments, not in the full year (Appendix B.7) |
| Hydrogen storage smooths wind's synoptic variability (Victoria et al. 2019) | 0.0 GW of hydrogen turbine and 0.0 TWh burnt back, at every budget | **Disagrees**, and Section 8.4 says why. Our footprint is small, and biomethane at one price all year balances the seasons |
| Cross-border electricity capacity grows and overtakes gas (Chyong et al. 2024) | 17.6 of 30 GW of offered interconnection bought, and only once the budget binds | **Agrees in direction.** We cannot speak to the comparison with gas |
| Electrolyser investment tracks the hydrogen price more than the carbon price (Lüth et al. 2023) | Electrolysers enter as the *carbon* budget tightens, against a fixed import price | **Apparent tension**, and an untested one: We never varied the hydrogen price |
| A hydrogen network is worth up to 26 bn€/yr (Neumann, Zeyen, et al. 2023) | No hydrogen trade between zones at all | **Untested.** Known sign: We understate the electrolytic route |
| Offshore integration reaches 420 GW with meshed grids and offshore hydrogen (Glaum et al. 2024) | Offshore lands in its zone; no offshore hubs exist | **Untested.** We cannot represent the question |
| Flexible loads pay below-average prices; heat pumps almost double (Odenweller et al. 2026) | Not computed: We report system prices, not per-technology ones | **Not computed.** Everything needed for it is in the solved model |

flexibility. Where it disagrees outright, the merit order says why: The hydrogen turbine is the gas turbine burning a dearer fuel (Section 8.4). And where it is silent, it is silent because an ingredient is missing, not because the question is hard: Three of the eight rows are omissions with a known direction of error, listed in Appendix C. A small model will not tell you the cost of decarbonising Europe. It will tell you which ingredient produces which conclusion, and that is what lets you read the studies that do.

## D.4 Questions this leaves open

Each of the following is a claim above that this note's models could be made to test, in roughly increasing order of effort. They are here because they are the natural next things to do with the code, and because a reader who works through one of them will have done something closer to research than to an exercise.

i. *Find the break-even fuel price.* Section 8.4 argues that a hydrogen turbine is built only where

hydrogen is cheaper than the low-carbon fuel the same turbine could burn instead, in the hours when power is dear, and in Section 8 it never is. Raise the price of biomethane, or lower its ceiling, until German hydrogen storage and a turbine are built, and compare the electrolysers' running hours then with the 4,200 the industrial customer delivers now.

ii. *Separate the two prices.* Lüth et al. (2023) find electrolyser investment more sensitive to the hydrogen price than to the carbon price. Vary each in turn, the import price at a fixed budget and the budget at a fixed import price, and see whether that holds here.

iii. *Who pays what?* Compute the load-weighted average price paid by each flexible block and by the electrolysers, against the simple average price, and test the asymmetry of Odenweller et al. (2026). Nothing new needs solving; the prices are already in the solution.

iv. *Price spatial flexibility.* Re-solve the baseline of Section 8 with the interconnectors fixed instead of buyable, and read the difference as the value of the option to build wire.

v. *Give hydrogen a wire.* Allow hydrogen to move between two zones and see whether Norway, which has cheap power, no domestic hydrogen demand, and therefore no hydrogen sector at all, builds an electrolyser. This is the smallest edit in the list with the largest payoff against Neumann, Zeyen, et al. (2023).

# E The extensions of Section 8 written out

Sections 8.3 and 8.4 describe in words what the cars and the hydrogen sector add to the baseline model of Section 8.2. This appendix writes both extensions out as constraints and derives the optimality conditions that the section reads in words. As in the body, what is new relative to the model of Section 7 is in blue.

## E.1 The cars as storage

**Smart charging.** The fleet of a zone is described by the numbers that described the battery of Section 4, with three differences. First, the power rating is that of the cars' home chargers together, $\bar{s}_z^{EV}$, and only the share $\gamma_{z,h}^{EV}$ of the cars that is plugged in at hour $h$ can use it, an availability like a generator's, which falls in the hours people drive. Second, the energy capacity $\overline{soc}_z^{EV}$ is the share $\chi$ of the cars' battery capacity that the owners make available to the system; we follow Victoria et al. (2019) and set $\chi$ to one half, which gives storage for more than a week of the fleet's driving. Third, the batteries have a drain and a floor: The driving draws $D_{z,h}^{EV}$ from them whatever the price, and they must hold at least $\underline{soc}_{z,h}^{EV}$ for the trips ahead, a share of their capacity that is highest in the morning. Of each MWh drawn from the grid, $\eta^{EV}$ reaches the batteries. The planner chooses the charging $g_{z,h}^{EV,c}$, which takes the place of the fixed charging profile on the

demand side of the zone's market clearing, and the batteries' state of charge:

$$\sum_{g\in\mathcal{G}_z} q_{g,h} + \sum_{s\in\mathcal{S}_z} g^d_{s,h} + \sum_{\ell:\, b_\ell = z} \eta_\ell\, f_{\ell,h}$$
$$= \sum_j d_{j,z,h} + \sum_{s\in\mathcal{S}_z} g^c_{s,h} + \sum_{\ell:\, a_\ell = z} f_{\ell,h} + g^{EV,c}_{z,h} \qquad (\lambda_{z,h}) \qquad \forall z,h \qquad (53a)$$

$$soc^{EV}_{z,h} \;=\; soc^{EV}_{z,h-1} + \eta^{EV} g^{EV,c}_{z,h} - D^{EV}_{z,h} \qquad (\theta^{EV}_{z,h}) \qquad \forall z,h \qquad (53b)$$

$$\underline{soc}^{EV}_{z,h} \;\le\; soc^{EV}_{z,h} \;\le\; \overline{soc}^{EV}_{z} \qquad (\underline{\psi}^{EV}_{z,h}, \overline{\psi}^{EV}_{z,h}) \qquad \forall z,h \qquad (53c)$$

$$0 \le g^{EV,c}_{z,h} \;\le\; \gamma^{EV}_{z,h}\, \bar{s}^{EV}_z \qquad (\underline{\phi}^{EV,c}_{z,h}, \overline{\phi}^{EV,c}_{z,h}) \qquad \forall z,h \qquad (53d)$$

with the batteries ending the year where they started, $soc^{EV}_{z,0} = soc^{EV}_{z,T}$. Neither the batteries nor the chargers cost anything: They are bought for driving, not for the power system.

Compare eq. (53b) with a battery's law of motion (40f): The car battery is the battery of Section 4 with a drain, which it must serve whatever the price, and without a way back to the grid. Differentiating the Lagrangian with respect to $g^{EV,c}_{z,h}$ gives the charging condition

$$-\lambda_{z,h} + \eta^{EV}\theta^{EV}_{z,h} + \underline{\phi}^{EV,c}_{z,h} - \overline{\phi}^{EV,c}_{z,h} \;=\; 0, \qquad (54)$$

so the cars charge flat out when the price is below the value $\eta^{EV}\theta^{EV}_{z,h}$ of what reaches their batteries, not at all when it is above, and anywhere in between when the two are equal, in which case they set the price. The condition for the batteries' state of charge is a battery's, $\theta^{EV}_{z,h+1} - \theta^{EV}_{z,h} = \overline{\psi}^{EV}_{z,h} - \underline{\psi}^{EV}_{z,h}$, so $\theta^{EV}_{z,h}$ stays put until the batteries reach their ceiling or their floor, and between two such hours the fleet buys its energy in the cheapest hours it can reach.

**Vehicle-to-grid.** Vehicle-to-grid lets the batteries discharge to the grid as well, through the chargers of the participating cars, $\chi\gamma^{EV}_{z,h}\bar{s}^{EV}_z$ measured at the batteries, of which $\eta^{EV}$ reaches the grid. The cars then deliver $g^{EV,d}_{z,h}$ to the supply side of eq. (53a), and their law of motion and discharge limit are

$$soc^{EV}_{z,h} = soc^{EV}_{z,h-1} + \eta^{EV} g^{EV,c}_{z,h} - g^{EV,d}_{z,h}/\eta^{EV} - D^{EV}_{z,h} \qquad (\theta^{EV}_{z,h}) \qquad \forall z,h \qquad (55a)$$

$$0 \le g^{EV,d}_{z,h} \;\le\; \eta^{EV}\chi\,\gamma^{EV}_{z,h}\,\bar{s}^{EV}_z \qquad (\underline{\phi}^{EV,d}_{z,h}, \overline{\phi}^{EV,d}_{z,h}) \qquad \forall z,h \qquad (55b)$$

A discharge carries the stationary battery's small operating charge $c^s$ in the objective, so the cars do not cycle between hours of equal price; nothing else in the model wears their batteries. The discharge condition is

$$-c^s + \lambda_{z,h} - \theta^{EV}_{z,h}/\eta^{EV} + \underline{\phi}^{EV,d}_{z,h} - \overline{\phi}^{EV,d}_{z,h} \;=\; 0, \qquad (56)$$

so the cars sell when the price is above $\theta^{EV}_{z,h}/\eta^{EV}$, plus the operating charge. Together with eq. (54), this is the arbitrage band of Result 3: The cars buy below $\eta^{EV}\theta^{EV}_{z,h}$, sell above $\theta^{EV}_{z,h}/\eta^{EV}$, and stand still in between.

Victoria et al. (2019) model smart charging the same way, with one difference that matters: Their cars must be three quarters full every morning, which keeps the shifting within a day, while ours keep only the grid operators' minimum charge, a fraction of that, and can shift their charging across days.

## E.2 The hydrogen sector

**The model.** Each zone with a hydrogen demand gets a second market, as Section 5 added a market for heat. Its demand $D_z^{H_2}$ is flat through the year. *Reformers* make hydrogen from natural gas at a marginal cost $c_g$ and emit $e_g$ per MWh of it; like the gas plants, they may burn biomethane instead, and their capacity is bought like any other candidate's. *Imports* are a step supply curve, a quantity at a price for each import route. Both are generators of the hydrogen market, $g \in \mathcal{G}_z^{H_2}$, so the generator rows eqs. (40d) to (40e) cover them: A reformer is a candidate, and an import step a generator with a fixed capacity and no emissions.

*Electrolysers* turn electricity $q_{z,h}^{EL}$ into $\eta^{EL} q_{z,h}^{EL}$ of hydrogen, and a *turbine* burns hydrogen $q_{z,h}^{T}$ back into $\eta^{T} q_{z,h}^{T}$ of electricity; both capacities, $\bar{q}_z^{EL}$ and $\bar{q}_z^{T}$, are bought, at annual fixed costs $F^{EL}$ and $F^{T}$ per MW. *Storage facilities* $s \in \mathcal{S}_z^{H_2}$ move hydrogen in time, in salt caverns where the geology allows and in tanks everywhere. A hydrogen storage facility is the battery of Section 4 without losses and without a power rating: What it releases, $g_{s,h}^{S}$, is negative when it fills, it ends the year where it began, and its energy capacity $\overline{soc}_s$ is bought at $F_s^{soc}$ per MWh, up to the geology's ceiling $\overline{soc}_s^{\max}$ in a cavern. The hydrogen scenario keeps the smart-charging cars of appendix E.1. A zone's two markets and the hydrogen chain are

$$\sum_{g \in \mathcal{G}_z^{H_2}} q_{g,h} + \eta^{EL} q_{z,h}^{EL} + \sum_{s \in \mathcal{S}_z^{H_2}} g_{s,h}^{S} = D_z^{H_2} + q_{z,h}^{T} \qquad (\lambda_{z,h}^{H_2}) \qquad \forall z, h \qquad (57a)$$

$$\sum_{g \in \mathcal{G}_z} q_{g,h} + \sum_{s \in \mathcal{S}_z} g_{s,h}^{d} + \sum_{\ell:\, b_\ell = z} \eta_\ell \, f_{\ell,h} + \eta^{T} q_{z,h}^{T} = \sum_{j} d_{j,z,h} + \sum_{s \in \mathcal{S}_z} g_{s,h}^{c} + \sum_{\ell:\, a_\ell = z} f_{\ell,h} + g_{z,h}^{EV,c} + q_{z,h}^{EL} \qquad (\lambda_{z,h}) \qquad \forall z, h \qquad (57b)$$

$$0 \le q_{z,h}^{EL} \le \bar{q}_z^{EL} \qquad (\underline{\mu}_{z,h}^{EL}, \overline{\mu}_{z,h}^{EL}) \qquad \forall z, h \qquad (57c)$$

$$0 \le q_{z,h}^{T} \le \bar{q}_z^{T} \qquad (\underline{\mu}_{z,h}^{T}, \overline{\mu}_{z,h}^{T}) \qquad \forall z, h \qquad (57d)$$

$$soc_{s,h} = soc_{s,h-1} - g_{s,h}^{S} \qquad (\theta_{s,h}) \qquad \forall s, h \qquad (57e)$$

$$0 \le soc_{s,h} \le \overline{soc}_s \qquad (\underline{\psi}_{s,h}, \overline{\psi}_{s,h}) \qquad \forall s, h \qquad (57f)$$

$$0 \le \bar{q}_z^{EL}, \quad 0 \le \bar{q}_z^{T}, \quad 0 \le \overline{soc}_s \le \overline{soc}_s^{\max} \qquad (\underline{\beta}_z^{EL}; \underline{\beta}_z^{T}; \underline{\beta}_s, \overline{\beta}_s) \qquad \forall z, s \qquad (57g)$$

and the objective gains their fixed costs, $-\sum_z \left(F^{EL} \bar{q}_z^{EL} + F^{T} \bar{q}_z^{T}\right) - \sum_s F_s^{soc} \overline{soc}_s$. As in Section 5, neither converter carries a fuel cost of its own: The electricity an electrolyser uses is priced by $\lambda_{z,h}$, and the hydrogen a turbine burns by $\lambda_{z,h}^{H_2}$.

**The price of hydrogen.** Three optimality conditions carry the economics of the sector. A

reformer that runs below its capacity has

$$\lambda^{H_2}_{z,h} \;=\; c_g + \tau e_g, \tag{58}$$

so whenever reforming is at the margin, the price of hydrogen is the cost of reforming plus the carbon tax on what reforming emits. A reformer burning biomethane emits nothing, so once $\tau$ is above the switch price, its condition reads $\lambda^{H_2}_{z,h} = p^F_{\text{bio}}/\eta_{\text{SMR}}$, with $p^F_{\text{bio}}$ the price of biomethane and $\eta_{\text{SMR}}$ the reformer's efficiency: the price of biomethane, grossed up for the losses of reforming. An electrolyser satisfies

$$-\lambda_{z,h} + \eta^{EL}\lambda^{H_2}_{z,h} + \underline{\mu}^{EL}_{z,h} - \overline{\mu}^{EL}_{z,h} \;=\; 0, \tag{59}$$

so it runs flat out whenever electricity is cheaper than the hydrogen it makes, $\lambda_{z,h} < \eta^{EL}\lambda^{H_2}_{z,h}$, and its capacity is worth building when those margins, summed over the hours it runs flat out, cover its annual fixed cost: $\sum_h \overline{\mu}^{EL}_{z,h} = F^{EL}$ for an electrolyser that is built, eq. (42) for a converter. The turbine's condition is the mirror image,

$$\eta^T\lambda_{z,h} - \lambda^{H_2}_{z,h} + \underline{\mu}^{T}_{z,h} - \overline{\mu}^{T}_{z,h} \;=\; 0, \tag{60}$$

so it runs when $\eta^T\lambda_{z,h} > \lambda^{H_2}_{z,h}$. Put together, a round trip through storage pays only when the price at which the electricity is sold back exceeds the price at which it was bought by the factor $1/(\eta^{EL}\eta^T)$, and only in hours that storage can connect.

**The carbon price set by an import step.** At a cut of 75%, the price of hydrogen is the price of the dearest import step in use, and that step sets the carbon price too. A reformer runs all year, so the hydrogen price must cover its fuel, its capital and its carbon, and eq. (58) with the capital added gives

$$\tau \;=\; \frac{\lambda^{H_2} - c_{\text{SMR}} - F_{\text{SMR}}/8760}{e_{\text{SMR}}} \;=\; \frac{69 - 25.08 - 2.95}{0.275} \;=\; 149\ €/\text{t}, \tag{61}$$

the solved tax to the euro. This is the plateau of Section 7.4 again: The marginal abatement is a step in a supply curve, here a step of hydrogen imports, and $\tau$ stays put until the step is used up.